

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

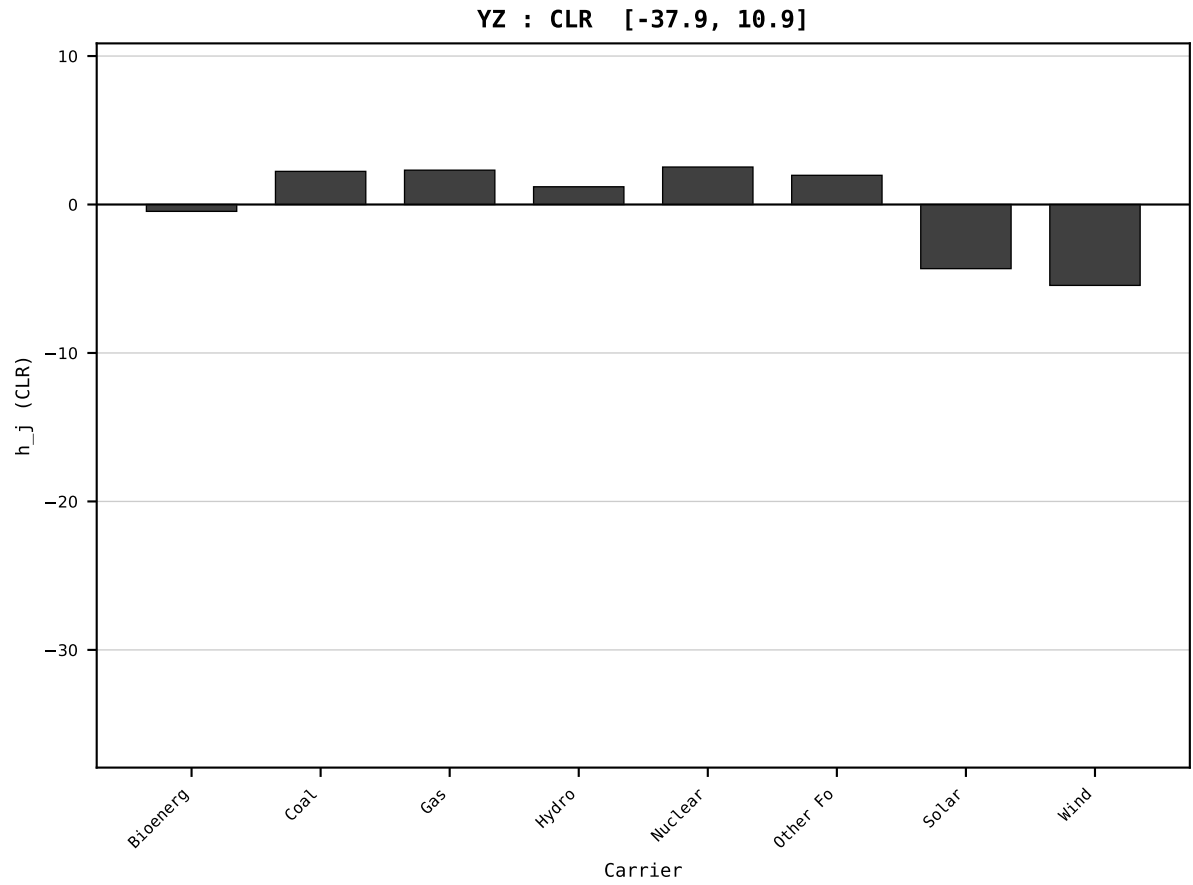
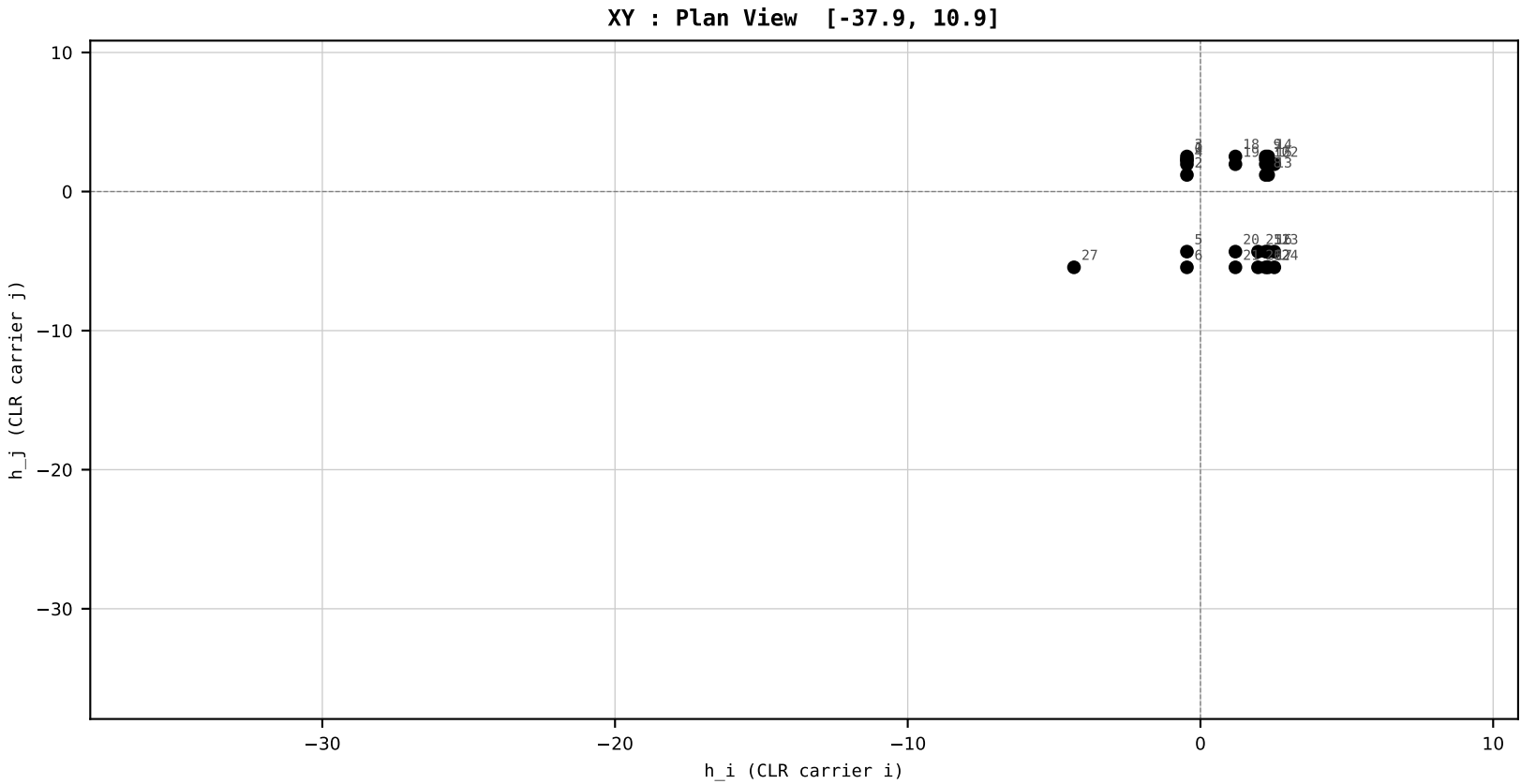
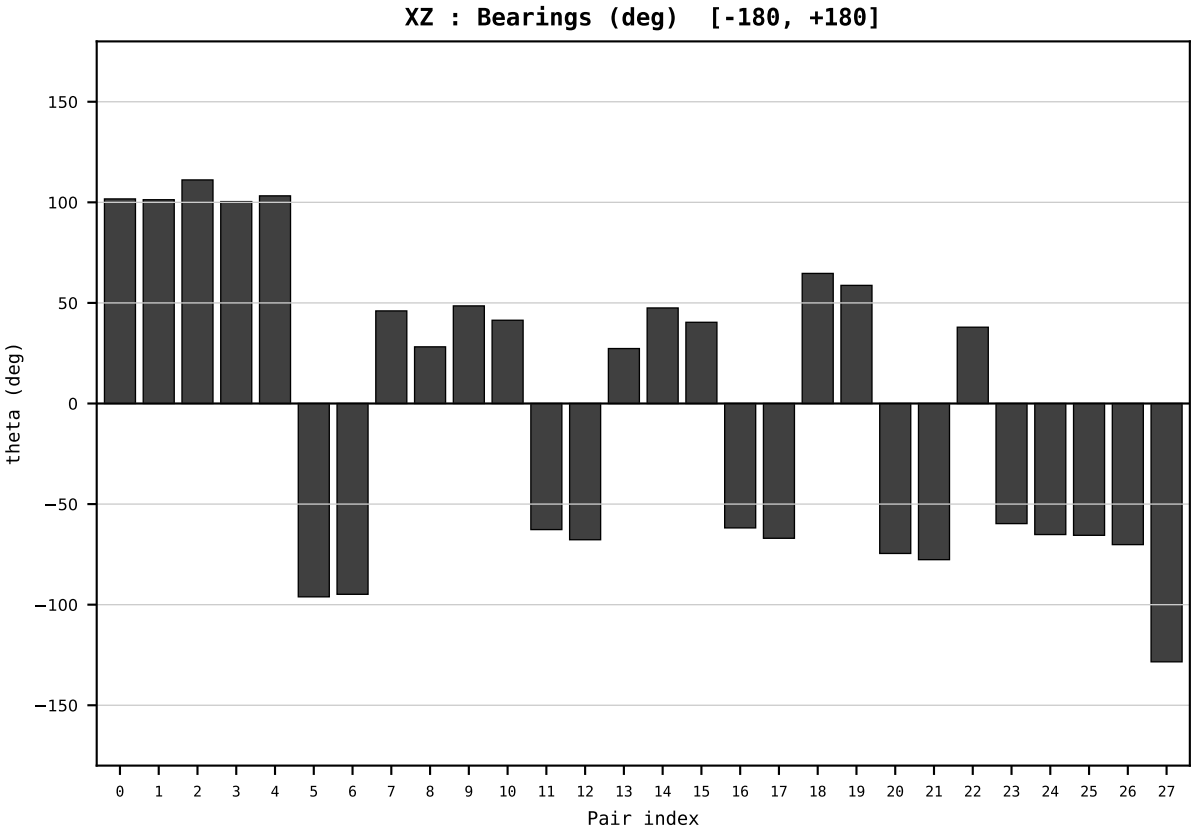
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234683
Ring = Hs-2
E_metric= 70.5673
kappa_HS= 2901.09
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 7.973 HLR
DR ratio= 2901.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

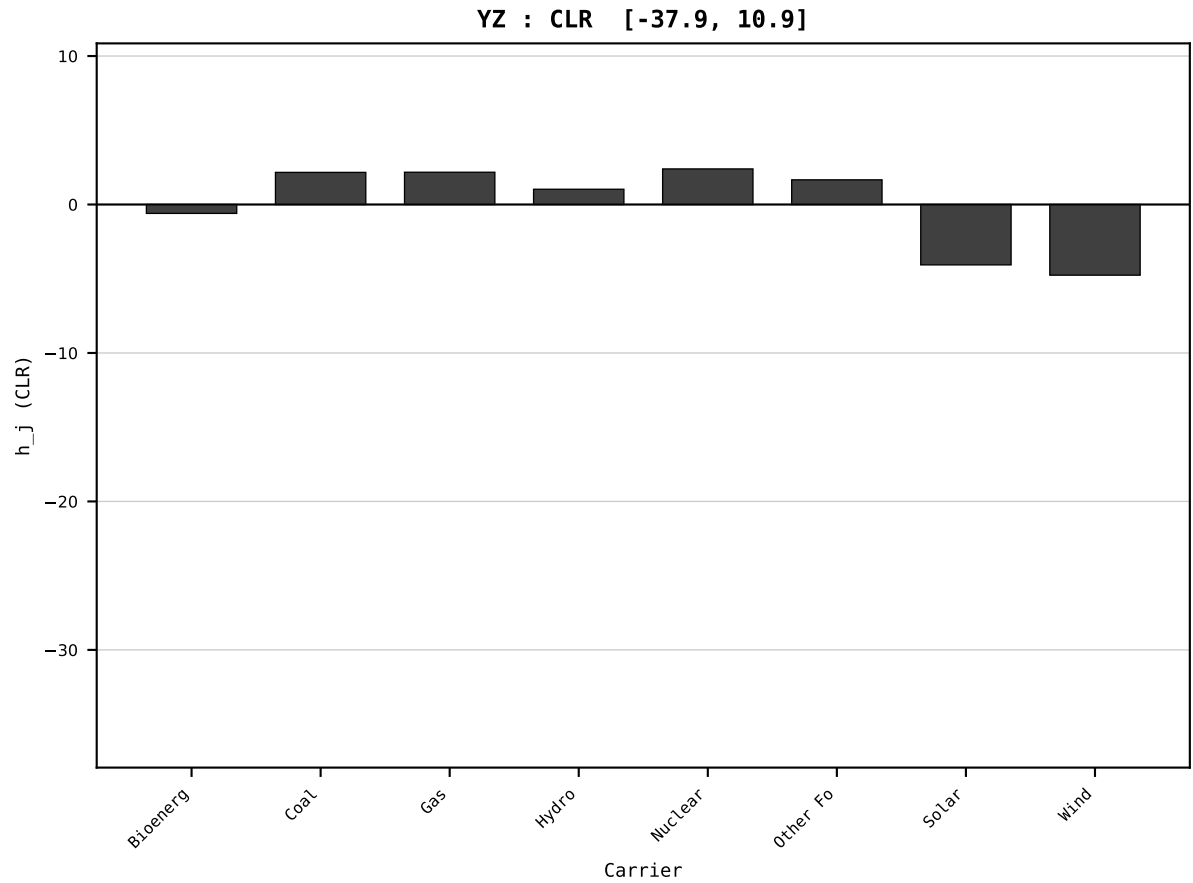
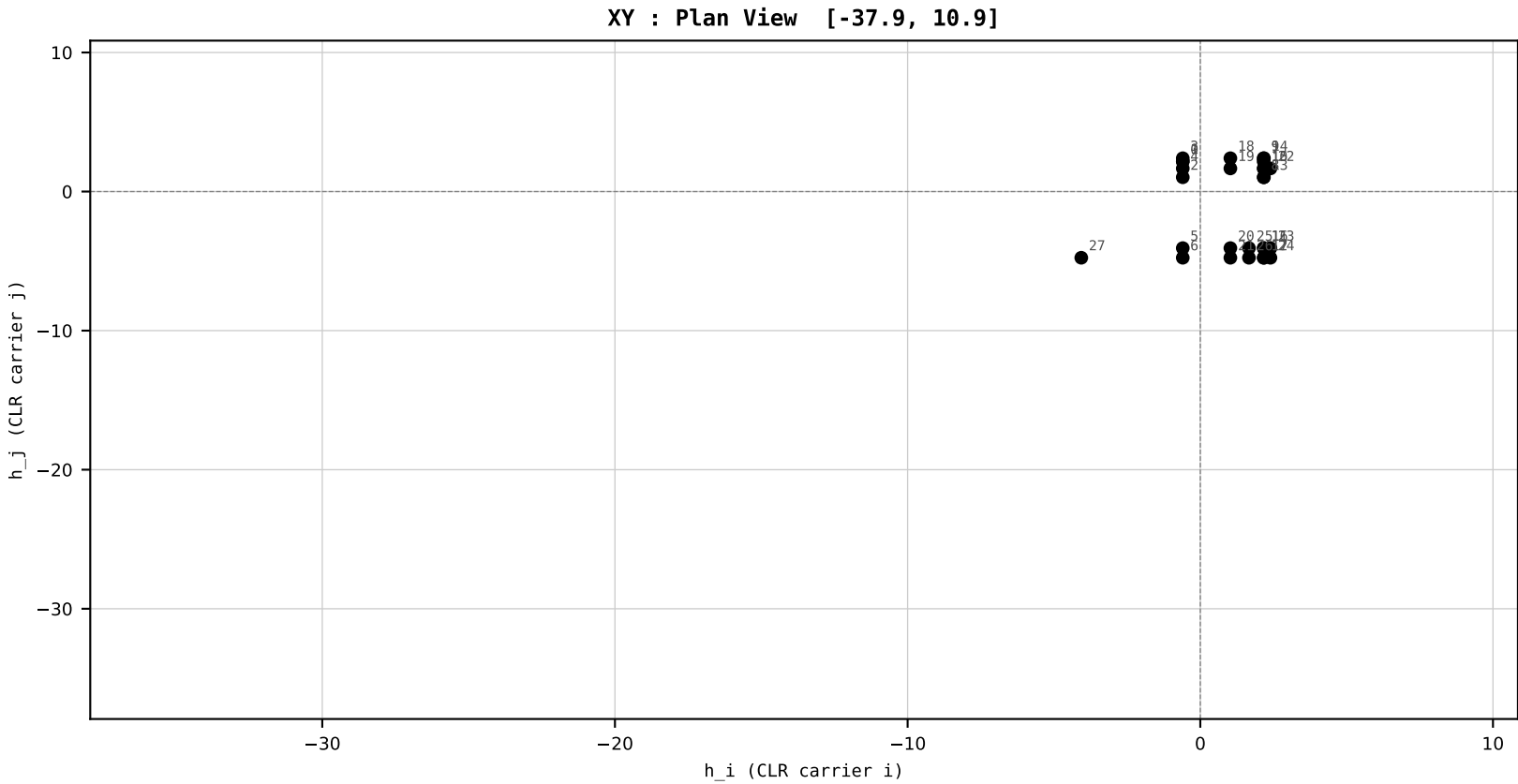
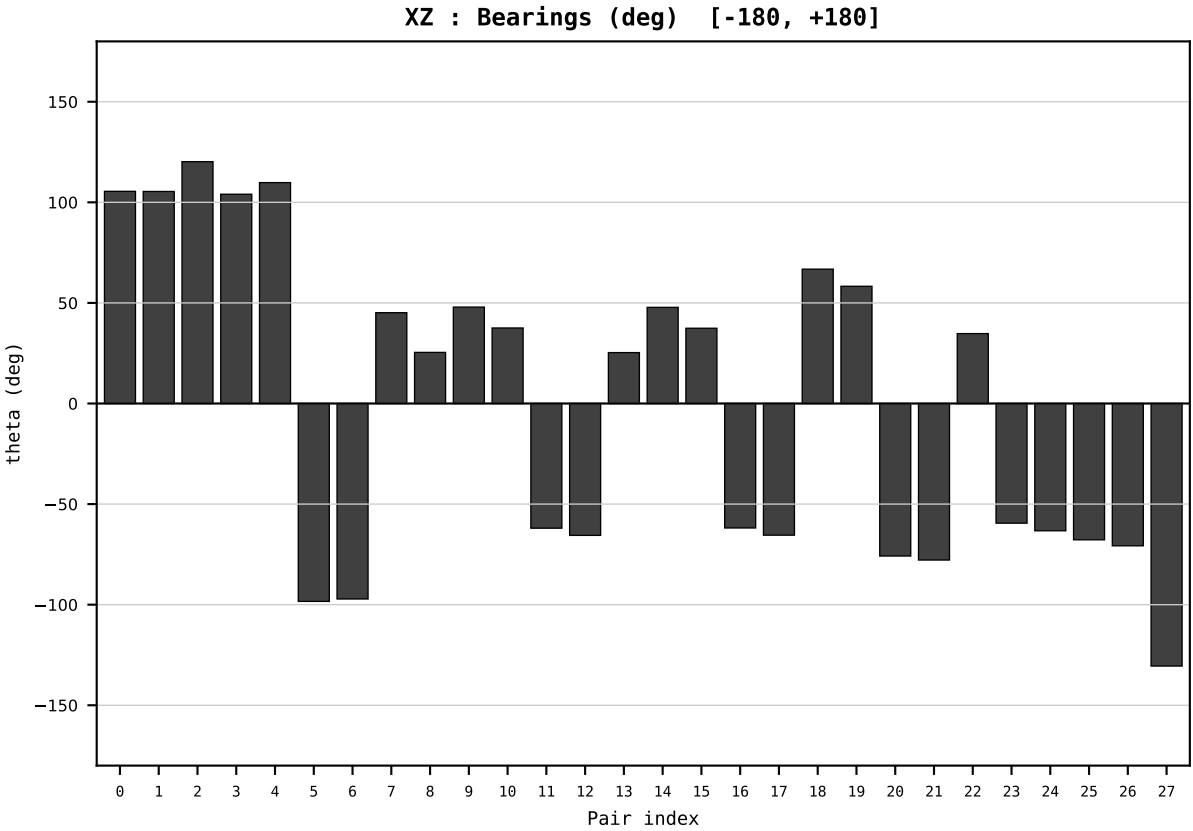
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.239561
Ring = Hs-2
E_metric= 58.5208
kappa_HS= 1282.16
omega = 2.8189 deg
d_A = 0.847846
Helm = Wind
Helm d = +0.689167
DR = 7.156 HLR
DR ratio= 1282.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

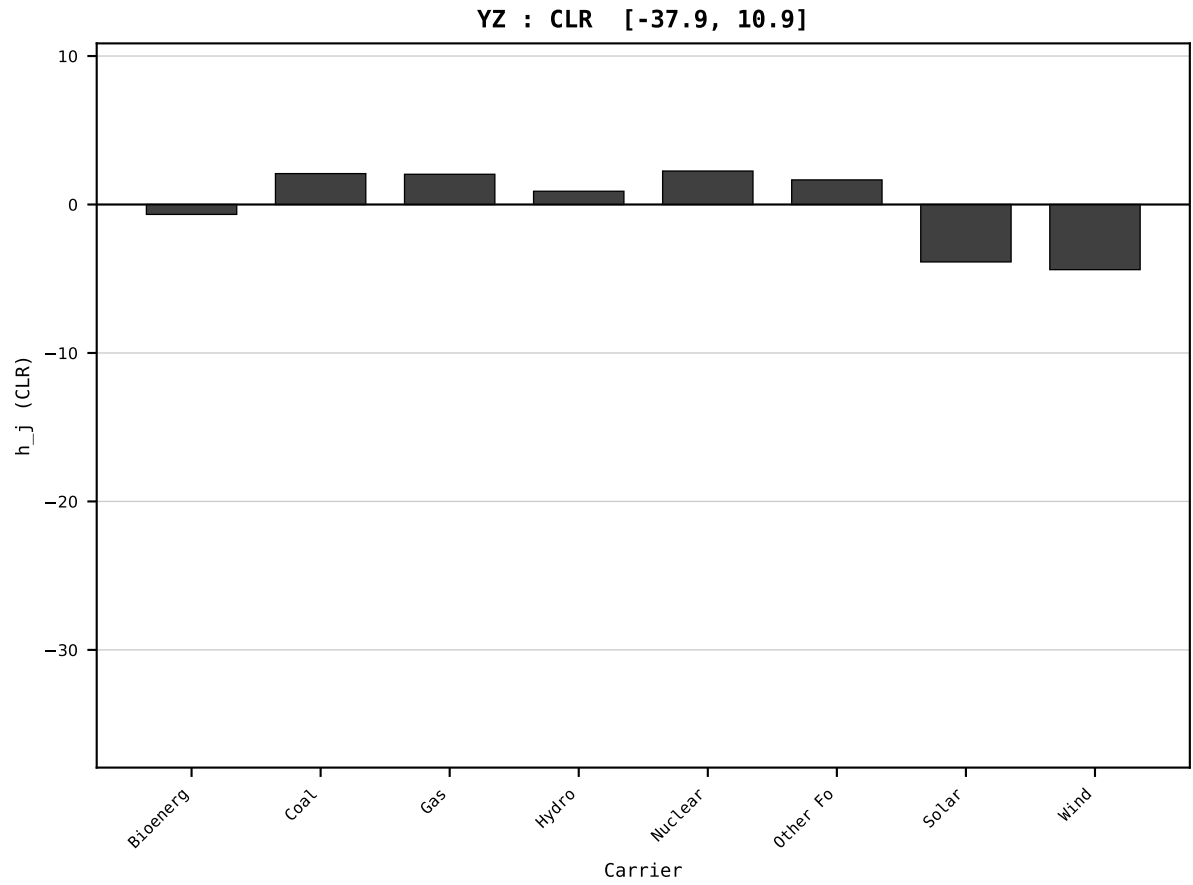
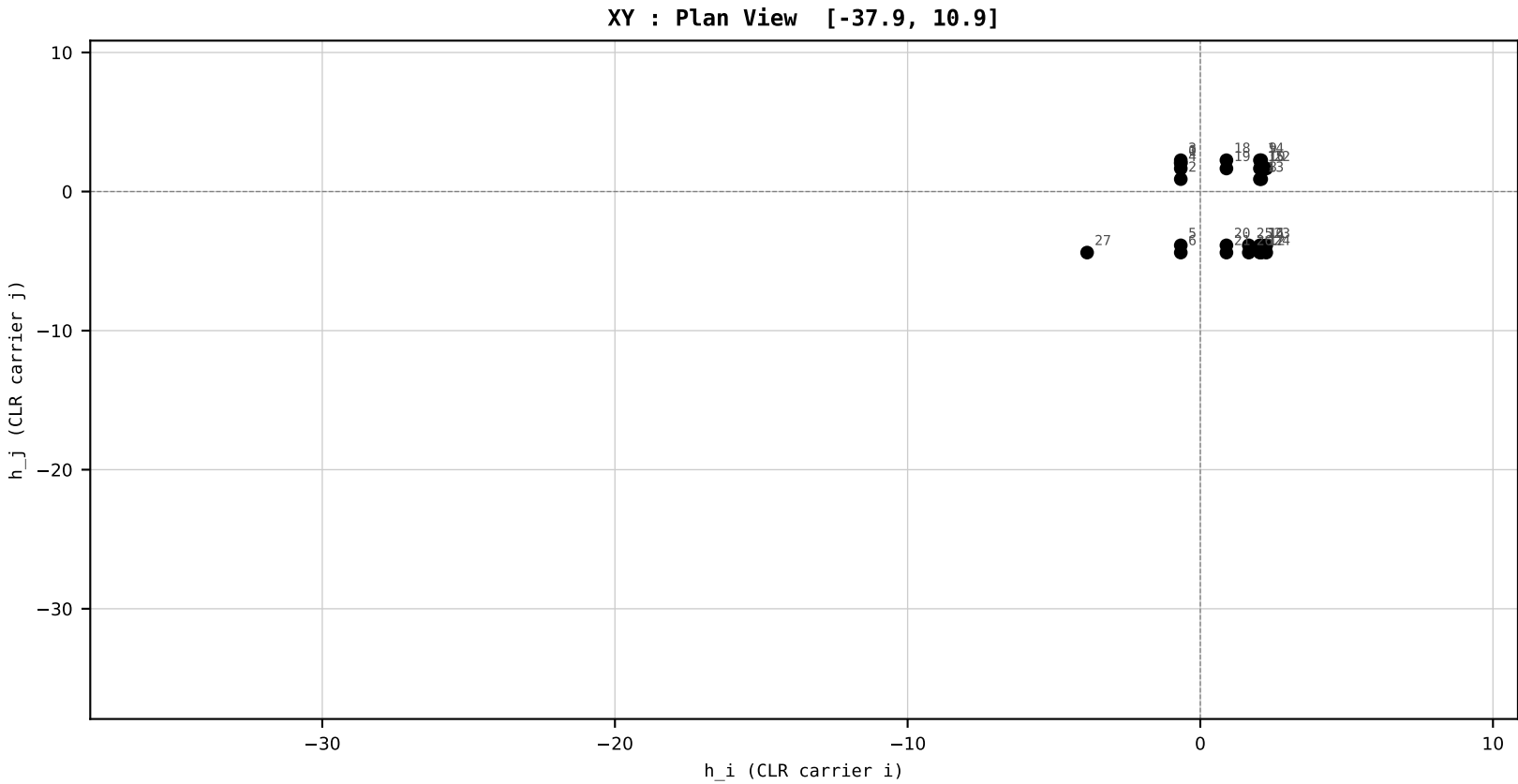
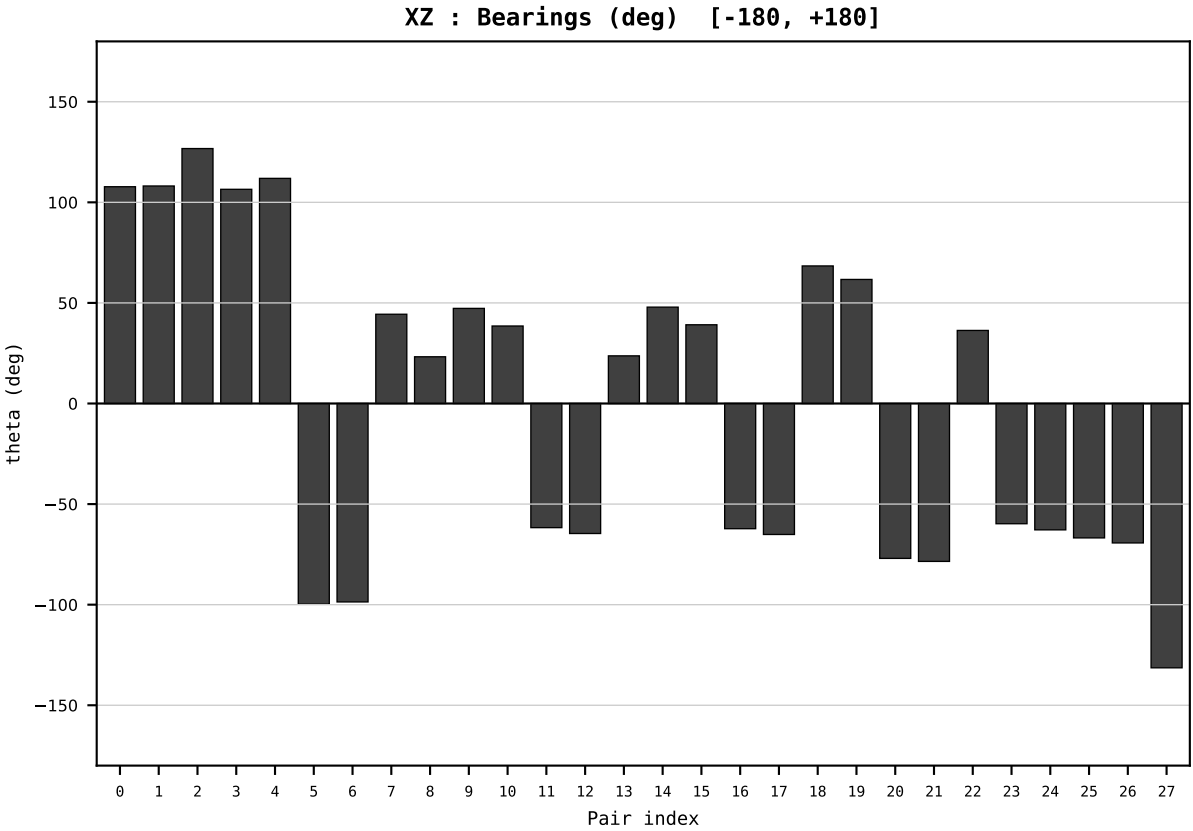
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234830
Ring = Hs-2
E_metric= 51.7554
kappa_HS= 766.49
omega = 1.5248 deg
d_A = 0.496689
Helm = Wind
Helm d = +0.371738
DR = 6.642 HLR
DR ratio= 766.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

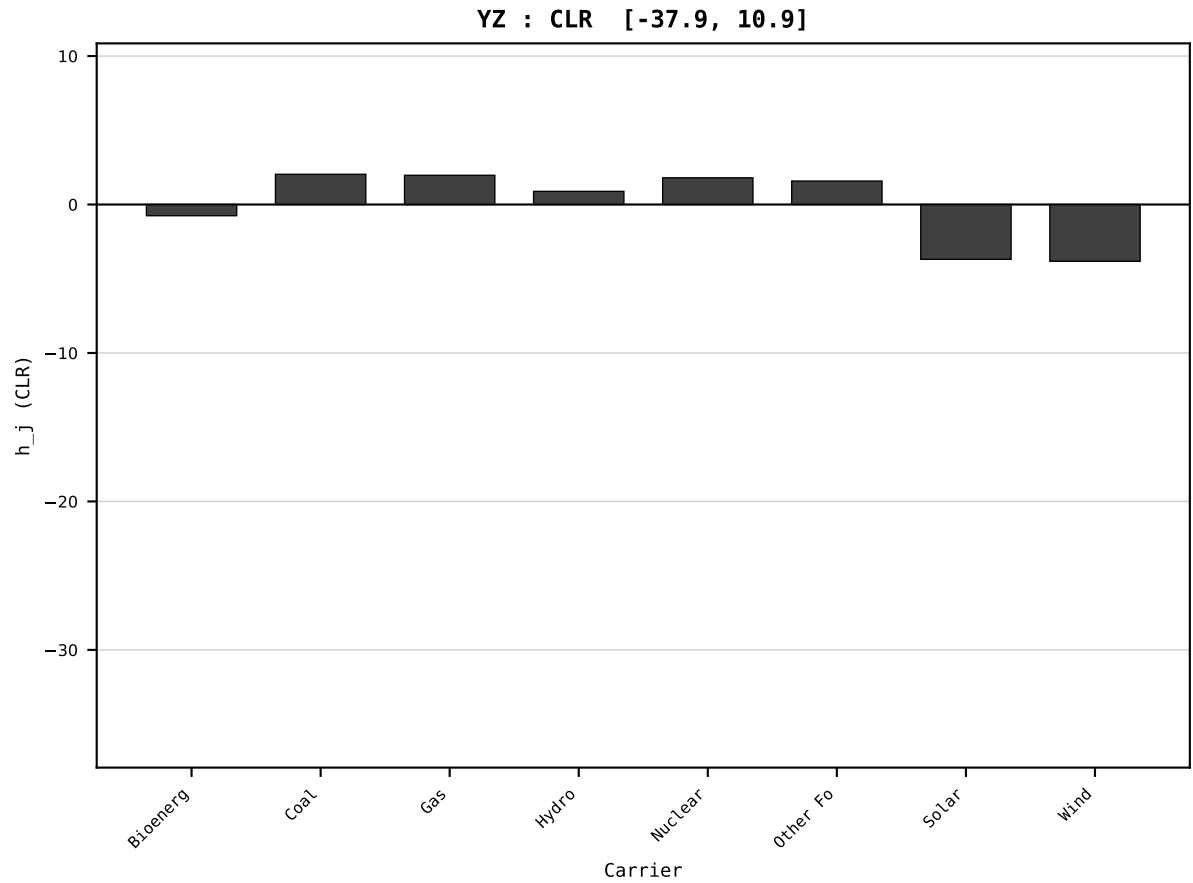
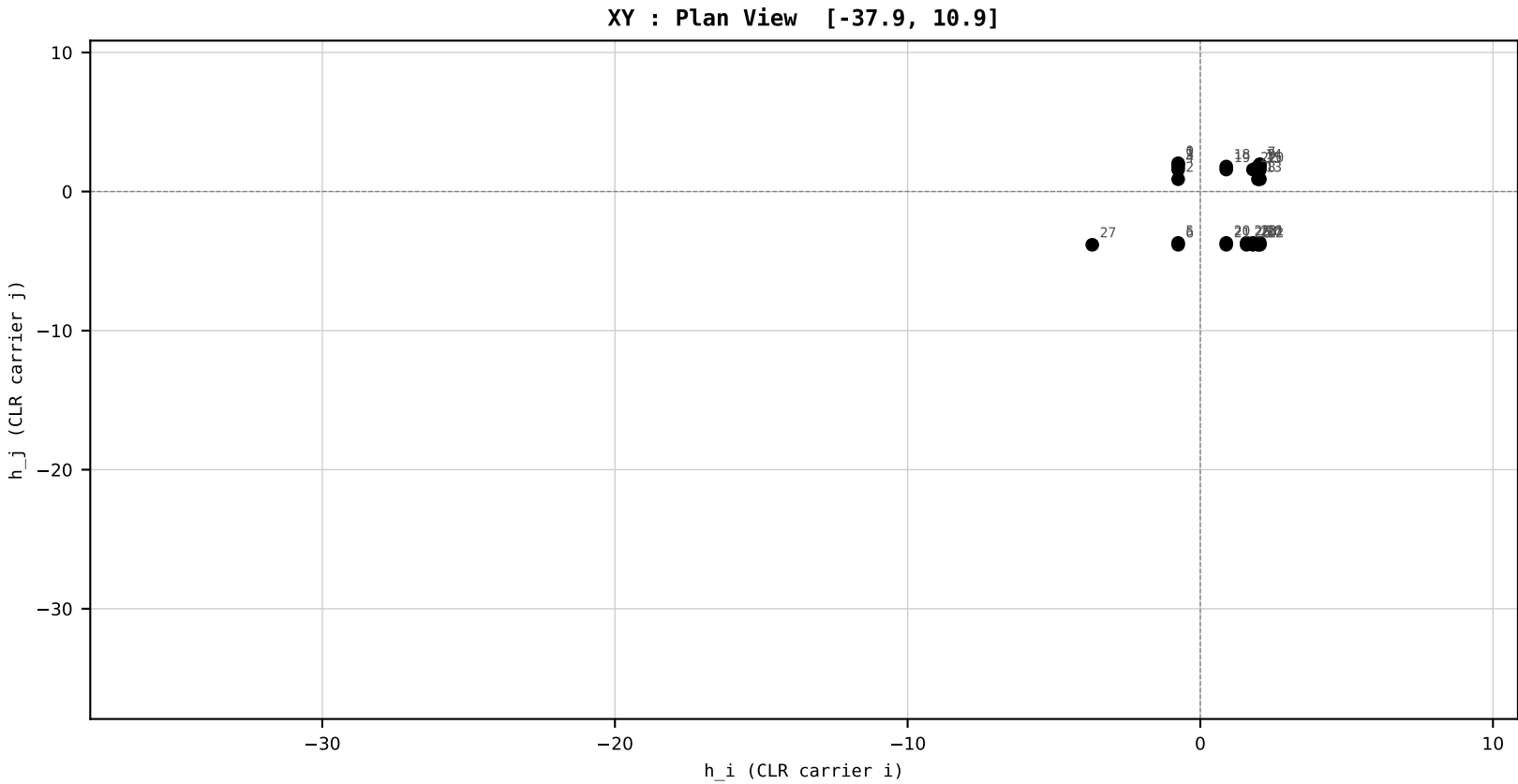
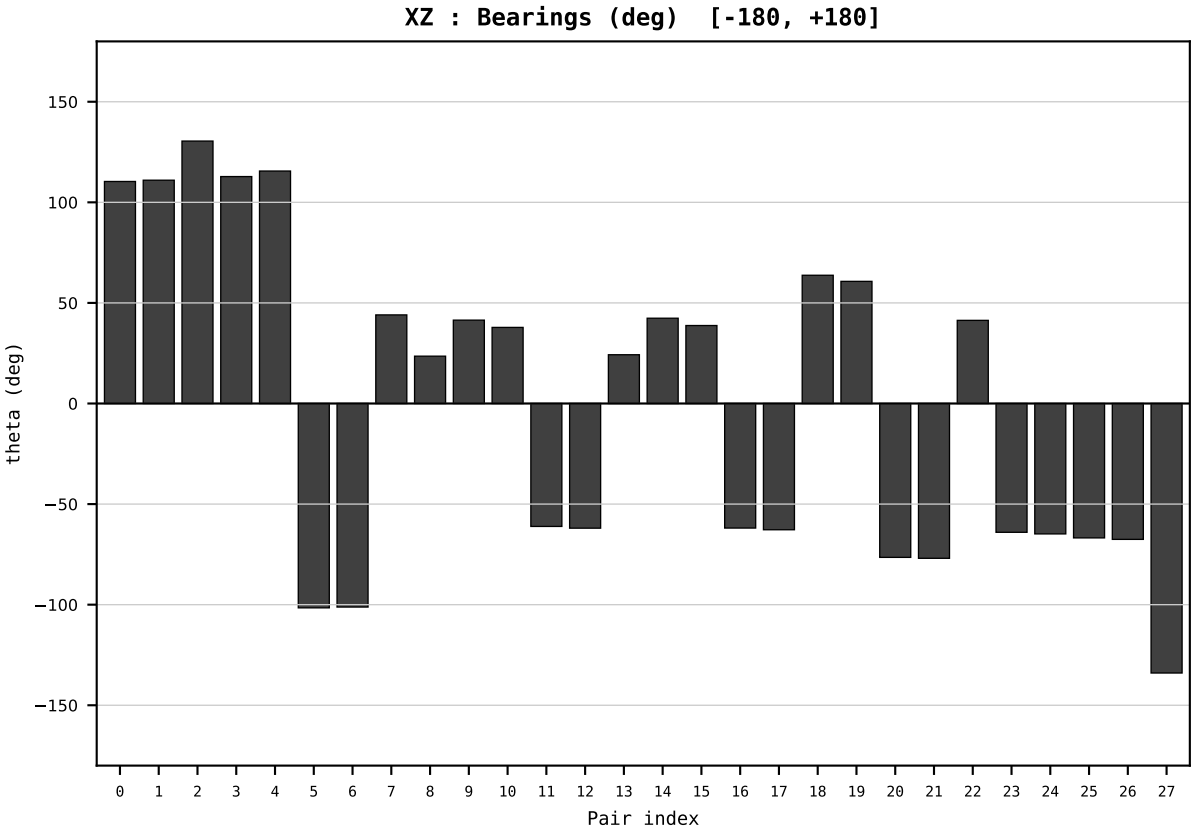
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.222849
Ring = Hs-2
E_metric= 43.3901
kappa_HS= 351.88
omega = 3.7738 deg
d_A = 0.757601
Helm = Wind
Helm d = +0.561548
DR = 5.863 HLR
DR ratio= 351.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

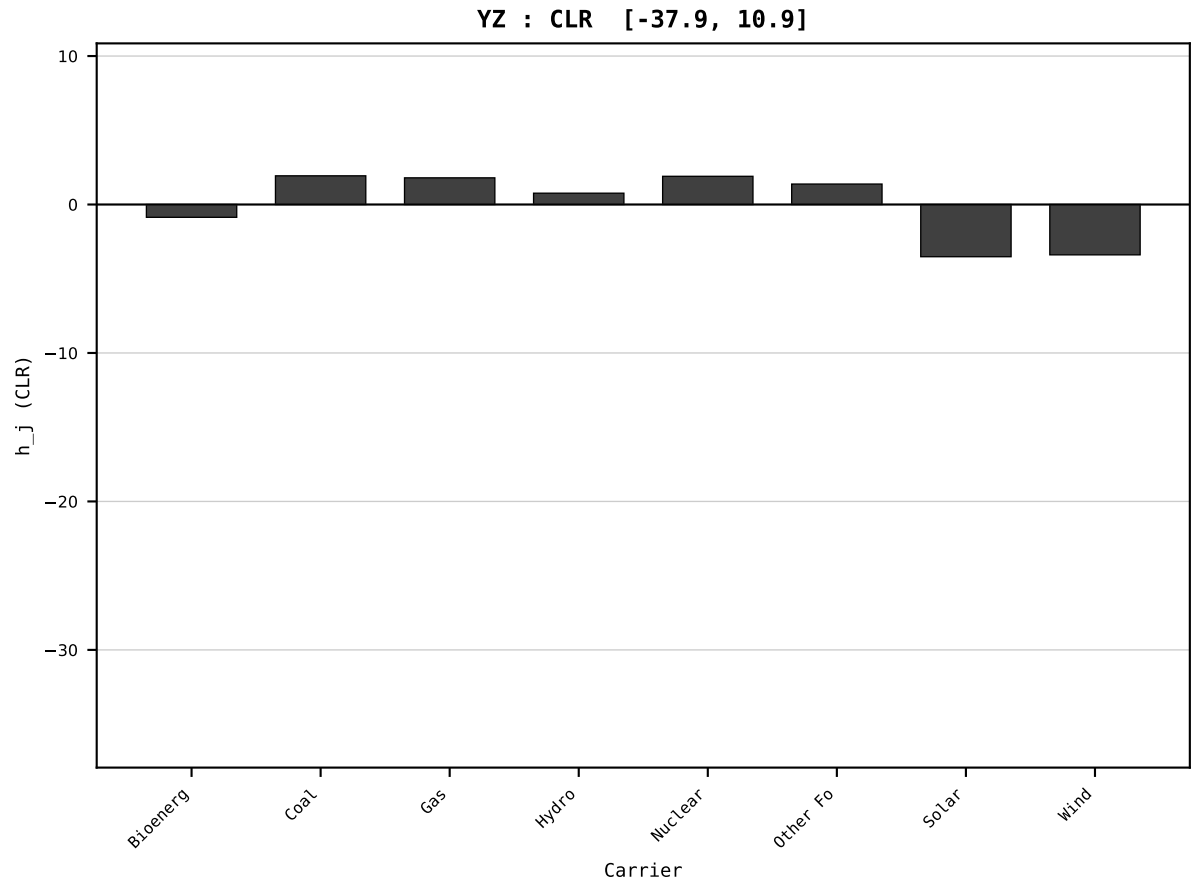
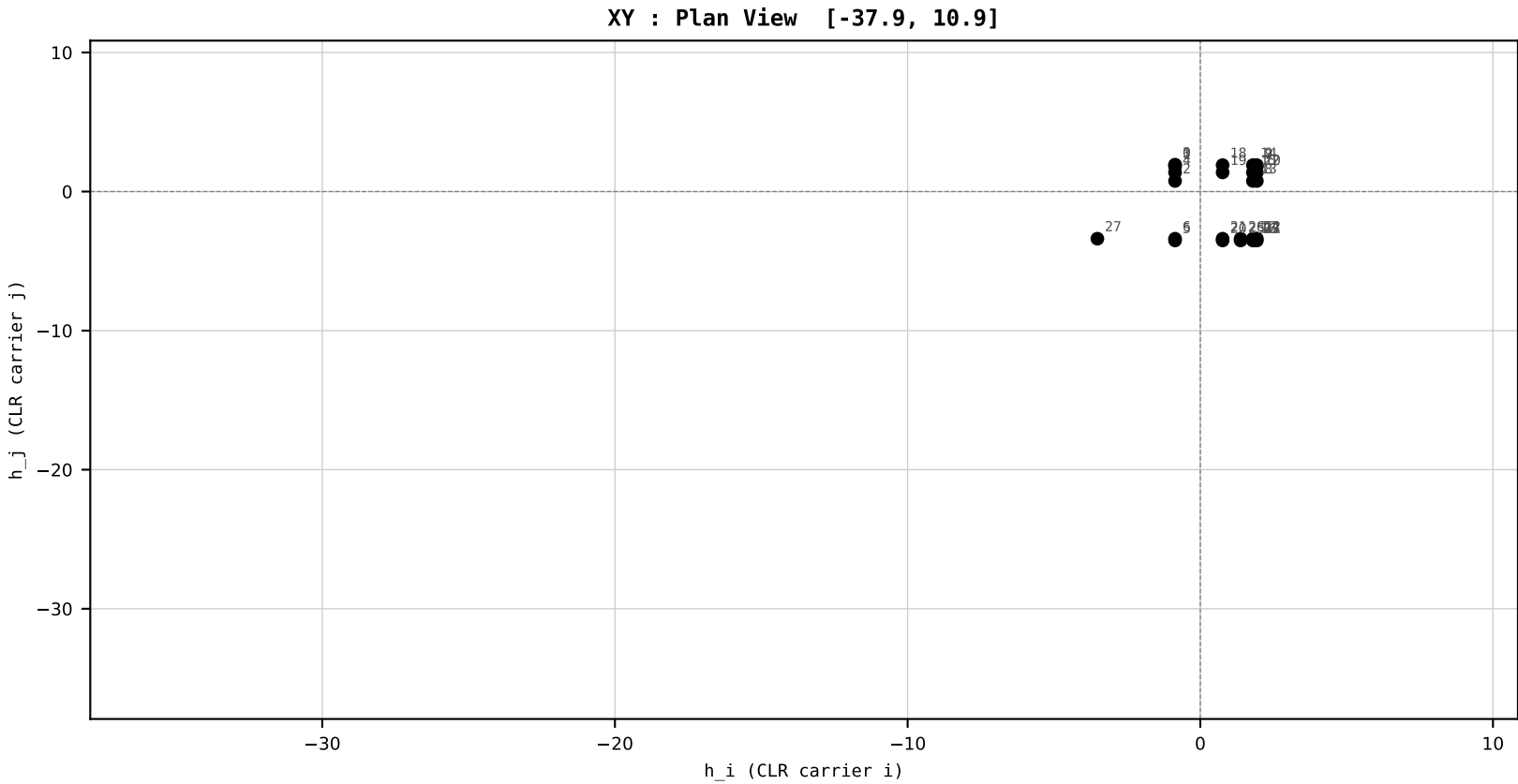
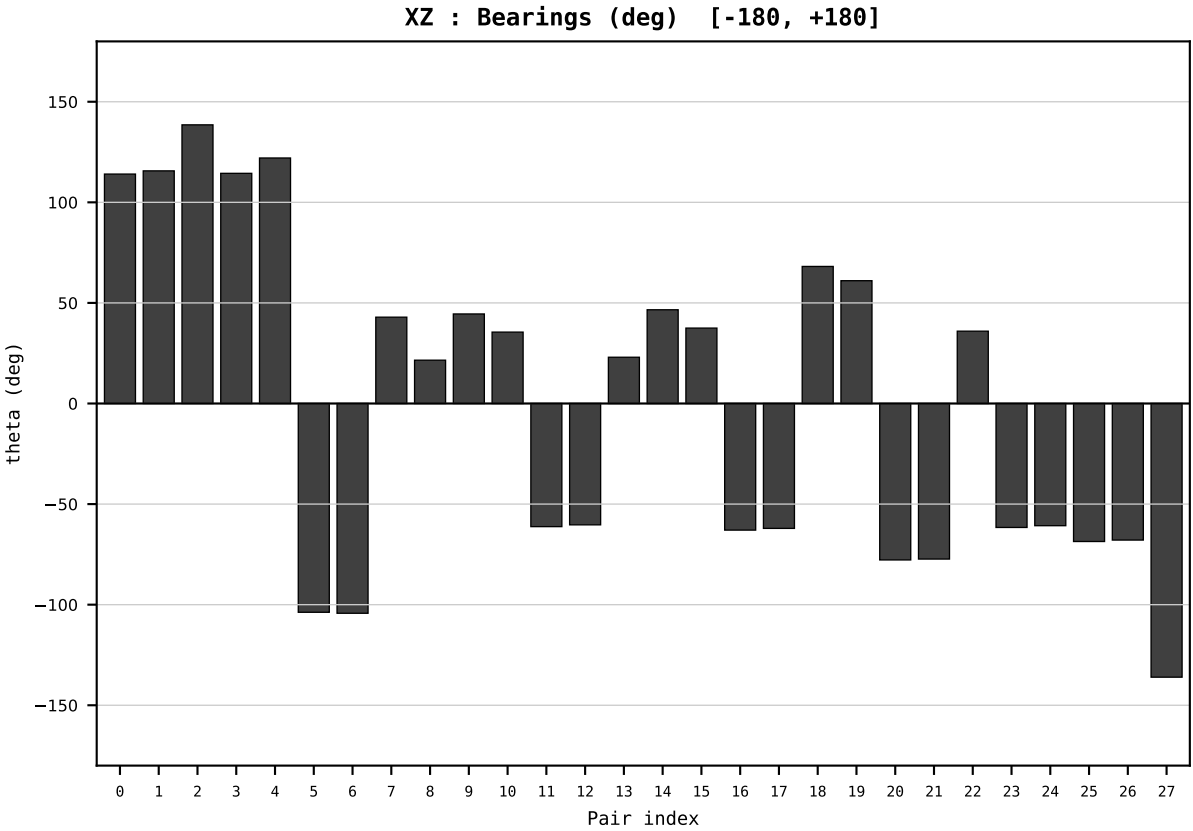
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.225250
Ring = Hs-2
E_metric= 37.6631
kappa_HS= 232.59
omega = 3.3324 deg
d_A = 0.582491
Helm = Wind
Helm d = +0.435321
DR = 5.449 HLR
DR ratio= 232.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

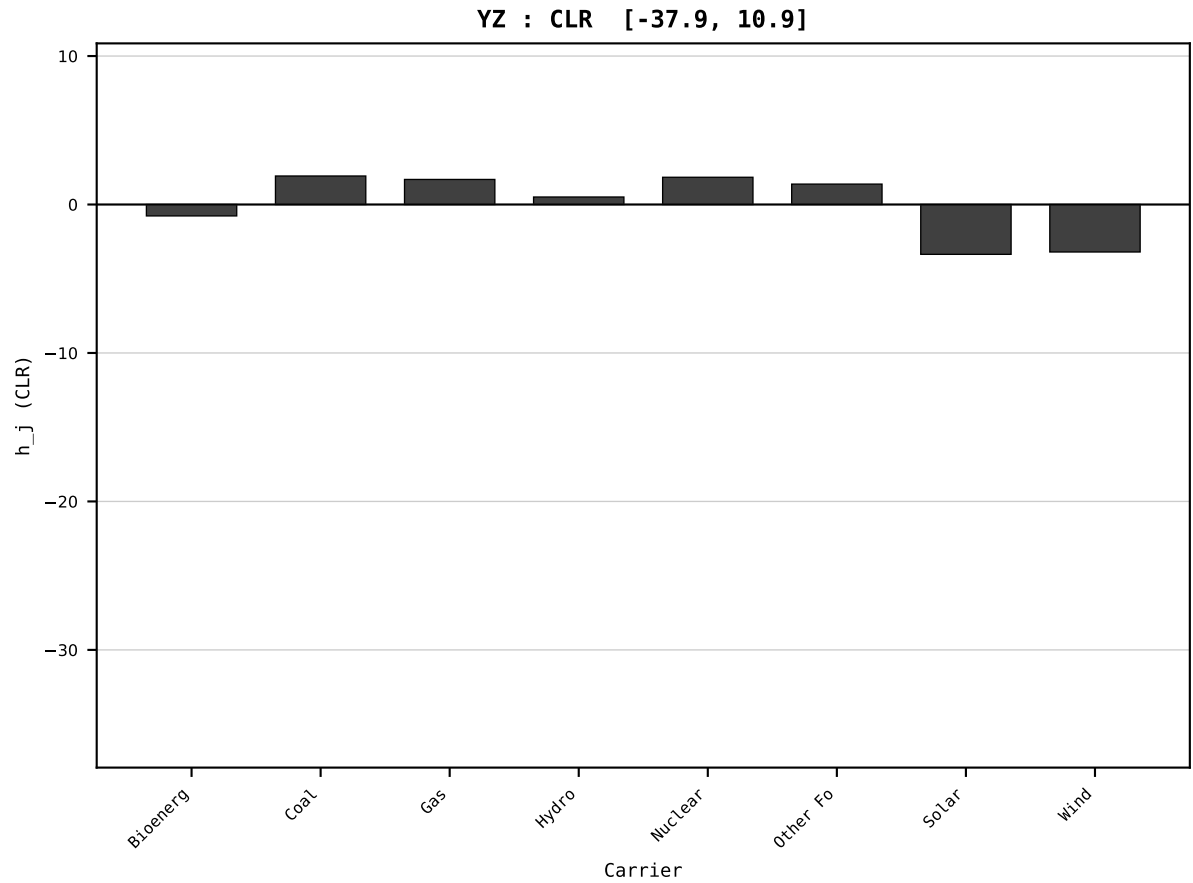
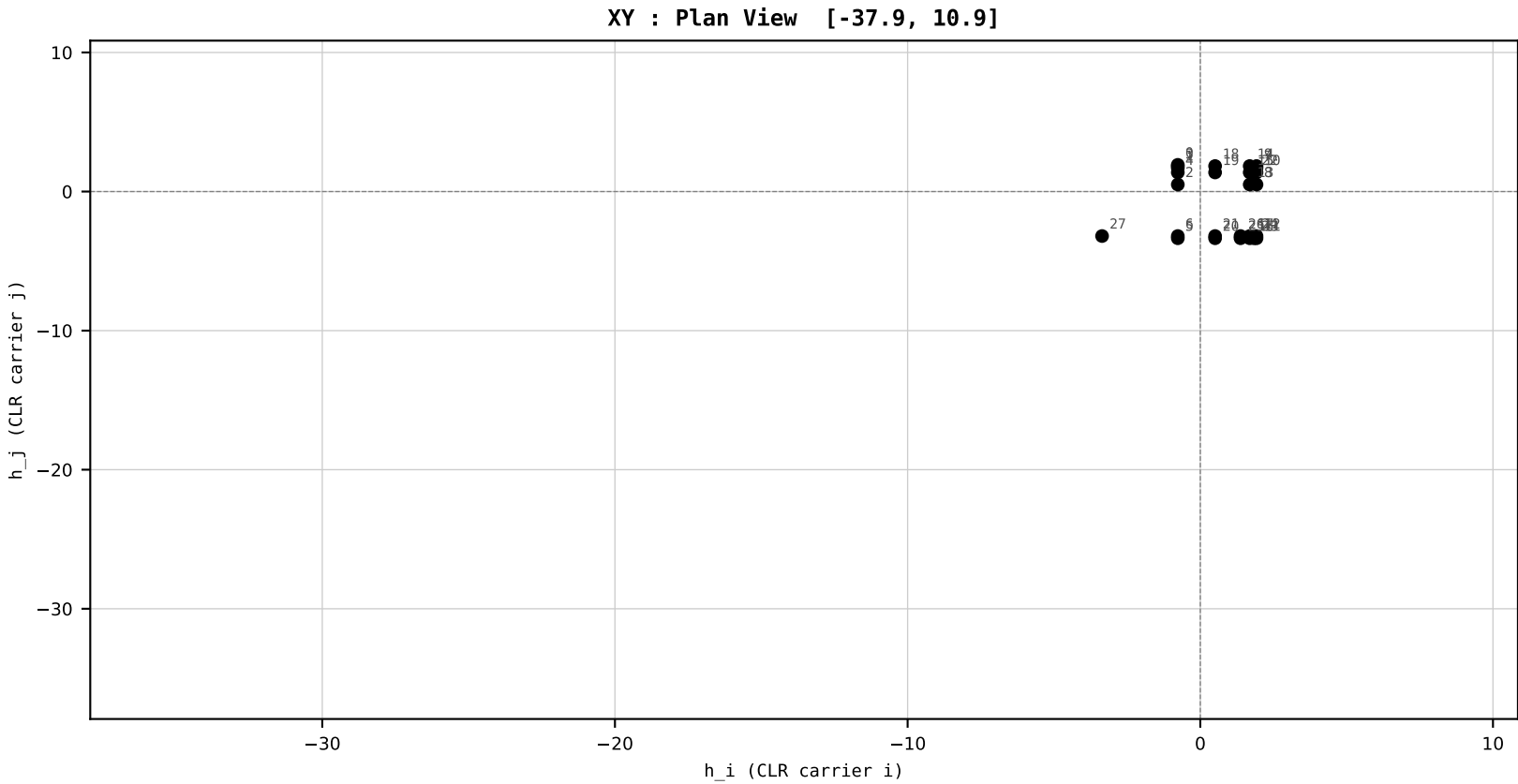
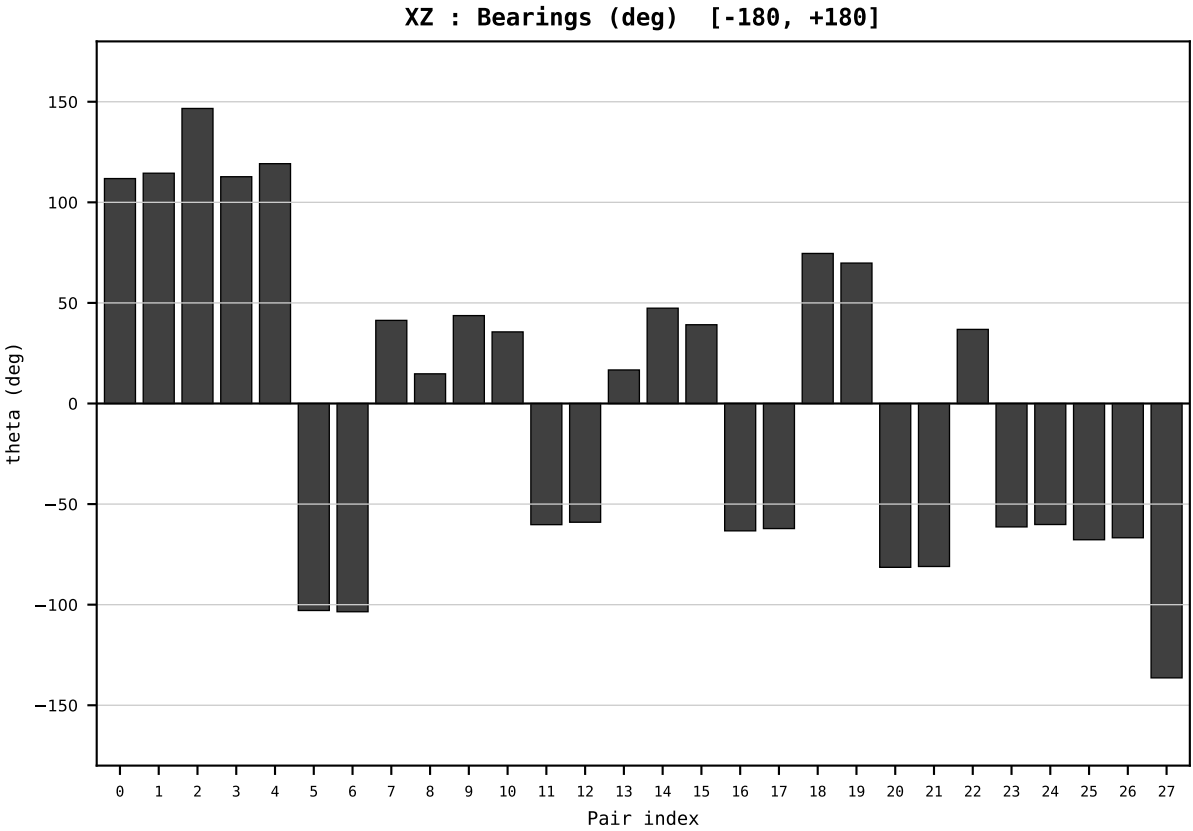
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227556
Ring = Hs-2
E_metric= 34.1389
kappa_HS= 195.83
omega = 2.4787 deg
d_A = 0.391966
Helm = Hydro
Helm d = -0.256970
DR = 5.277 HLR
DR ratio= 195.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

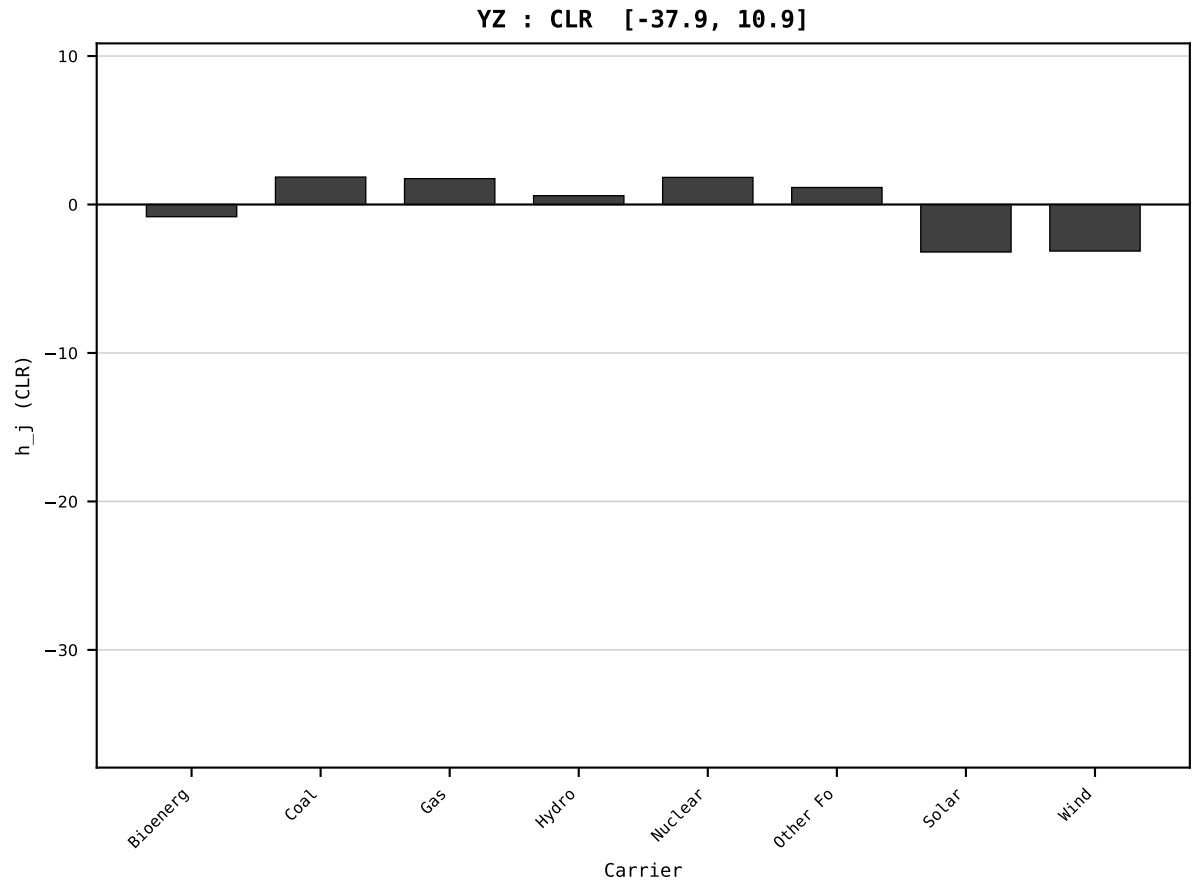
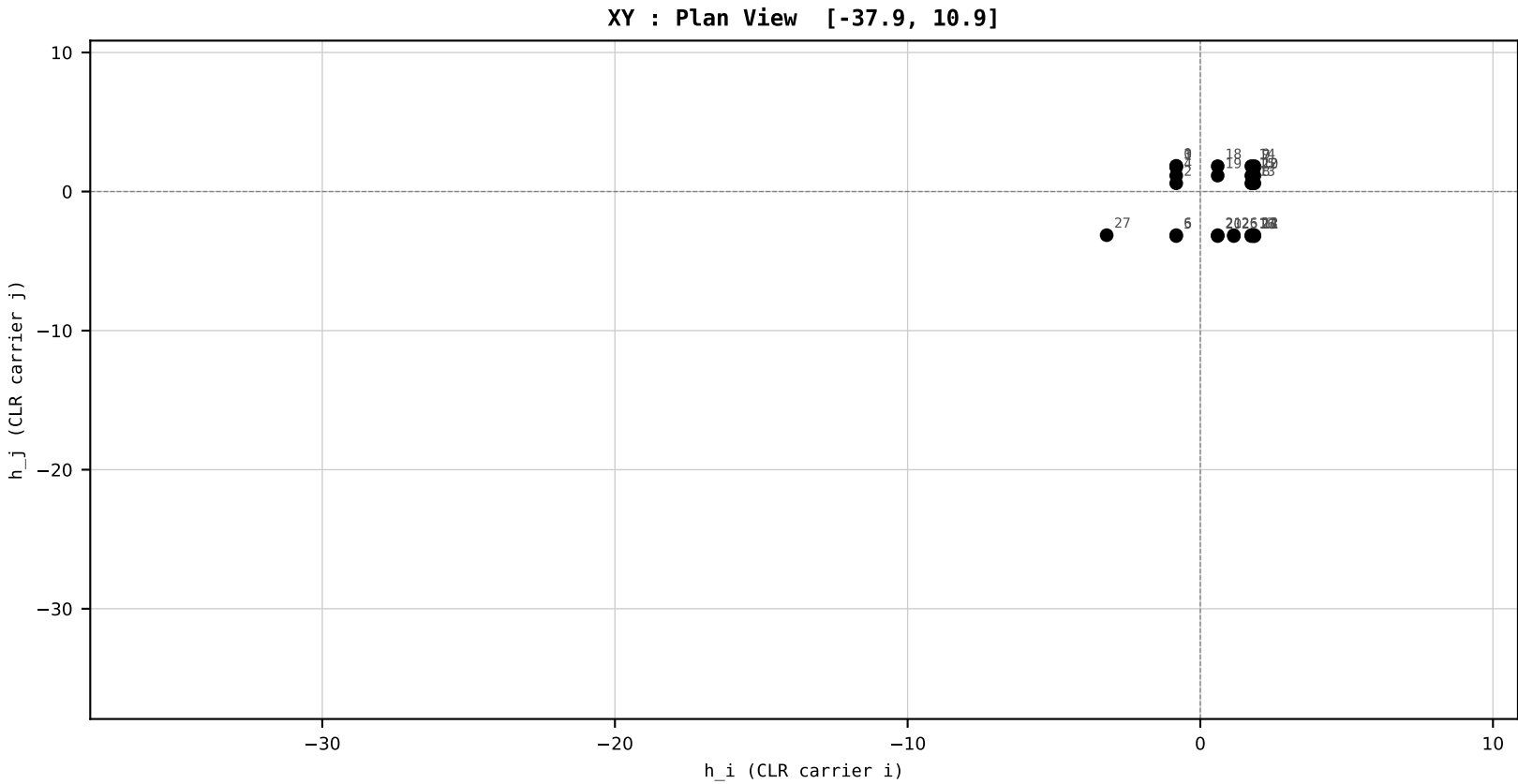
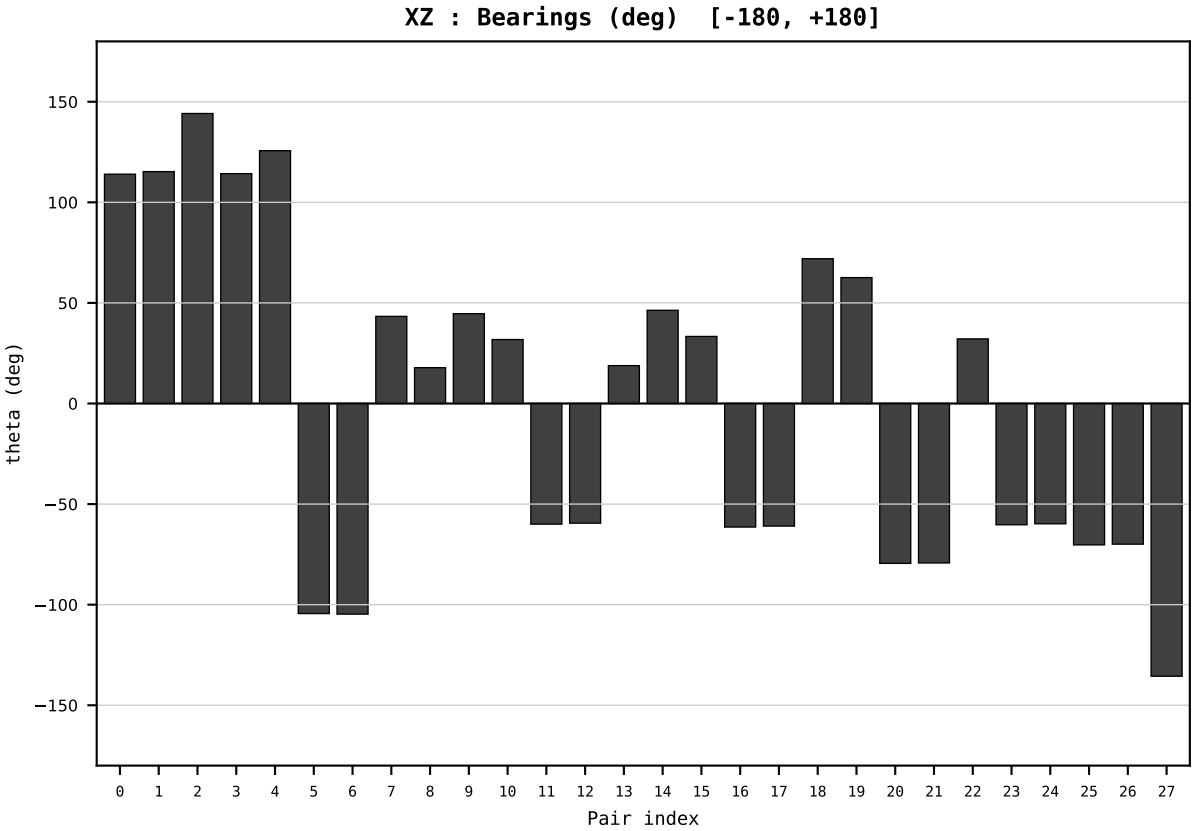
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227477
Ring = Hs-2
E_metric= 32.2009
kappa_HS= 155.67
omega = 2.6642 deg
d_A = 0.316211
Helm = Other Fossil
Helm d = -0.228493
DR = 5.048 HLR
DR ratio= 155.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

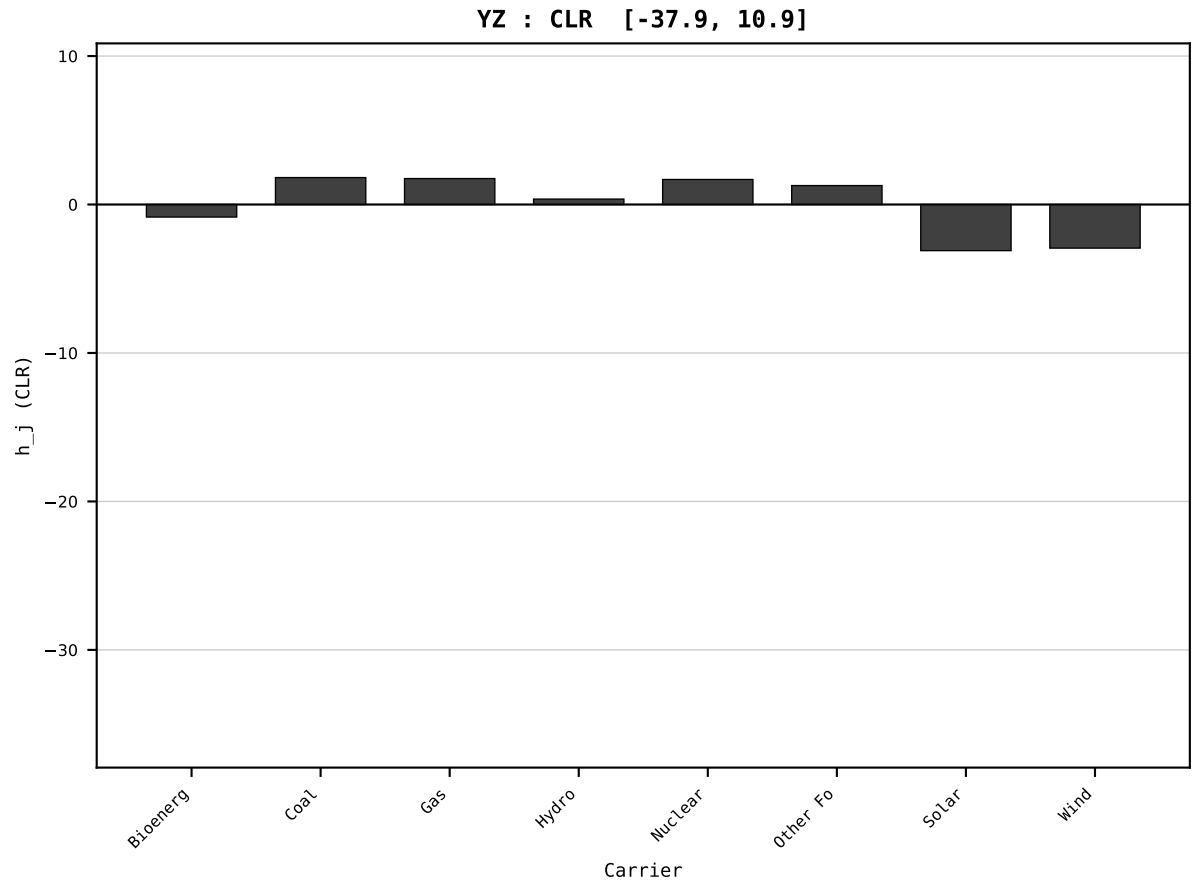
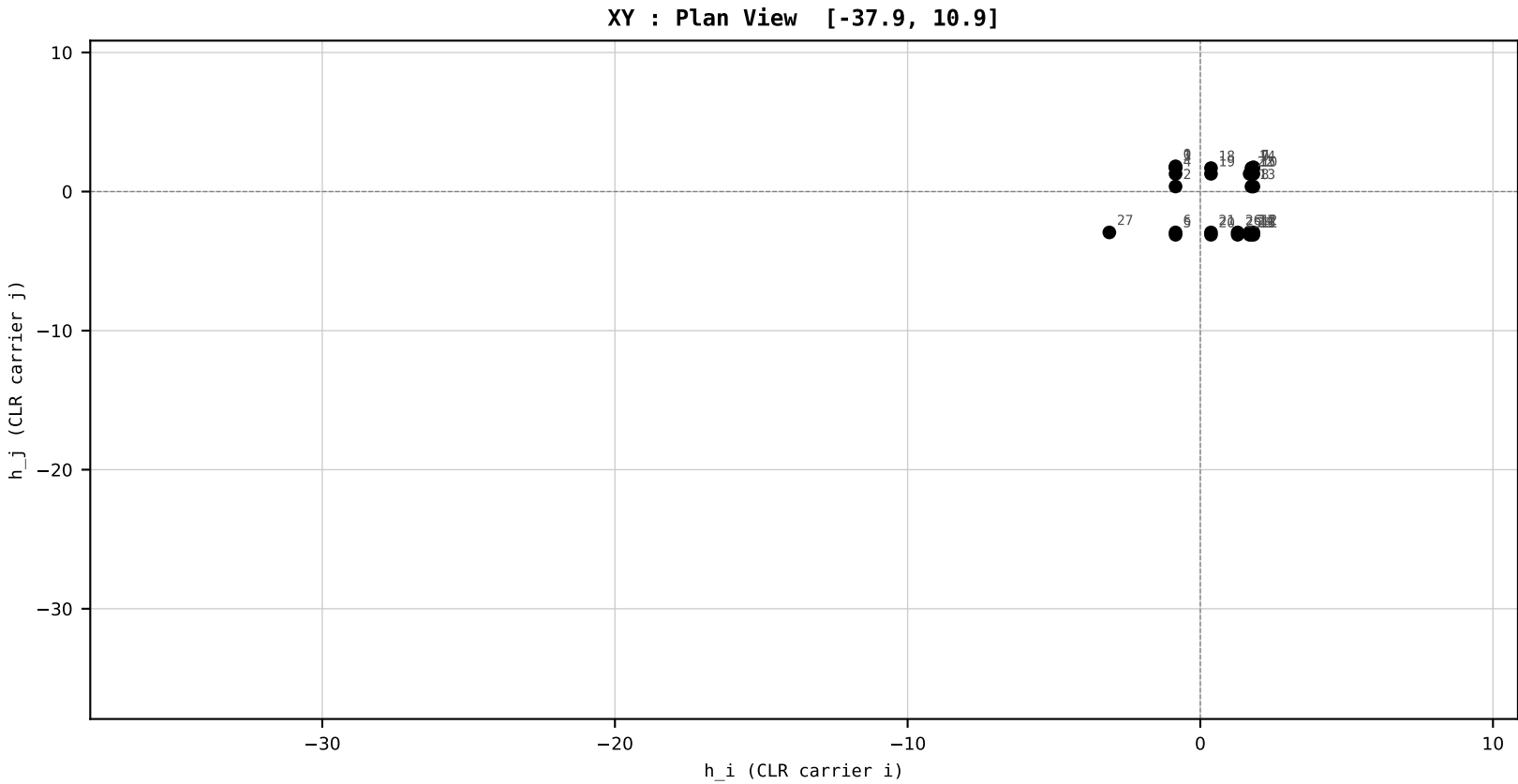
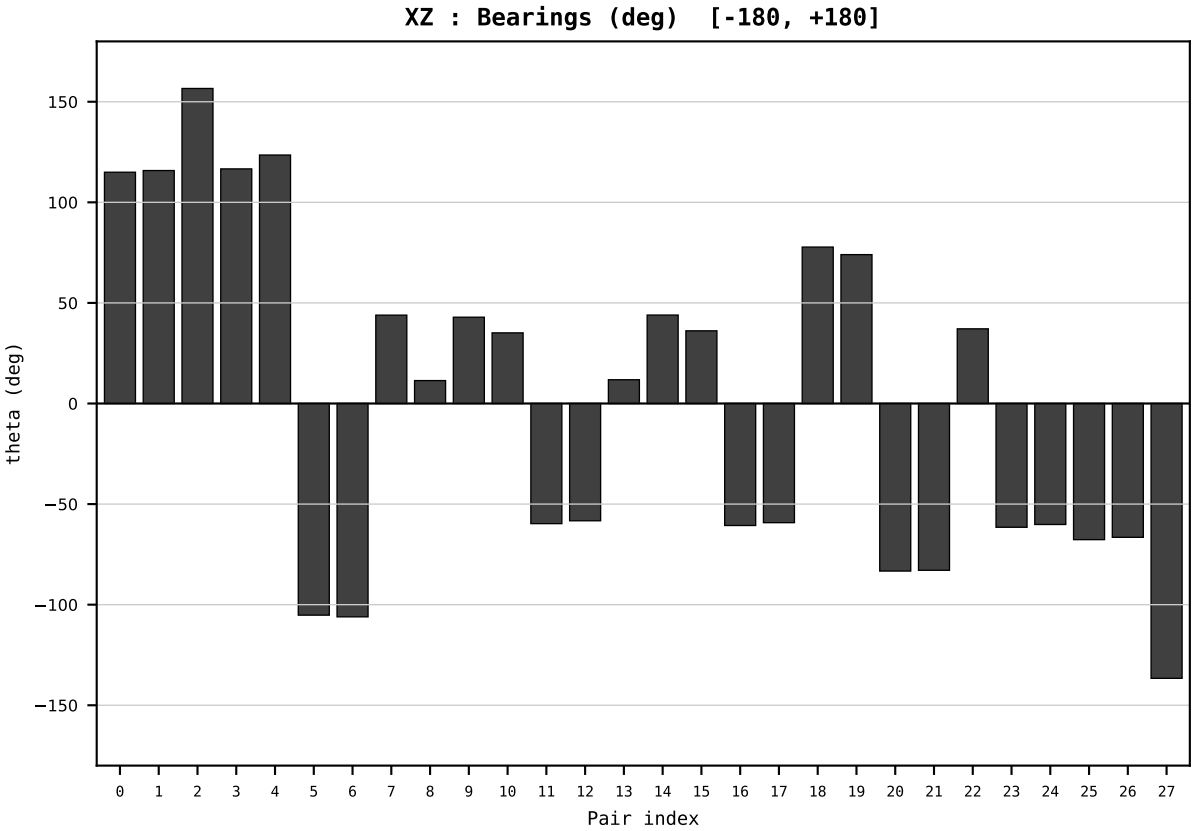
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227024
Ring = Hs-2
E_metric= 29.9546
kappa_HS= 137.36
omega = 3.2122 deg
d_A = 0.371740
Helm = Hydro
Helm d = -0.228639
DR = 4.923 HLR
DR ratio= 137.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



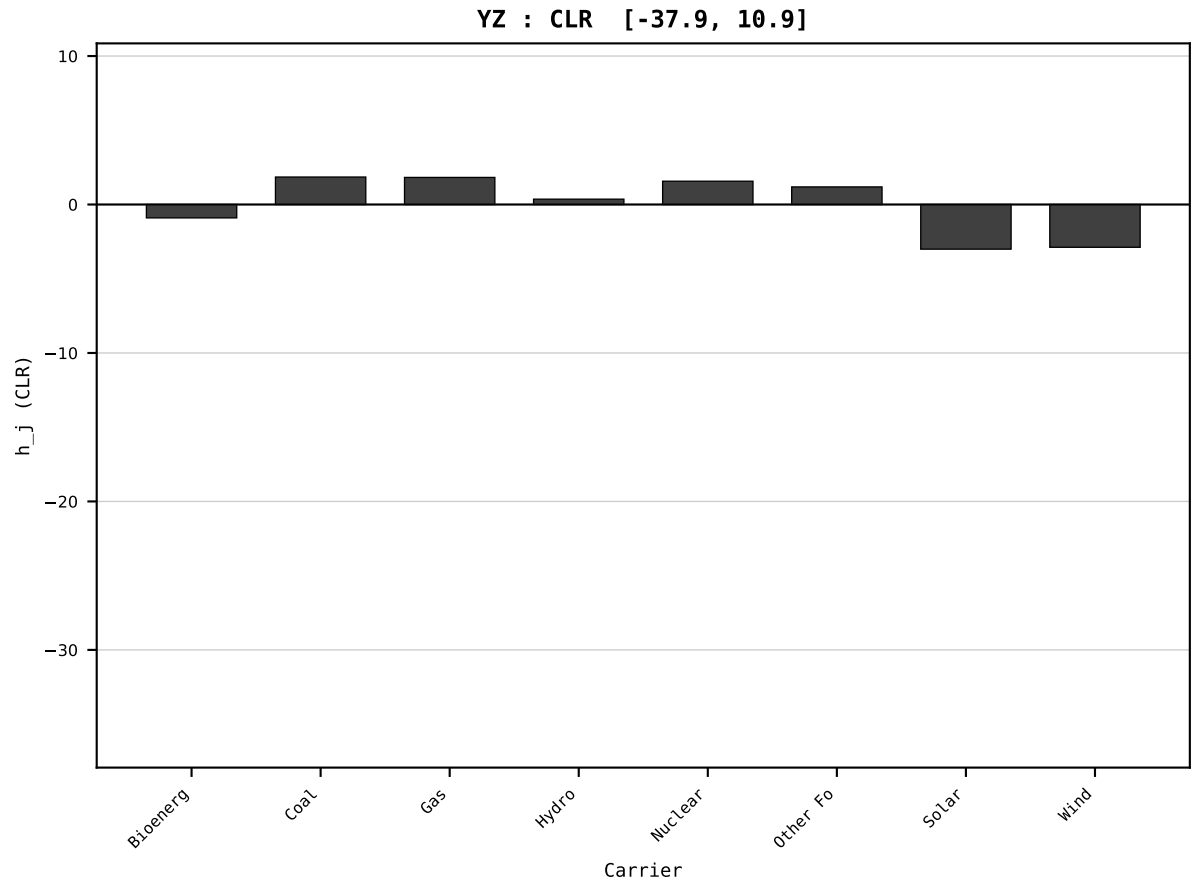
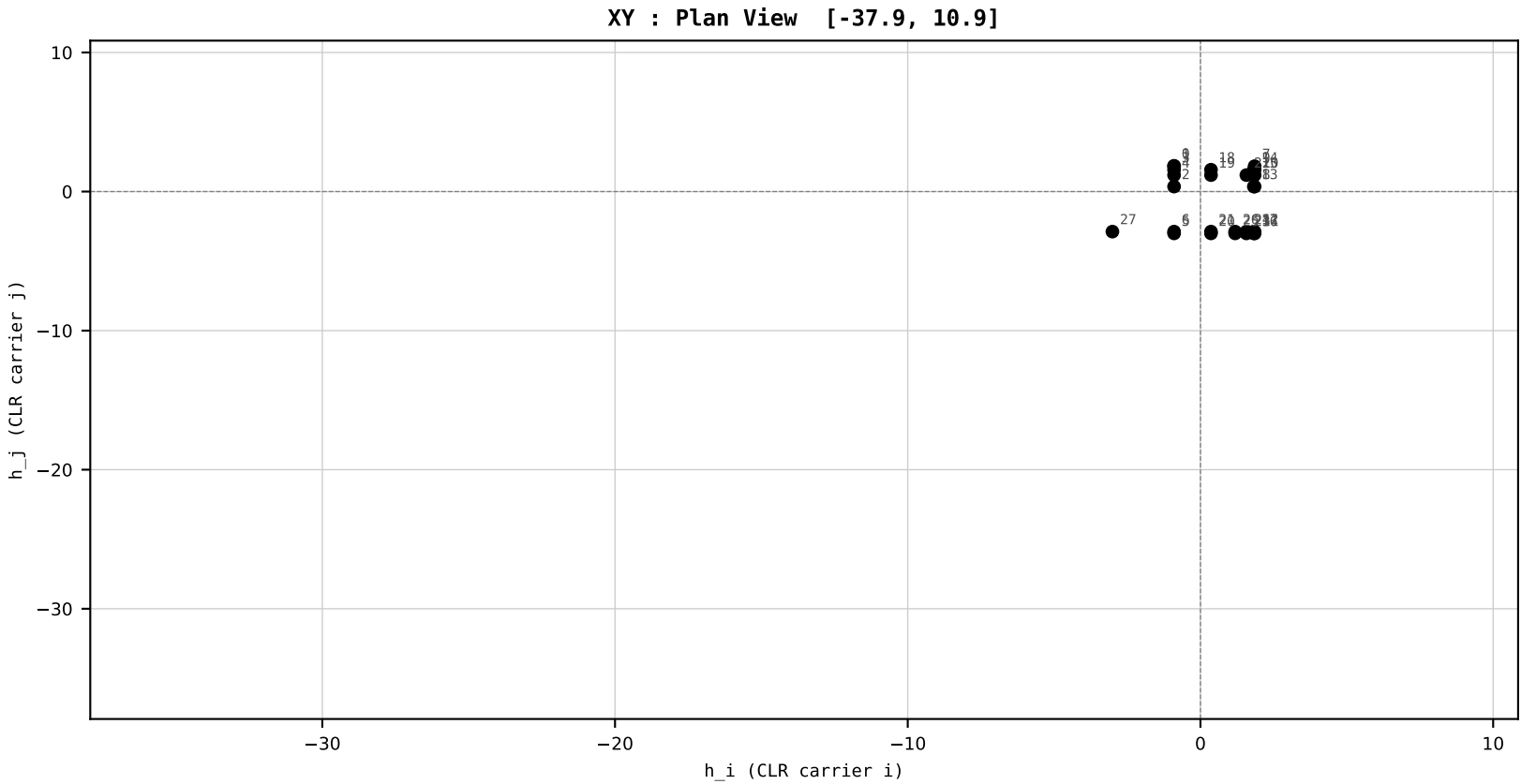
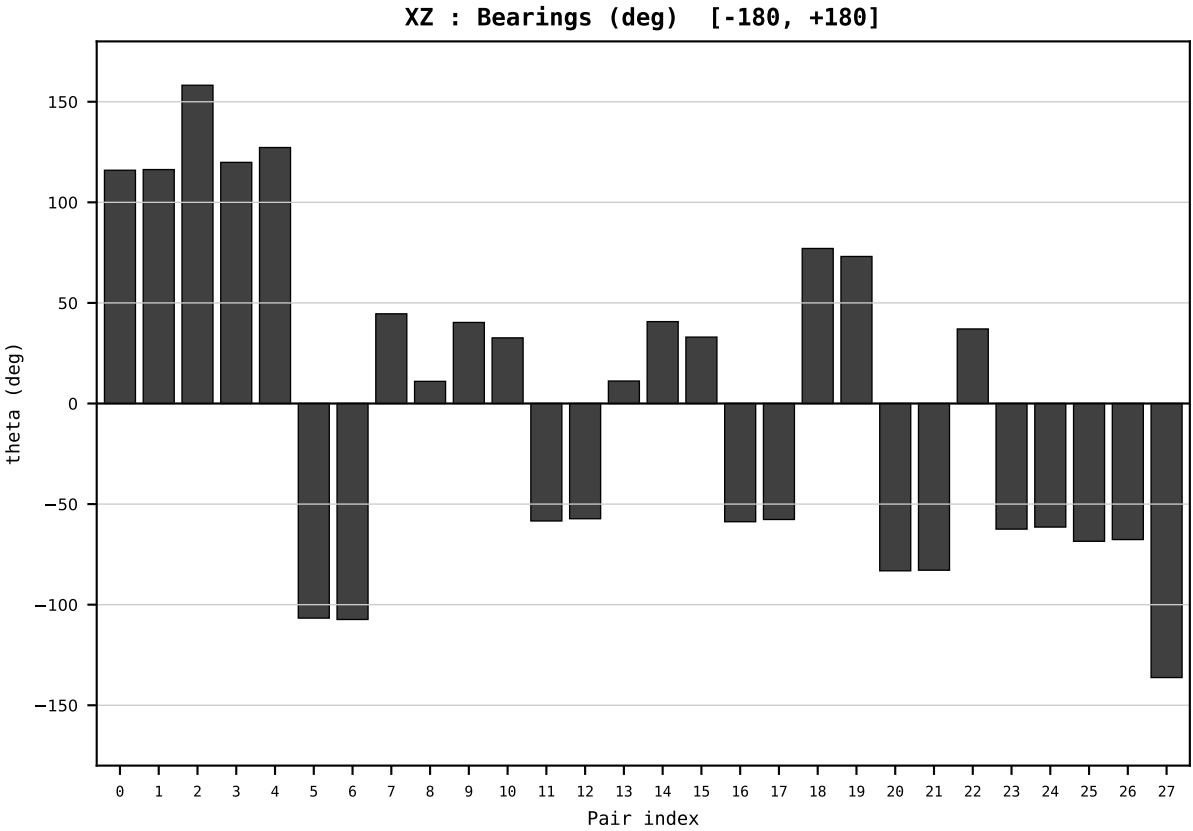
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.232256
Ring = Hs-2
E_metric= 28.9064
kappa_HS= 128.84
omega = 2.0079 deg
d_A = 0.213233
Helm = Nuclear
Helm d = -0.116881
DR = 4.859 HLR
DR ratio= 128.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



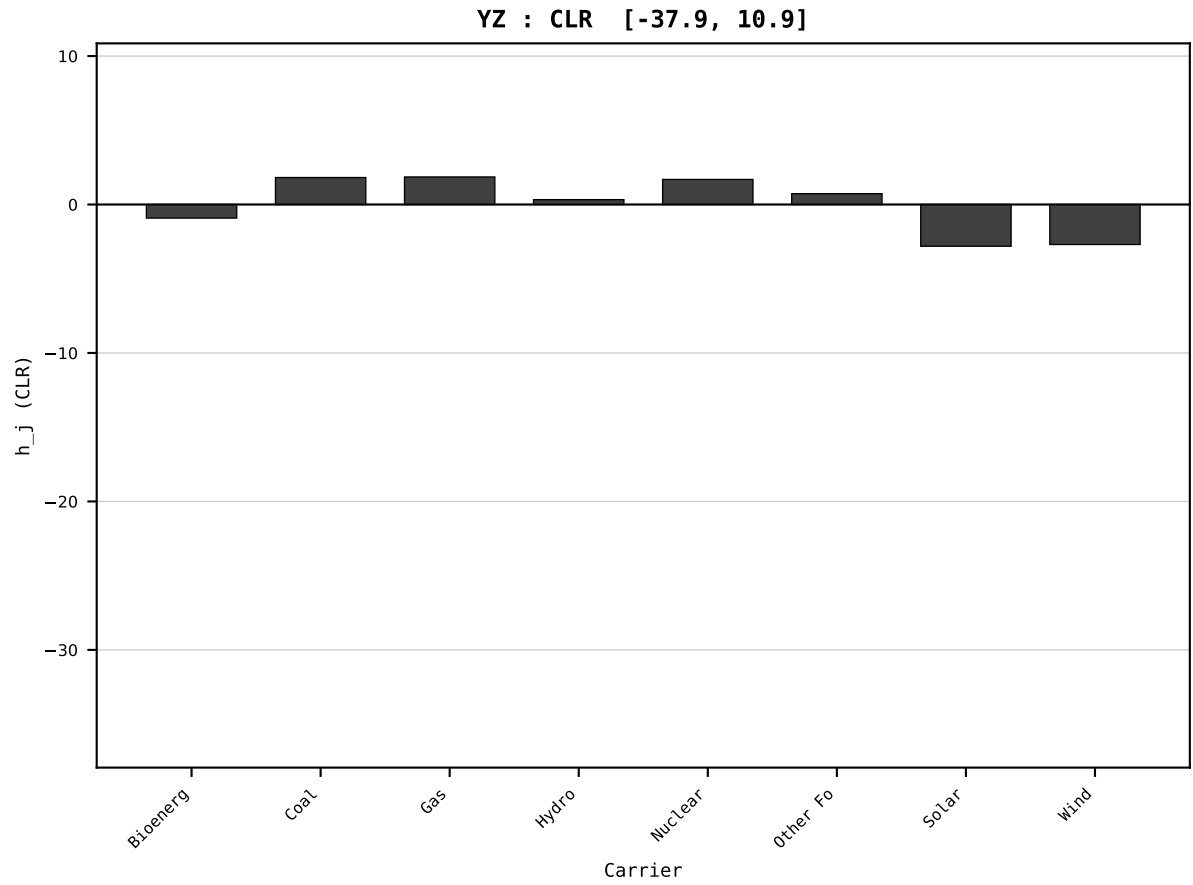
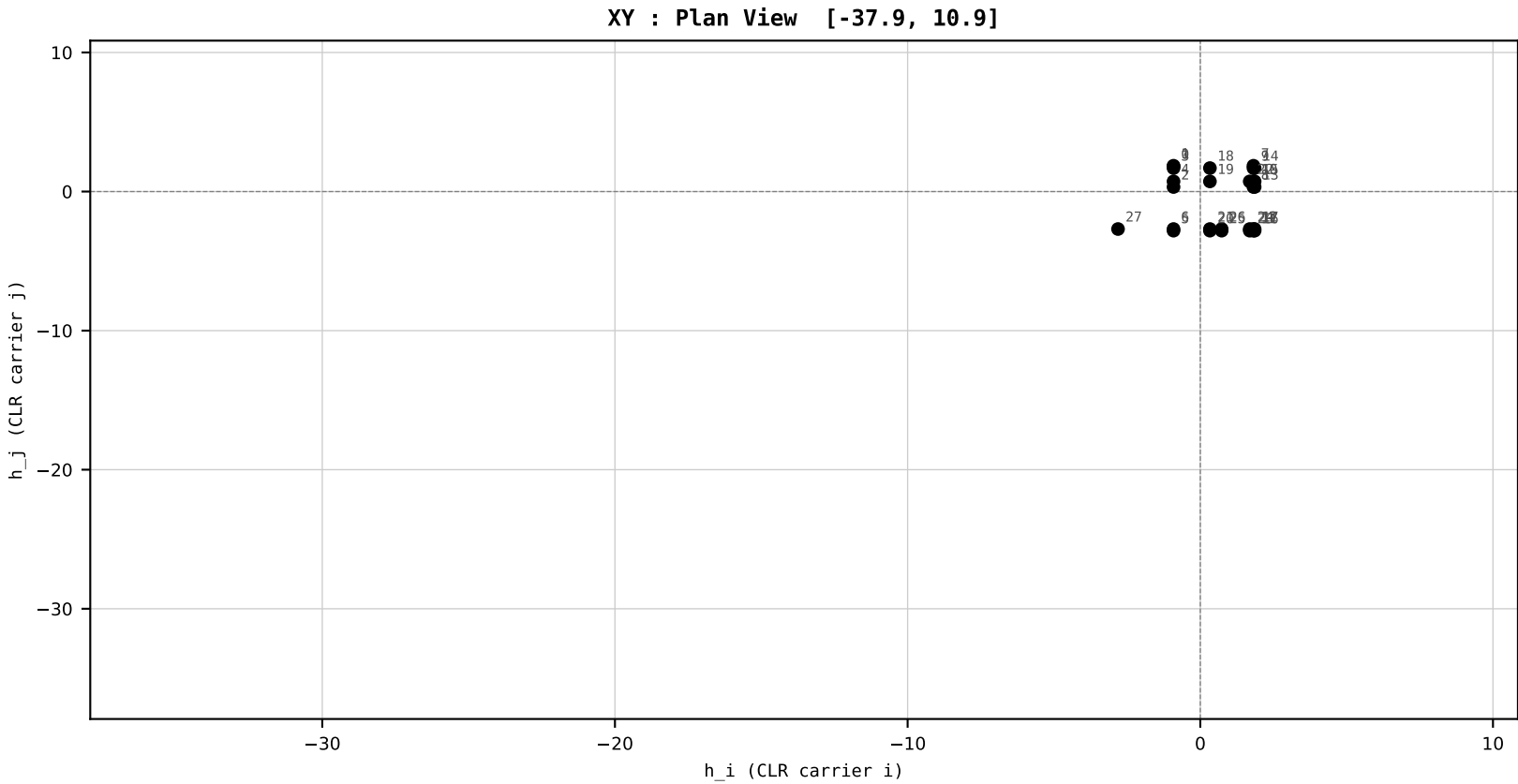
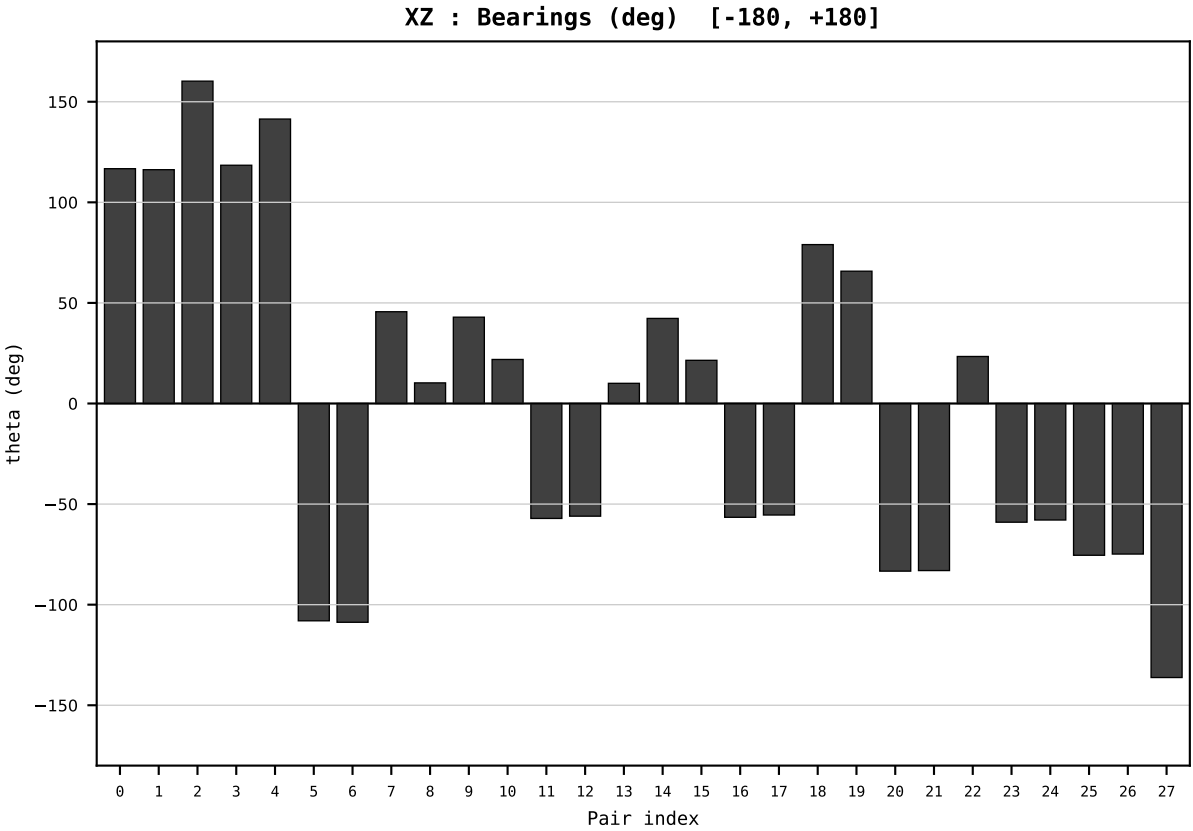
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.247095
Ring = Hs-2
E_metric= 26.2382
kappa_HS= 106.47
omega = 5.2774 deg
d_A = 0.545961
Helm = Other Fossil
Helm d = -0.454982
DR = 4.668 HLR
DR ratio= 106.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

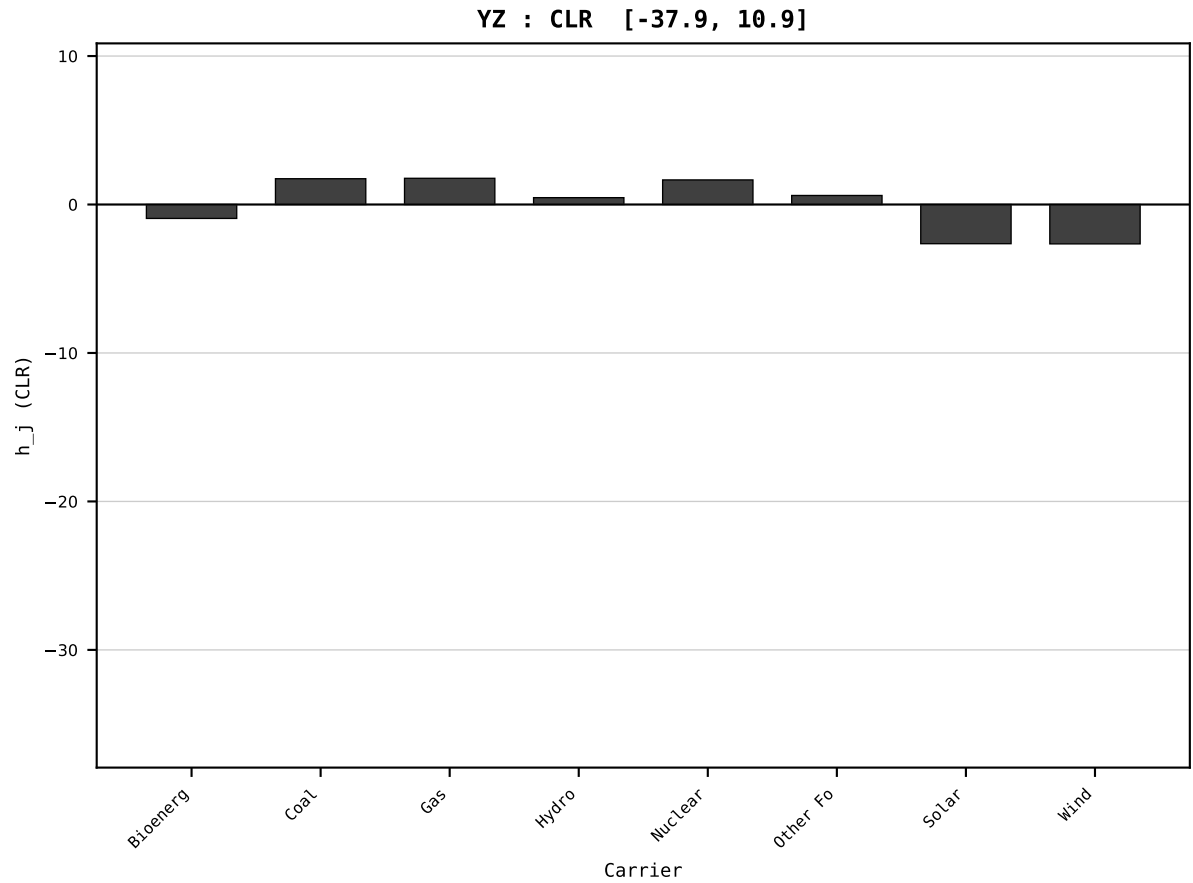
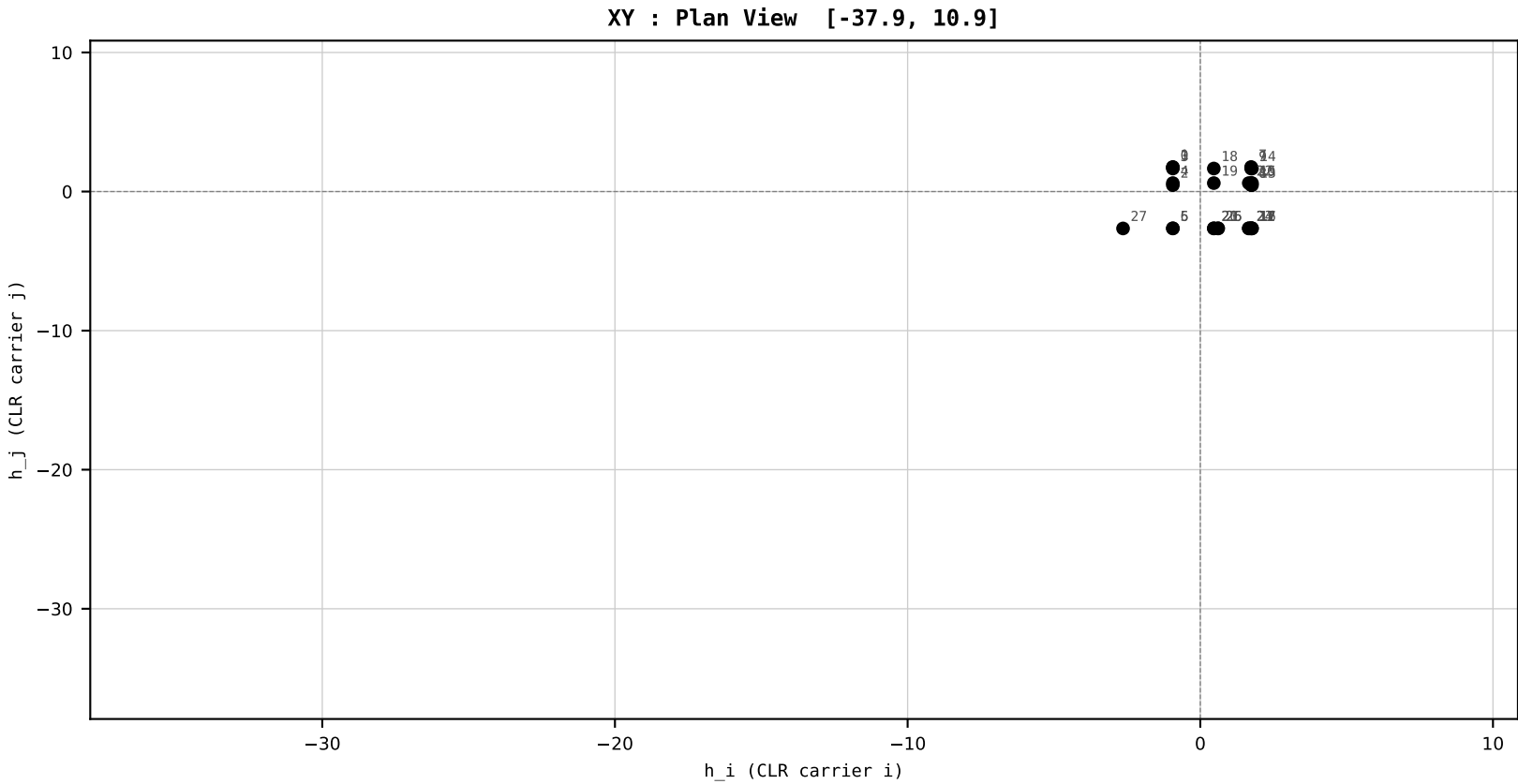
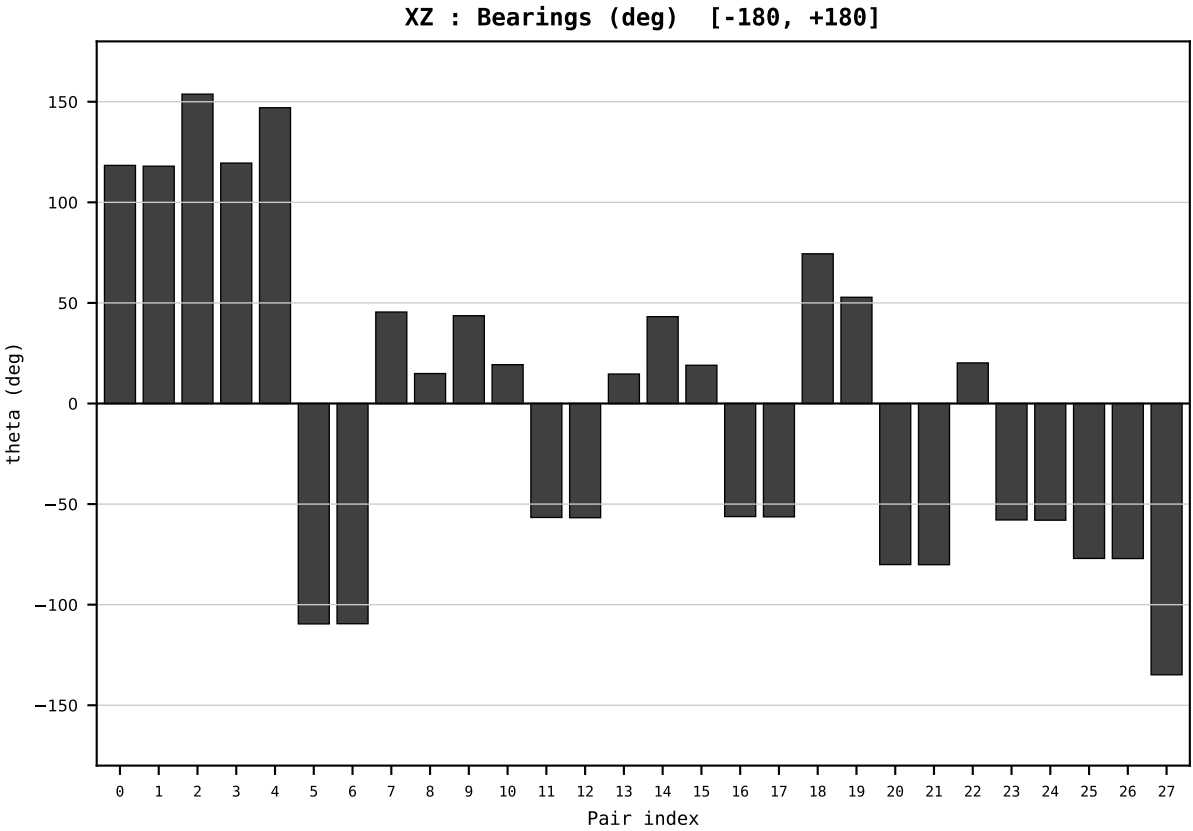
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.237884
Ring = Hs-2
E_metric= 24.3502
kappa_HS= 82.97
omega = 2.4032 deg
d_A = 0.282318
Helm = Solar
Helm d = +0.171611
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

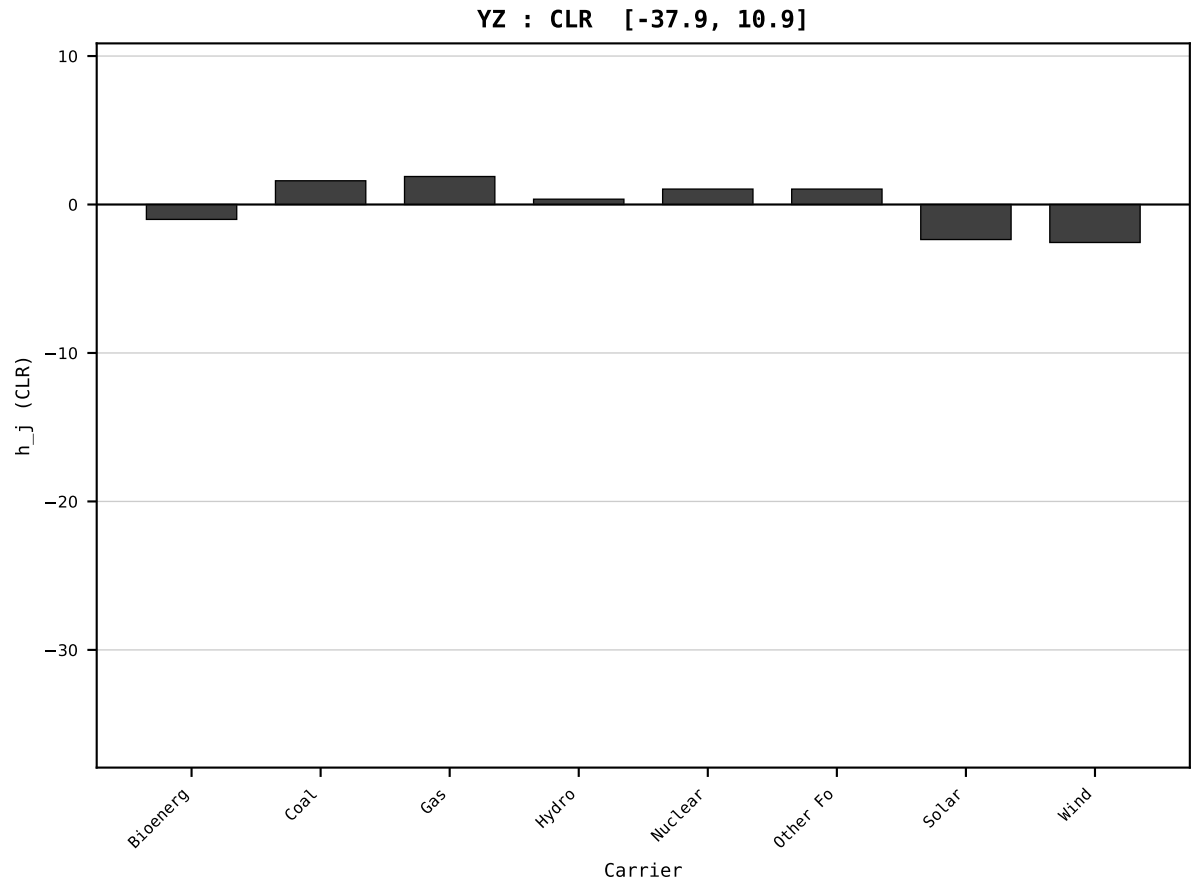
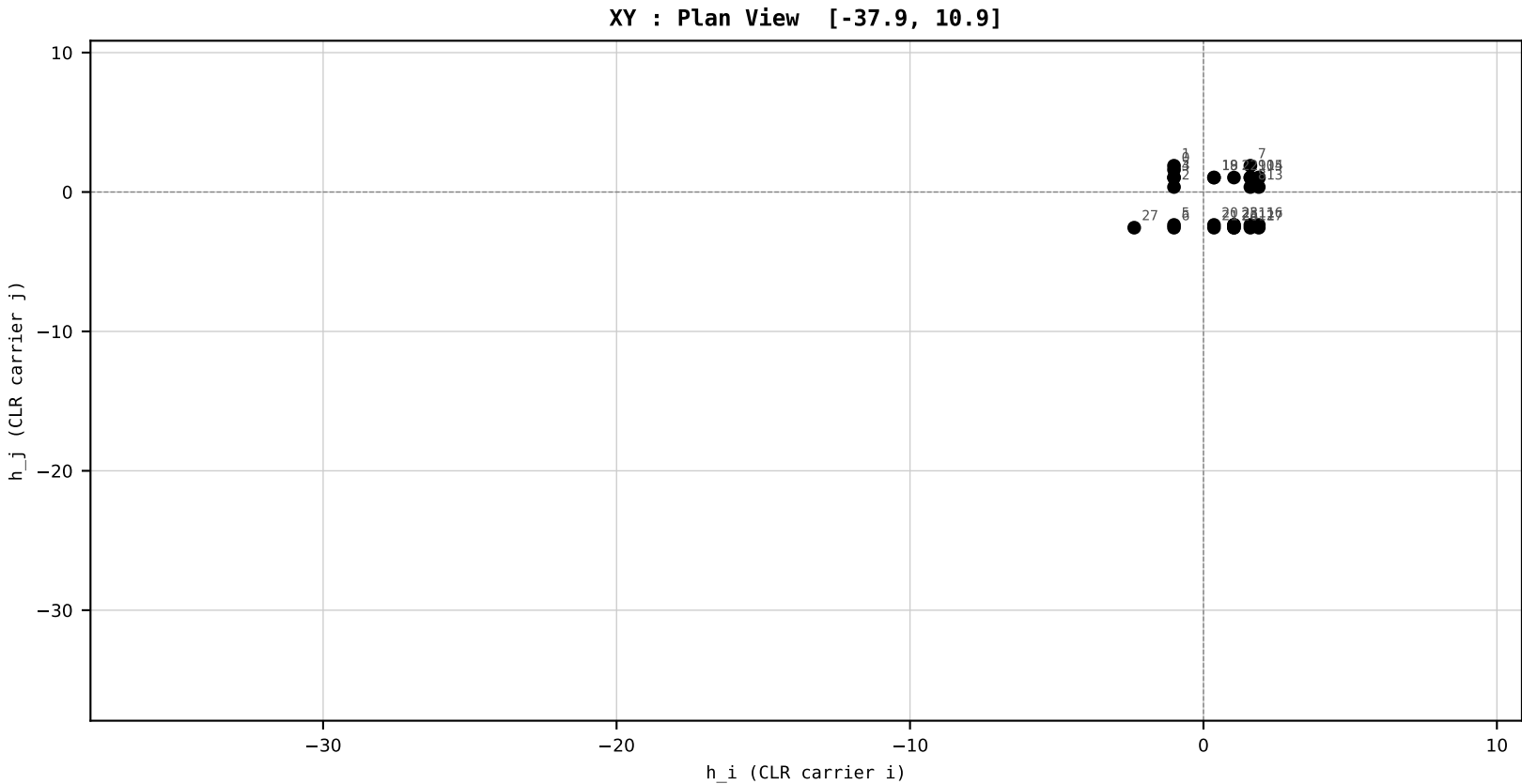
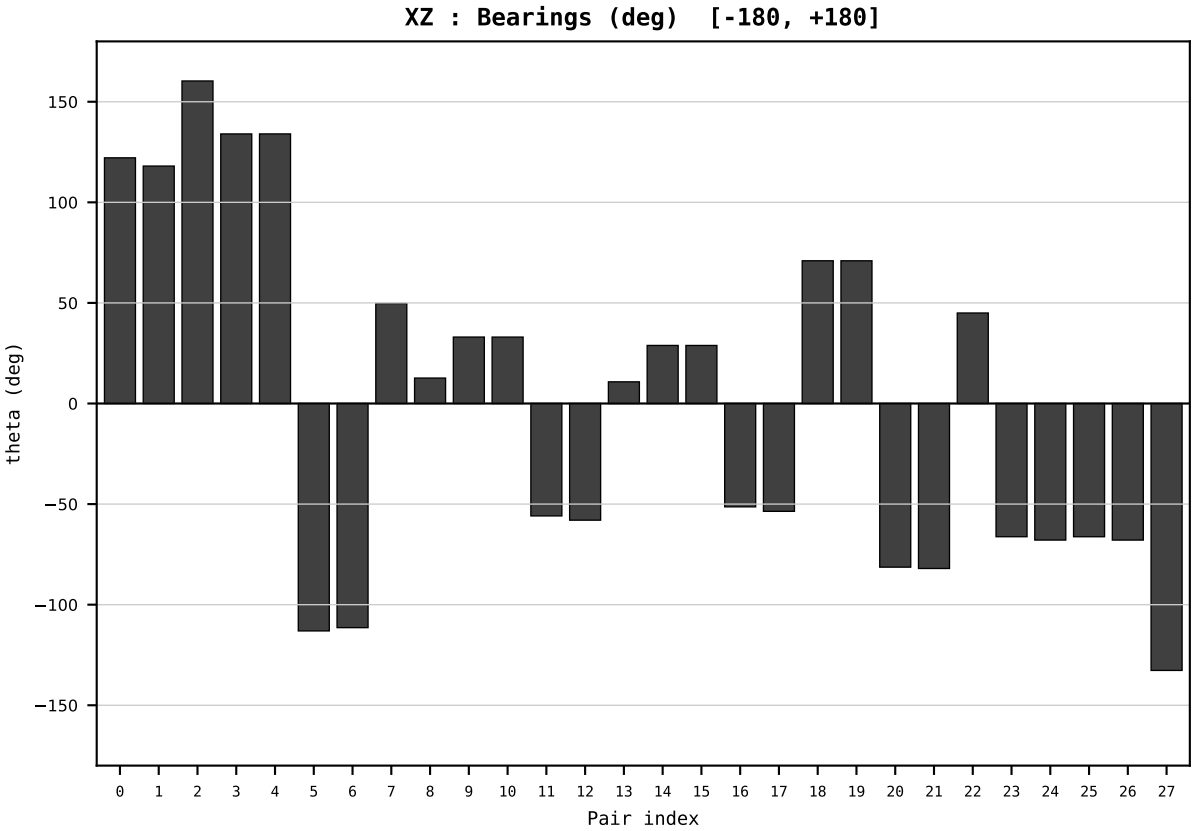
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.230931
Ring = Hs-2
E_metric= 21.5271
kappa_HS= 85.16
omega = 9.4013 deg
d_A = 0.837843
Helm = Nuclear
Helm d = -0.617218
DR = 4.445 HLR
DR ratio= 85.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

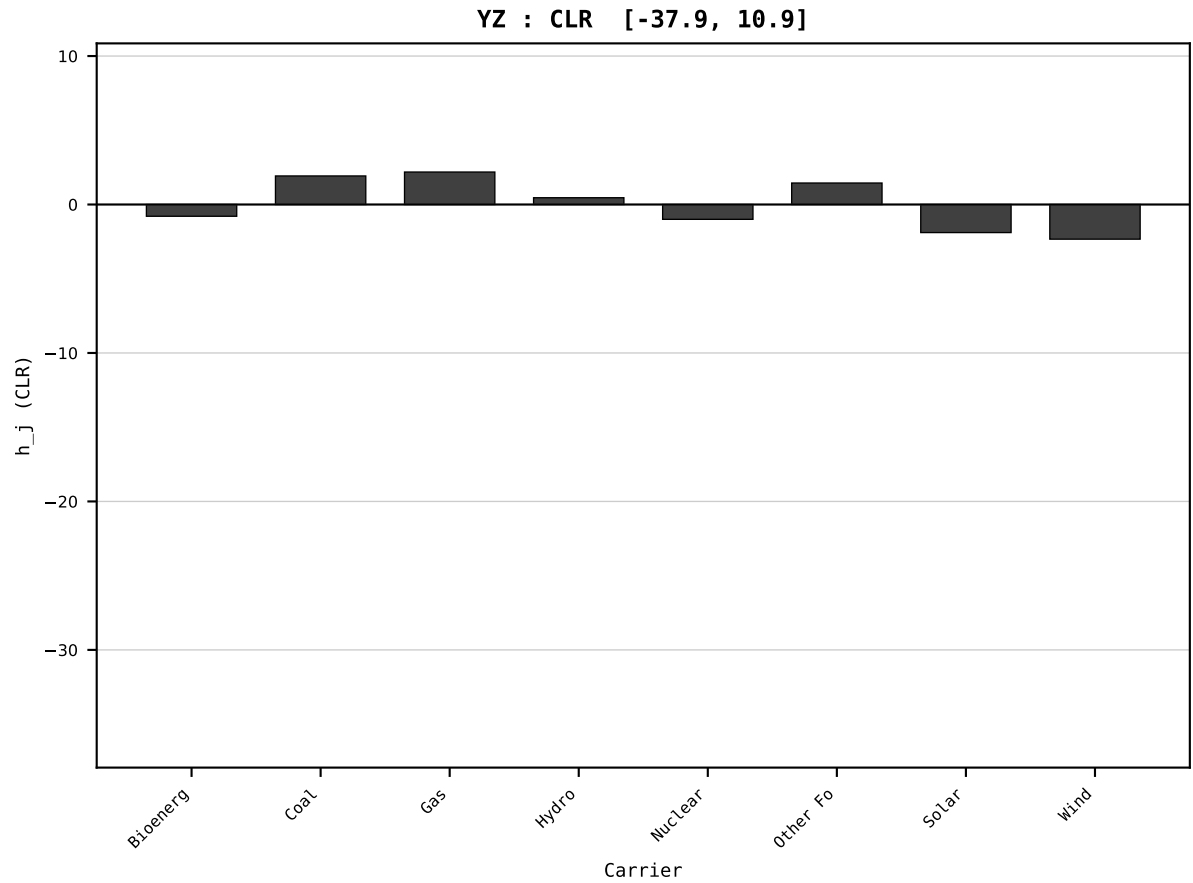
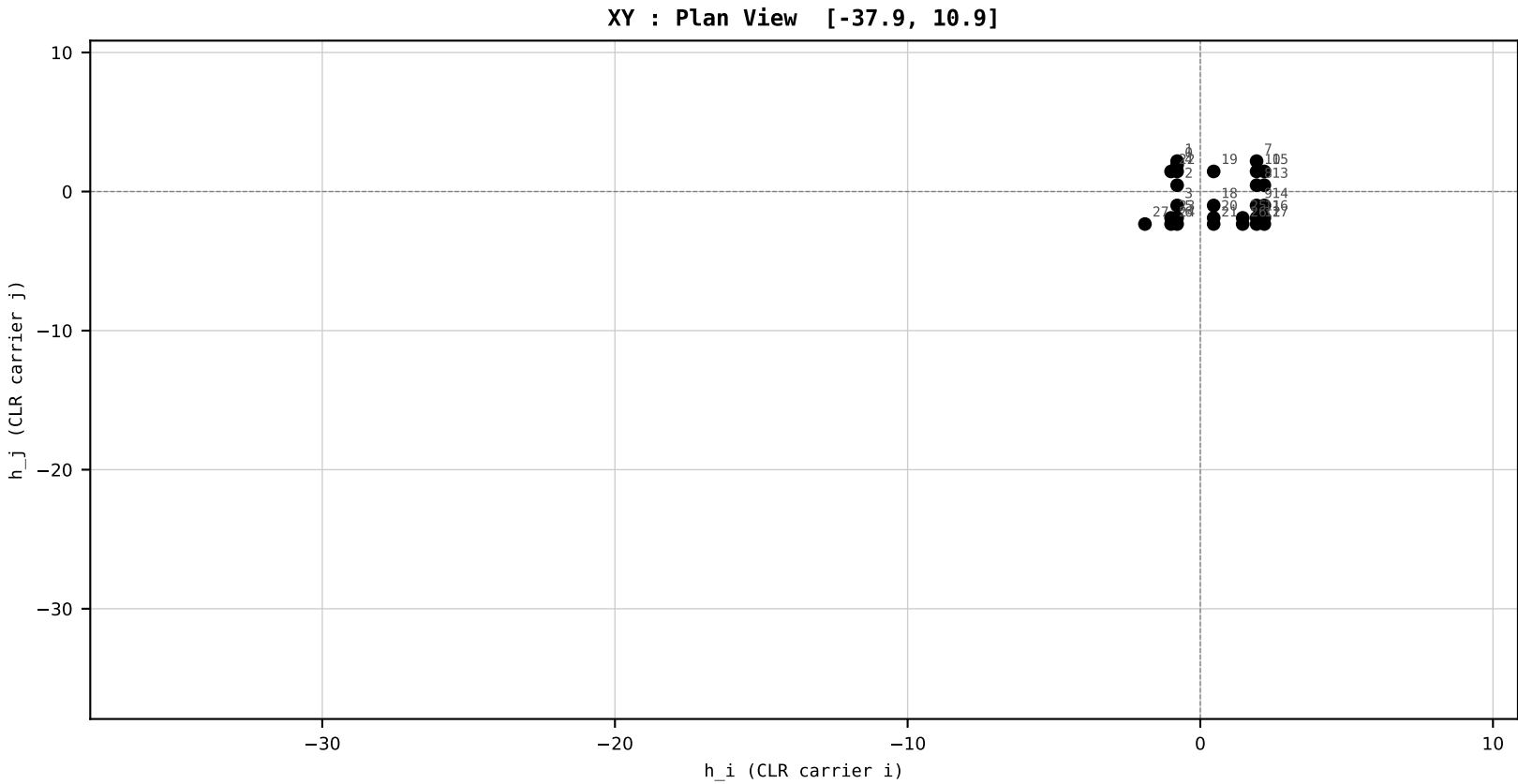
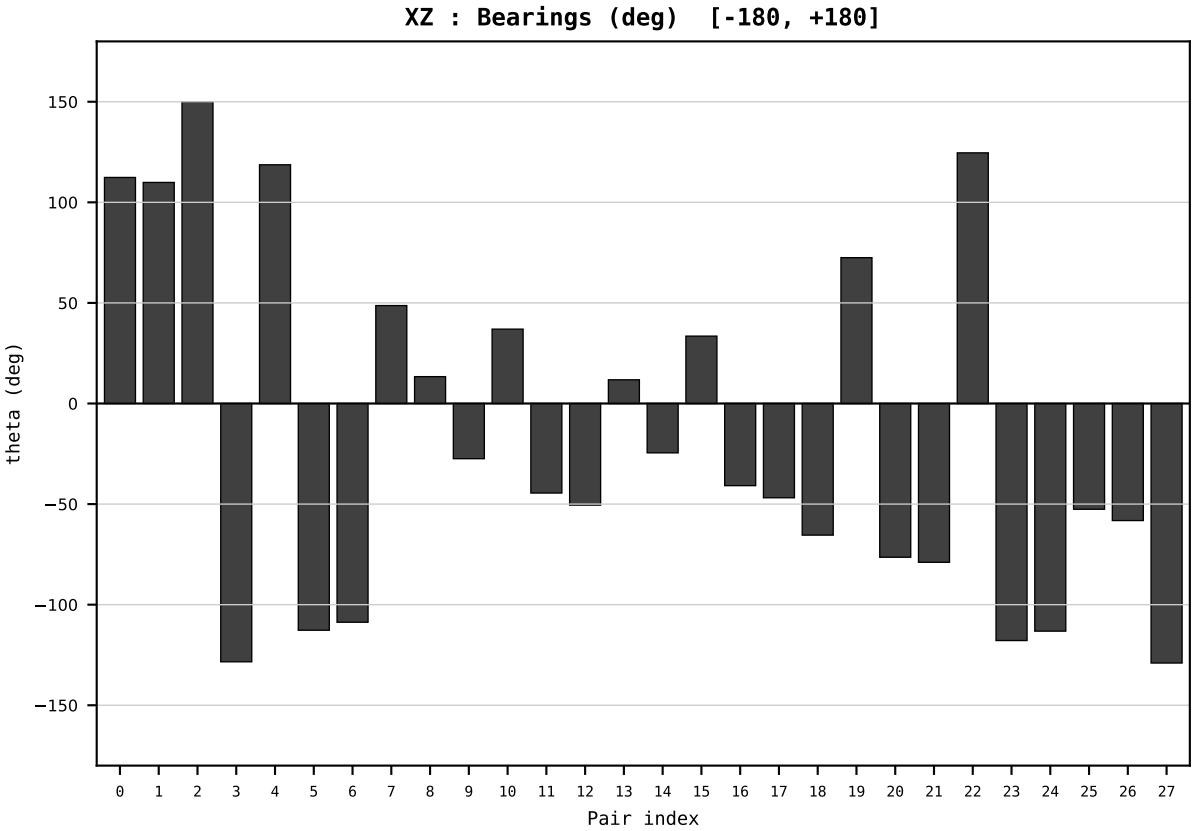
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312291
Ring = Hs-3
E_metric= 21.4426
kappa_HS= 92.07
omega = 27.4642 deg
d_A = 2.200635
Helm = Nuclear
Helm d = -2.038262
DR = 4.523 HLR
DR ratio= 92.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



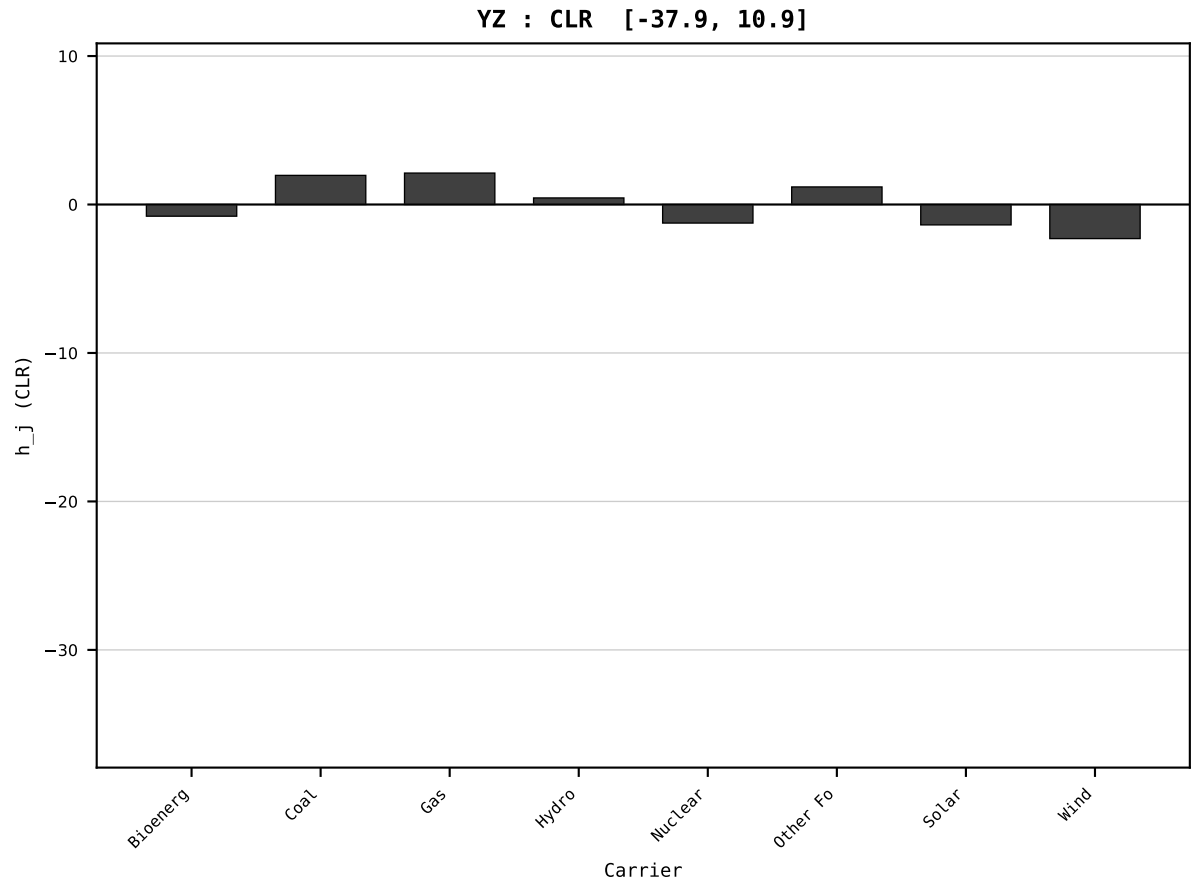
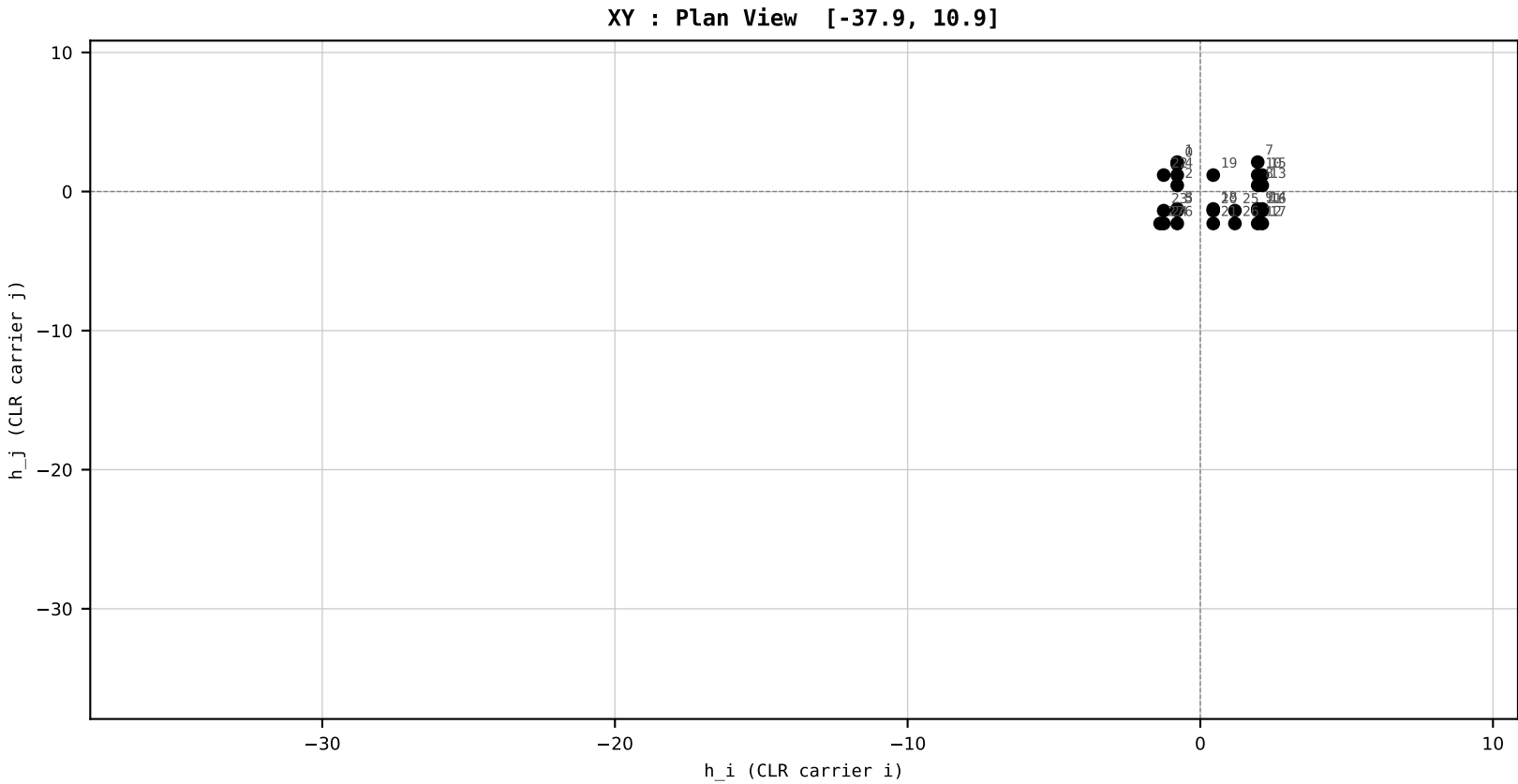
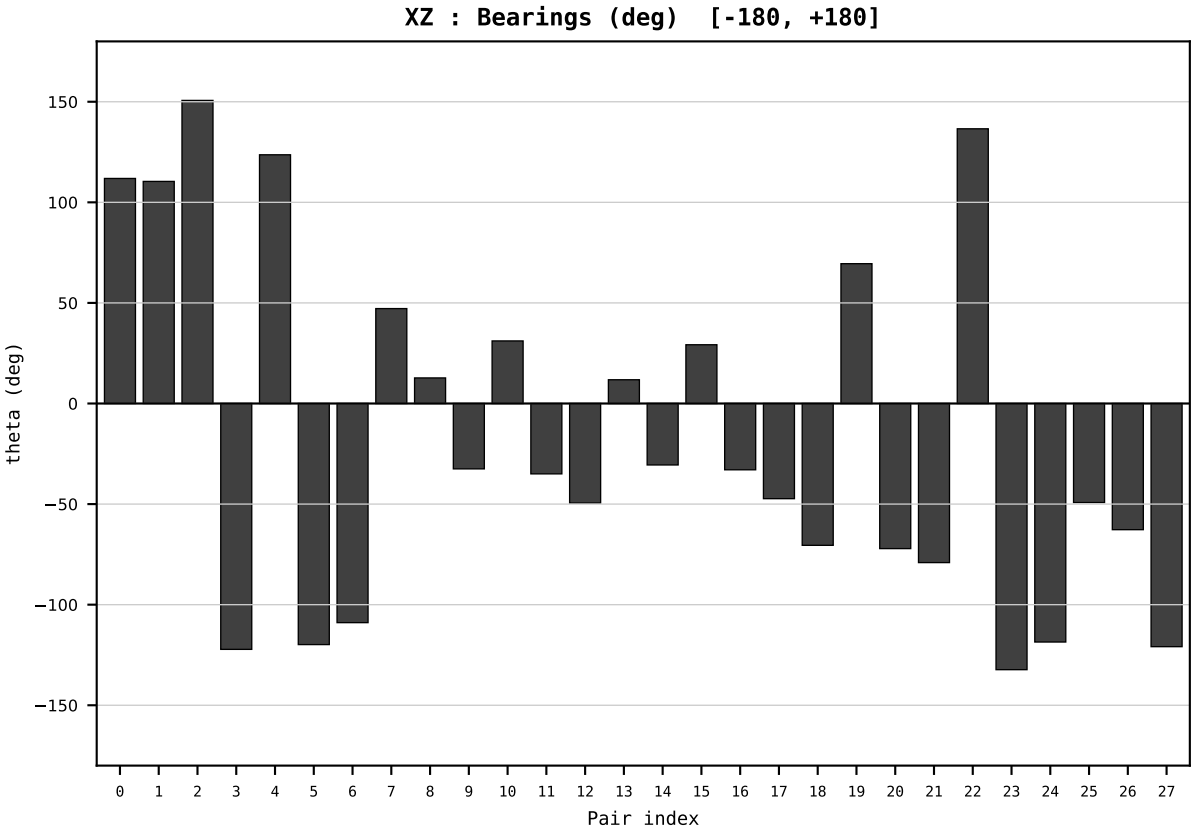
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312583
Ring = Hs-3
E_metric= 19.2716
kappa_HS= 82.57
omega = 7.5396 deg
d_A = 0.639857
Helm = Solar
Helm d = +0.518125
DR = 4.414 HLR
DR ratio= 82.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

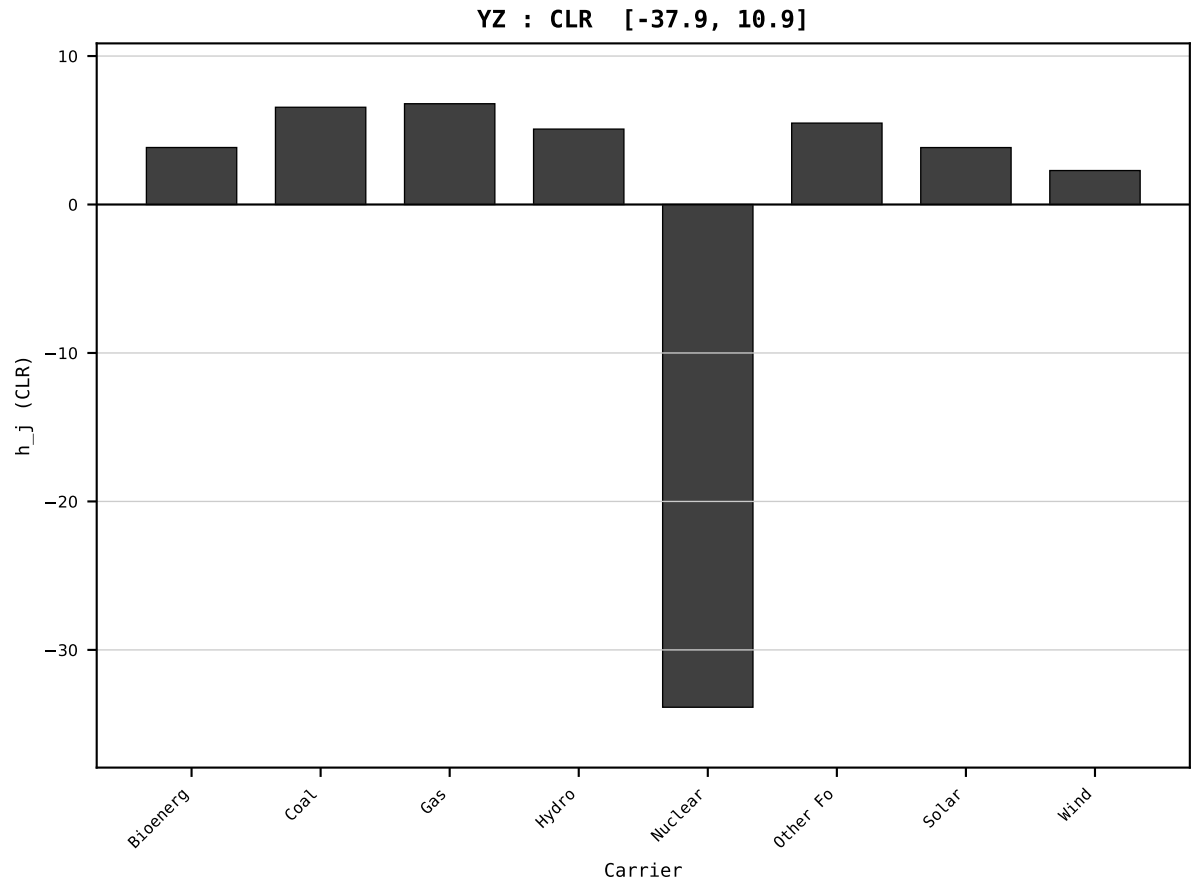
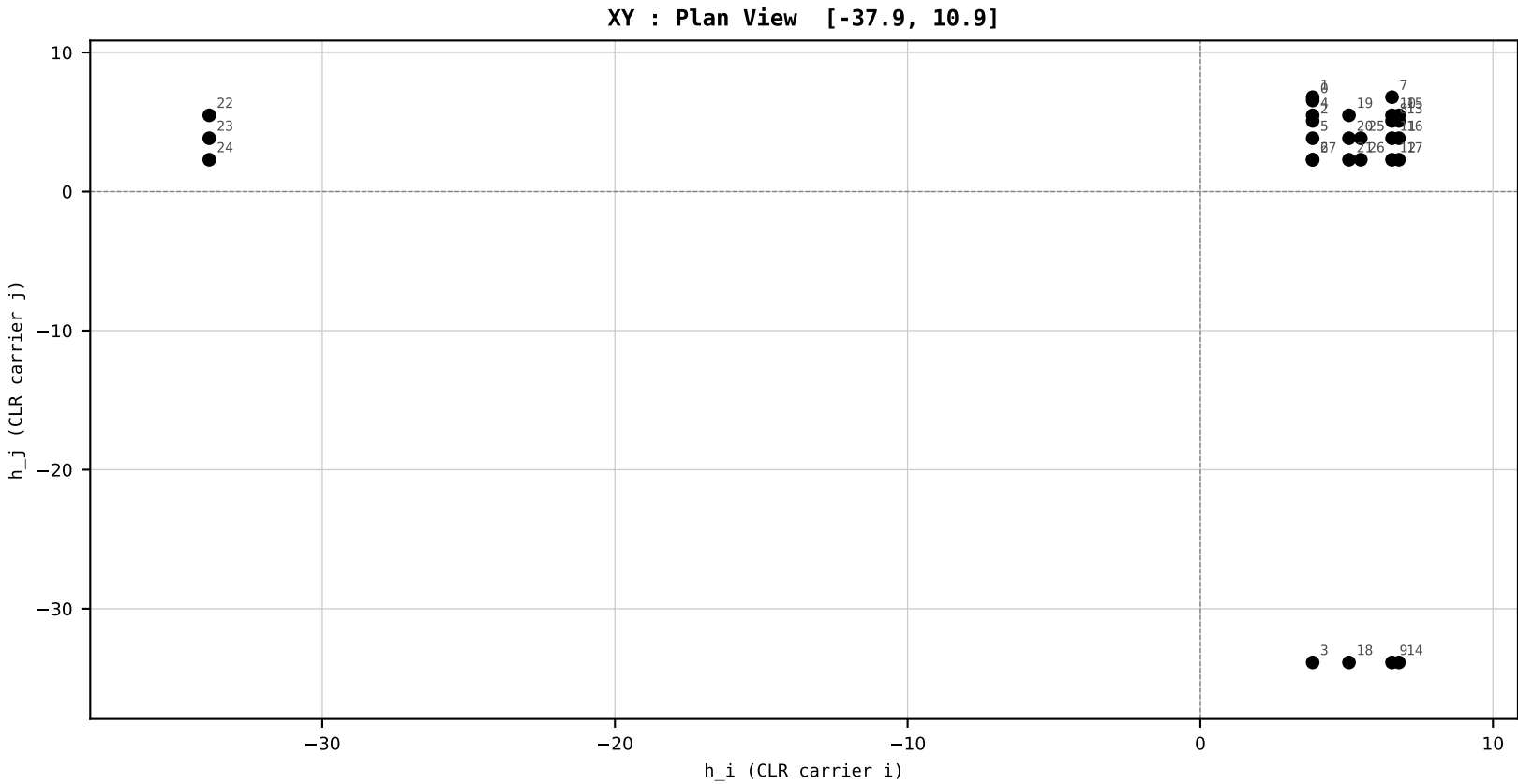
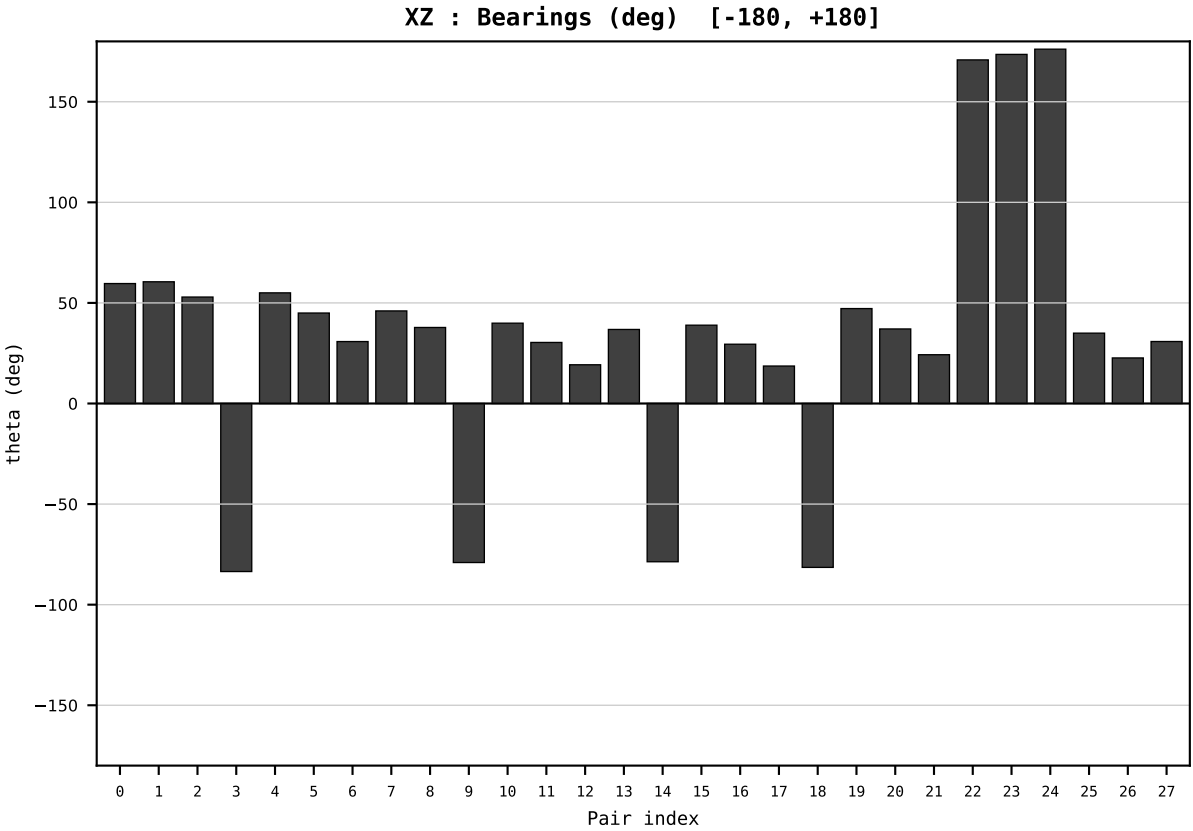
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.340926
Ring = Hs-3
E_metric= 896.7320
kappa_HS= 45087999999999936.00
omega = 66.1070 deg
d_A = 34.870573
Helm = Nuclear
Helm d = -32.612482
DR = 40.650 HLR
DR ratio= 450879998367363840.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

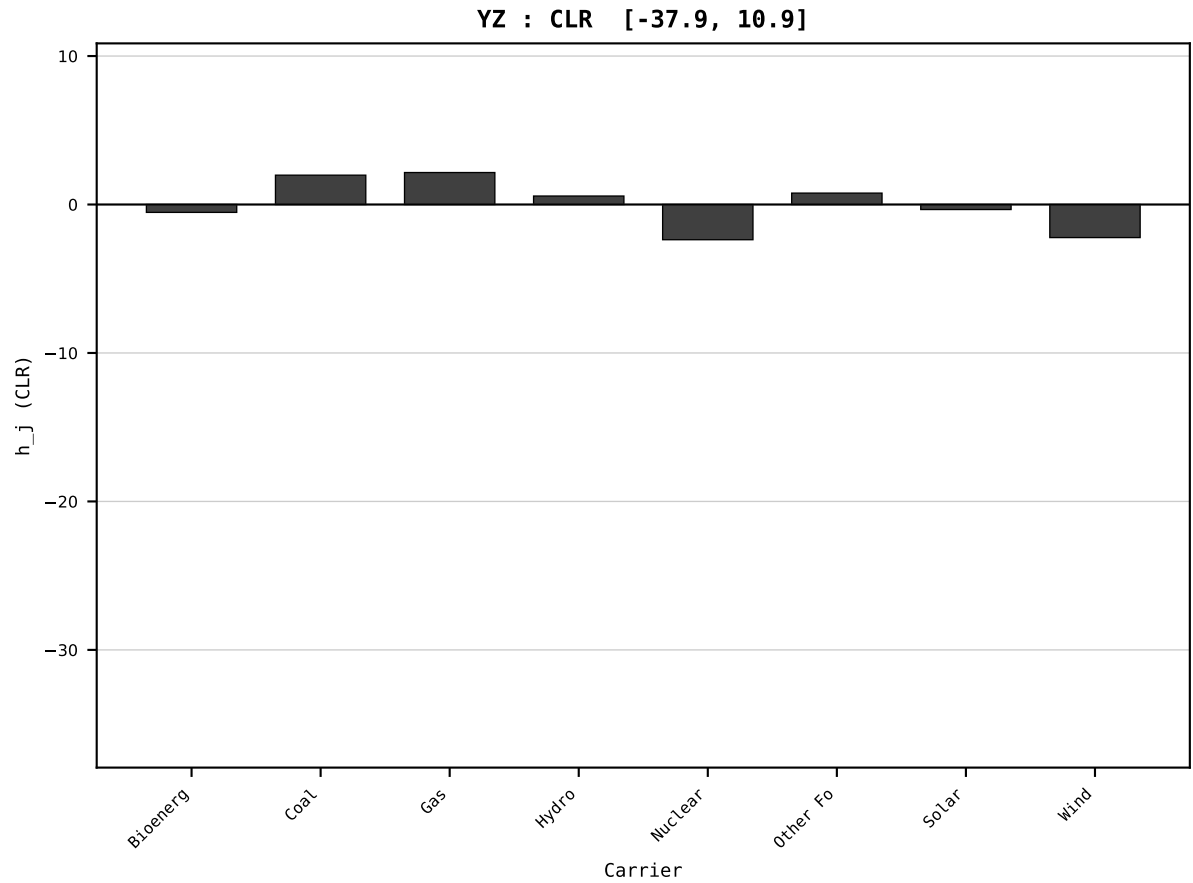
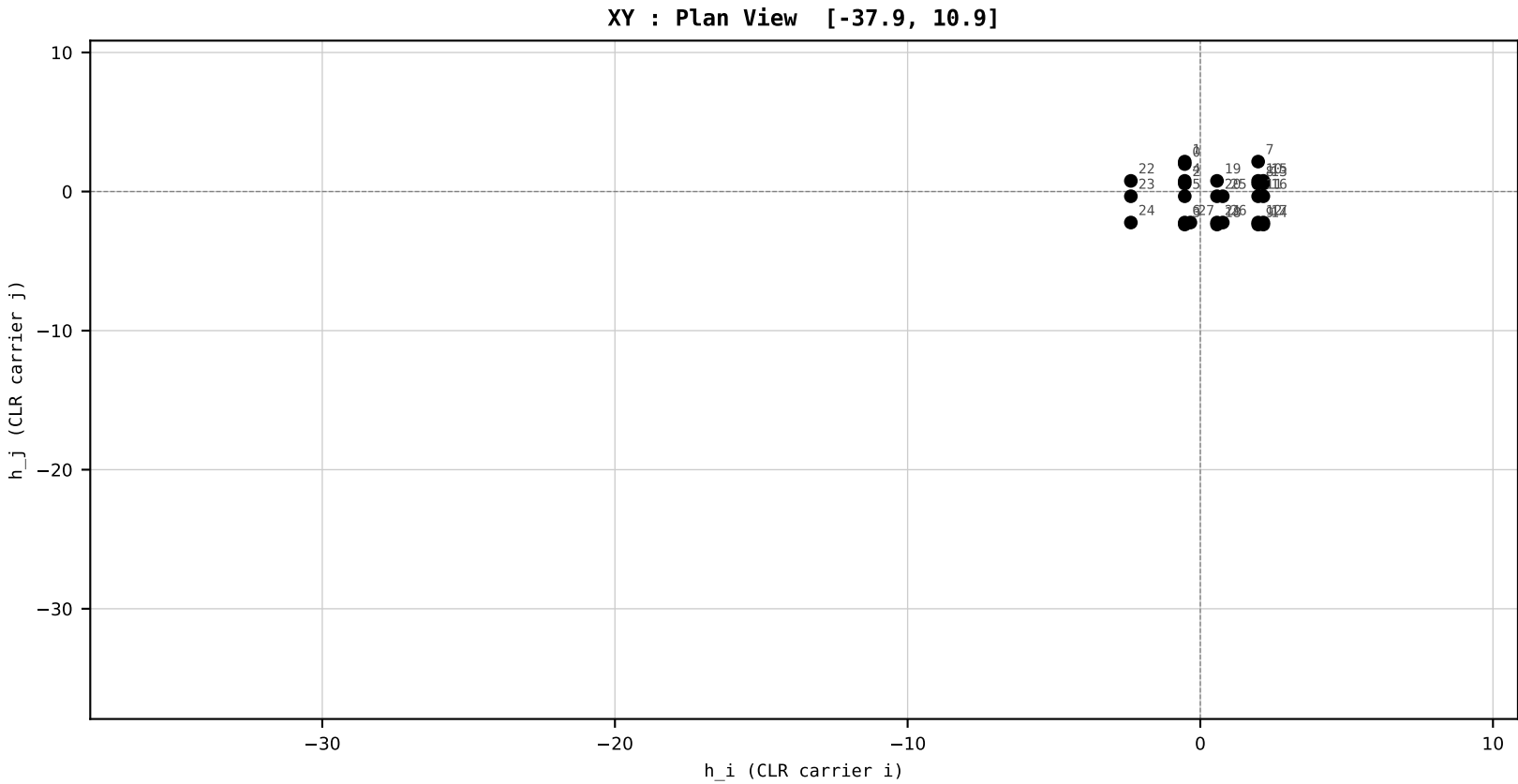
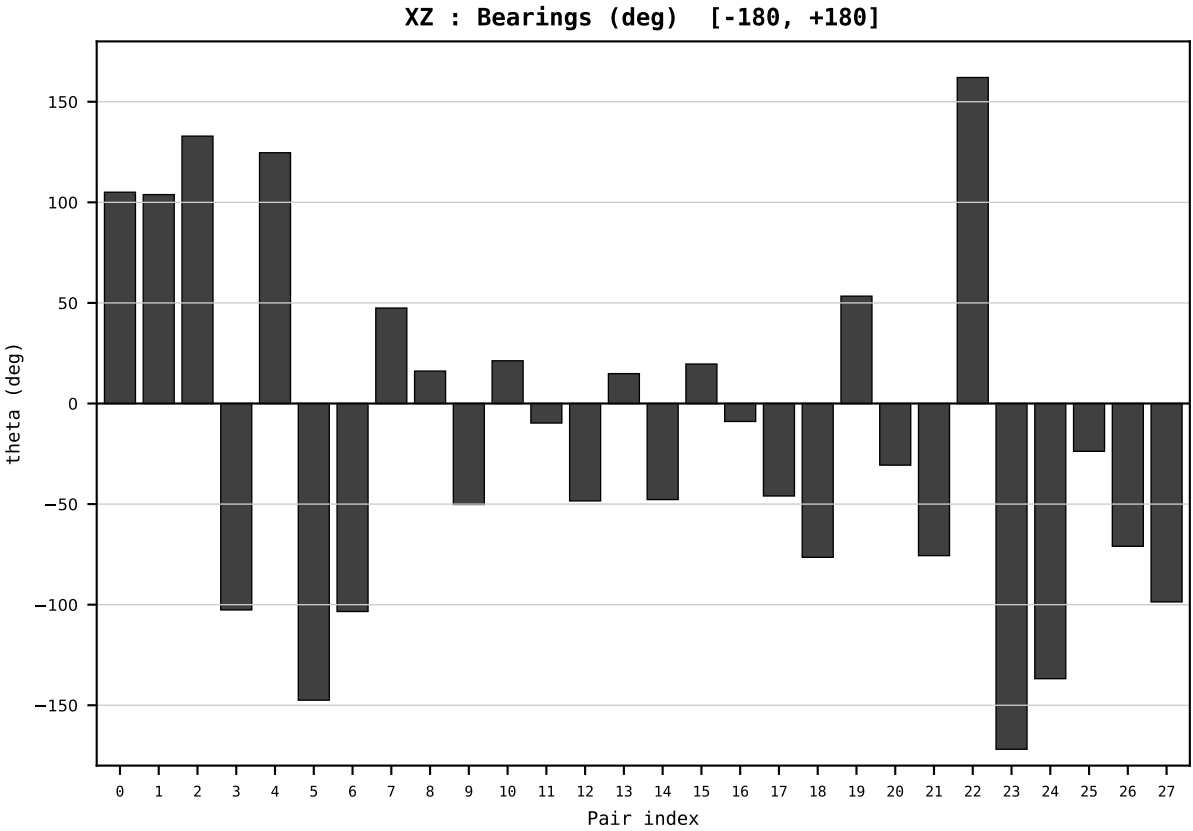
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.309676
Ring = Hs-3
E_metric= 20.4479
kappa_HS= 92.37
omega = 49.6746 deg
d_A = 33.667556
Helm = Nuclear
Helm d = +31.490431
DR = 4.526 HLR
DR ratio= 92.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



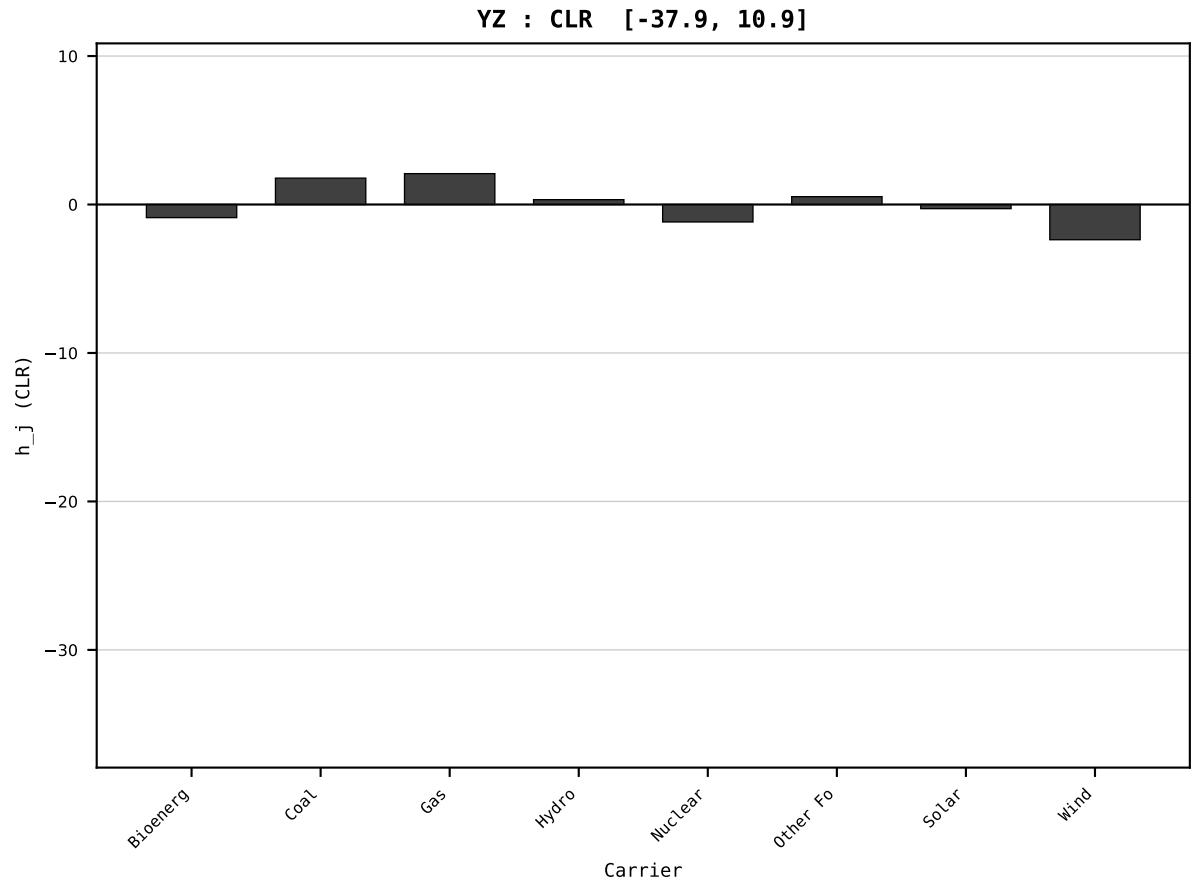
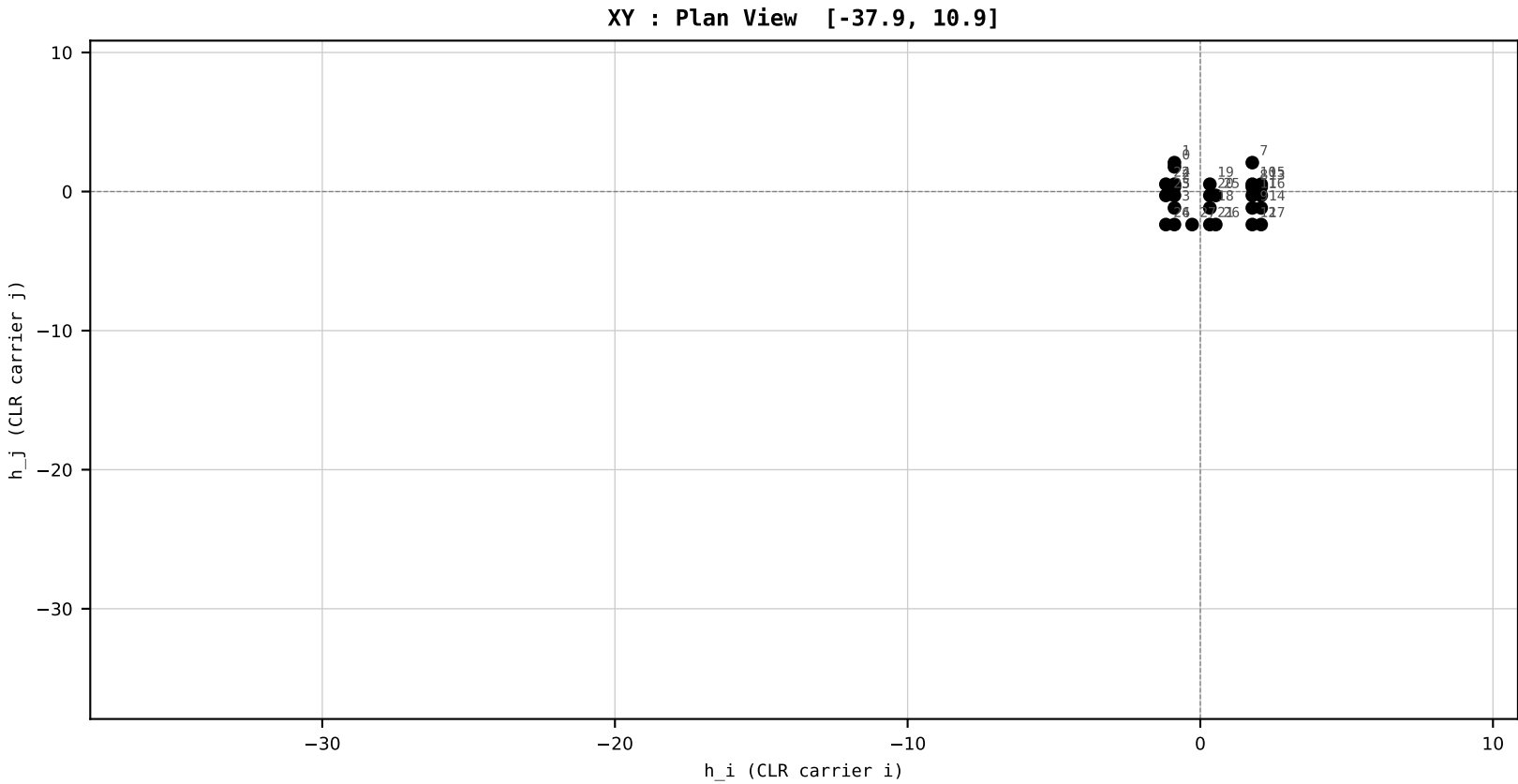
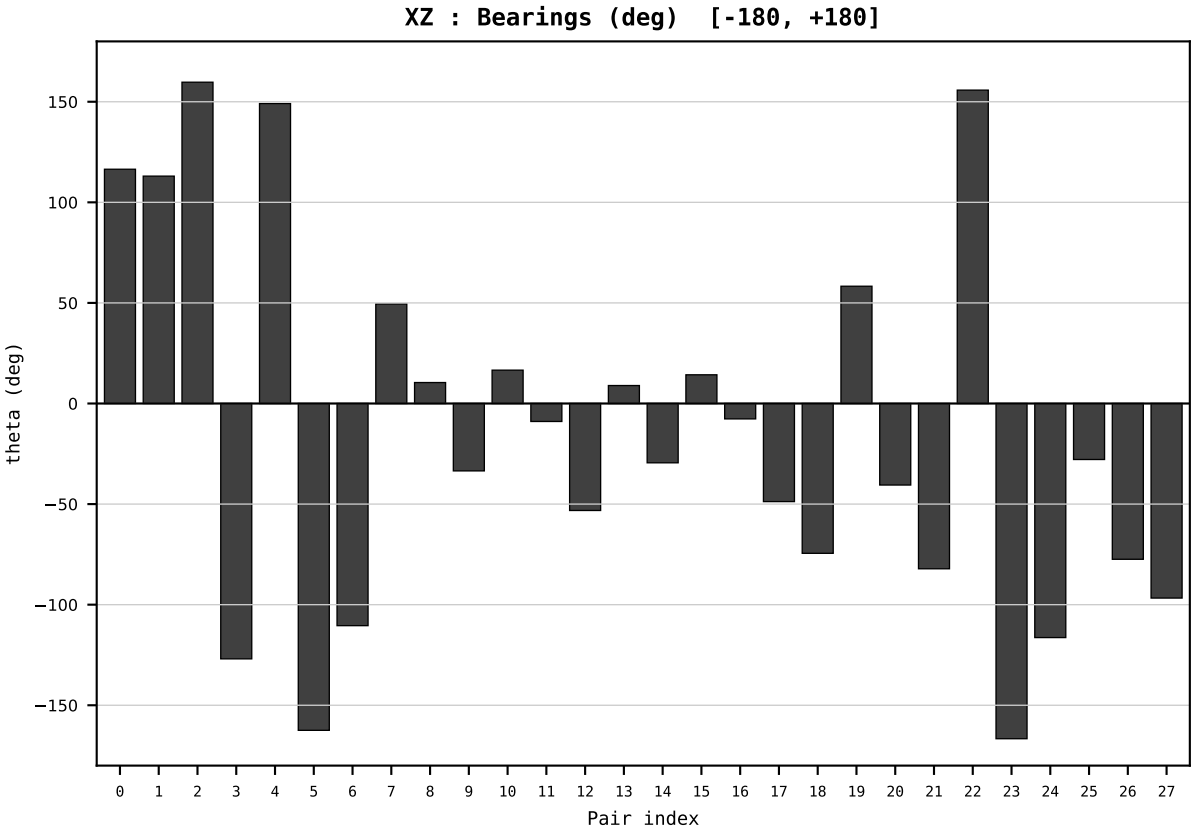
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.303033
Ring = Hs-3
E_metric= 15.7404
kappa_HS= 85.85
omega = 16.2591 deg
d_A = 1.320048
Helm = Nuclear
Helm d = +1.196283
DR = 4.453 HLR
DR ratio= 85.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

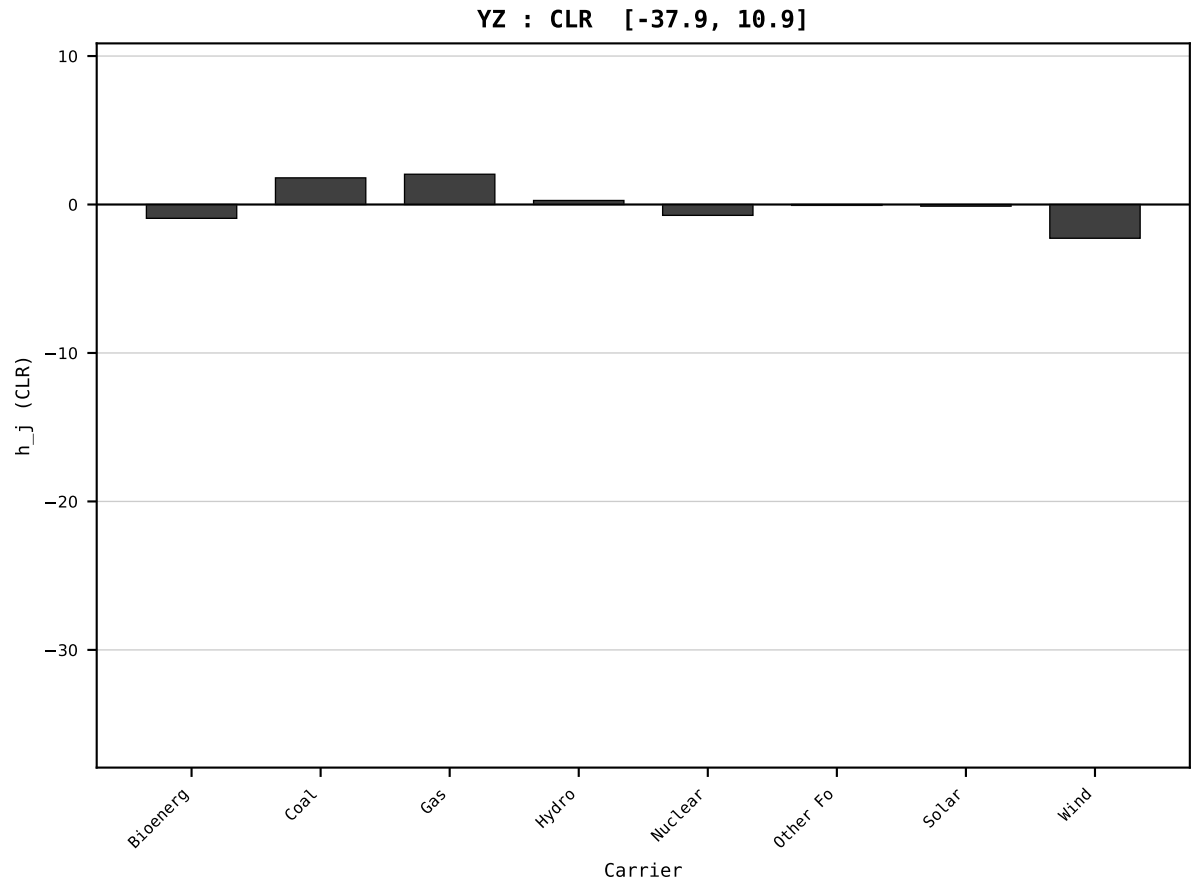
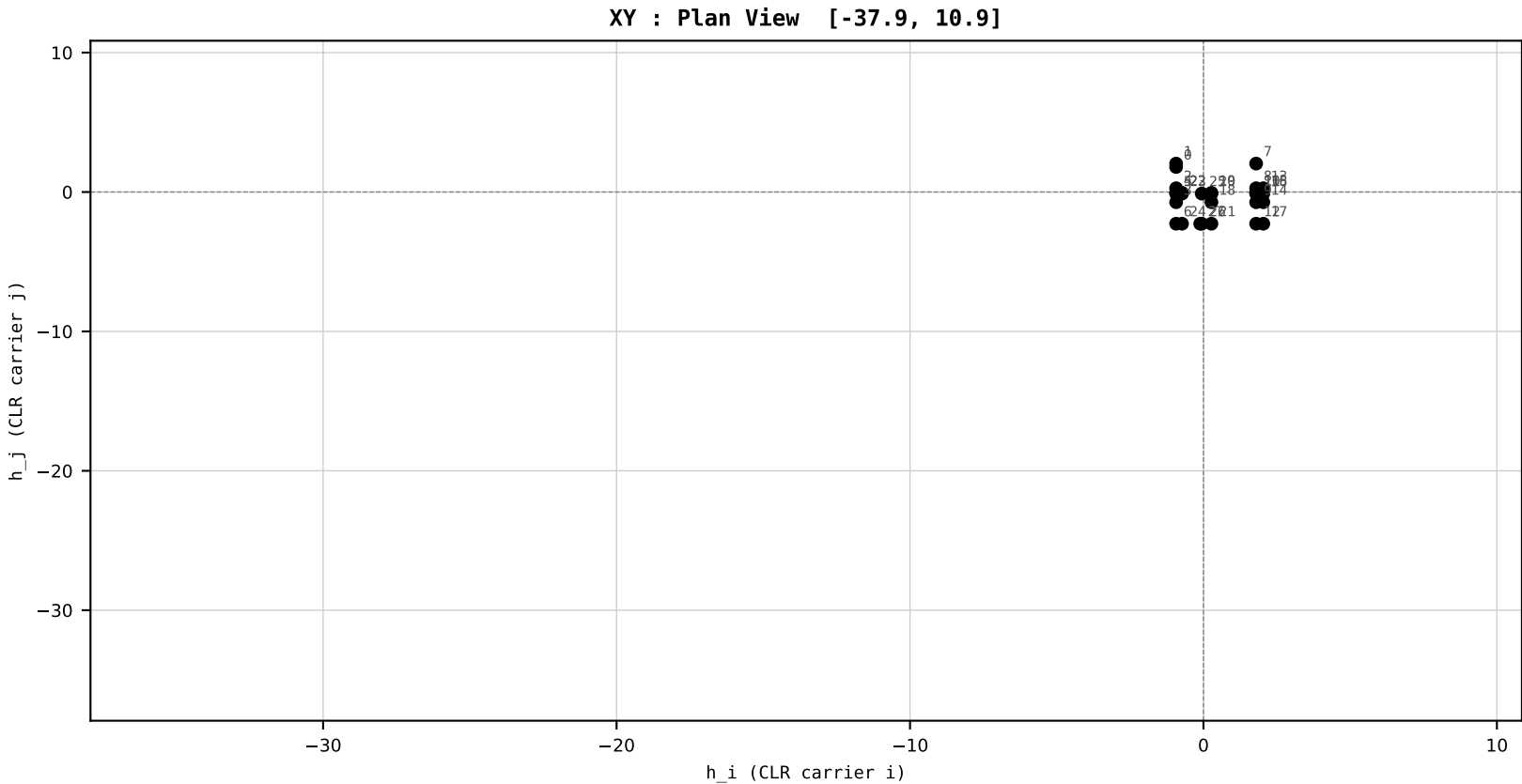
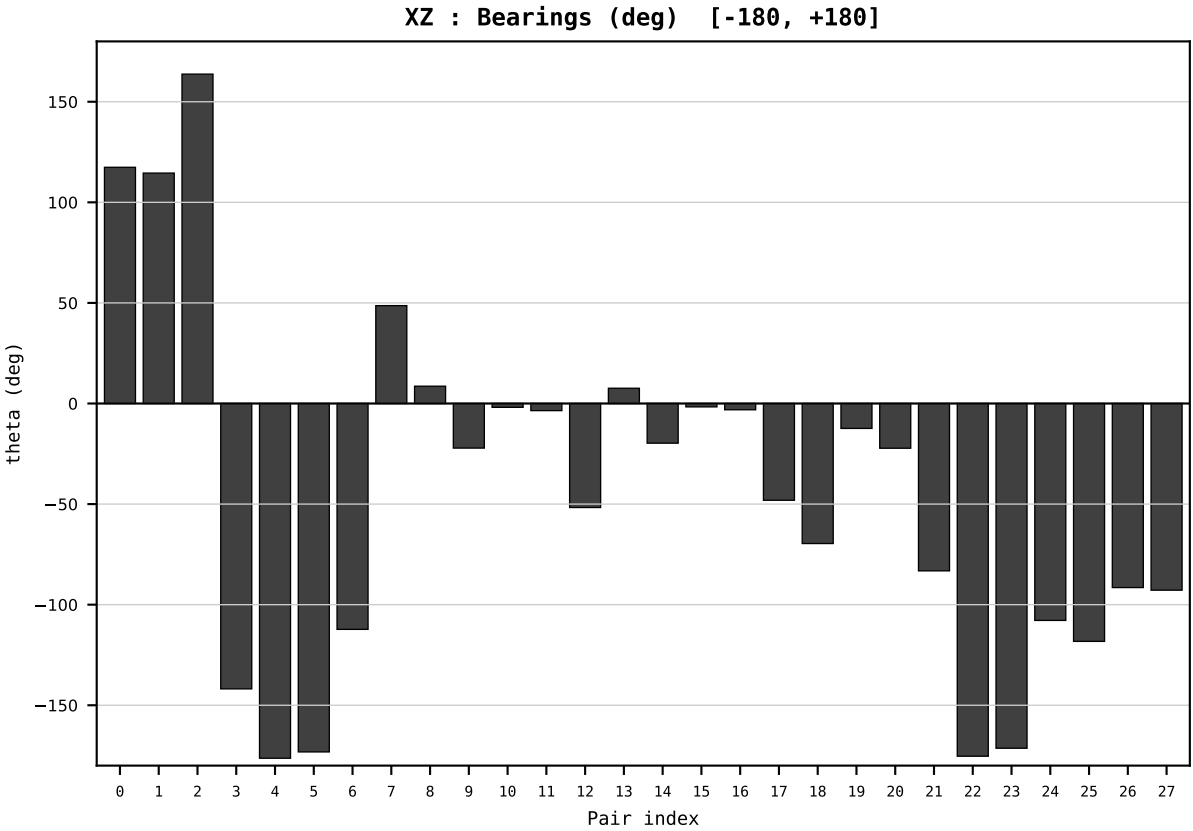
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.308283
Ring = Hs-3
E_metric= 14.0385
kappa_HS= 74.62
omega = 10.9468 deg
d_A = 0.767882
Helm = Other Fossil
Helm d = -0.588595
DR = 4.312 HLR
DR ratio= 74.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

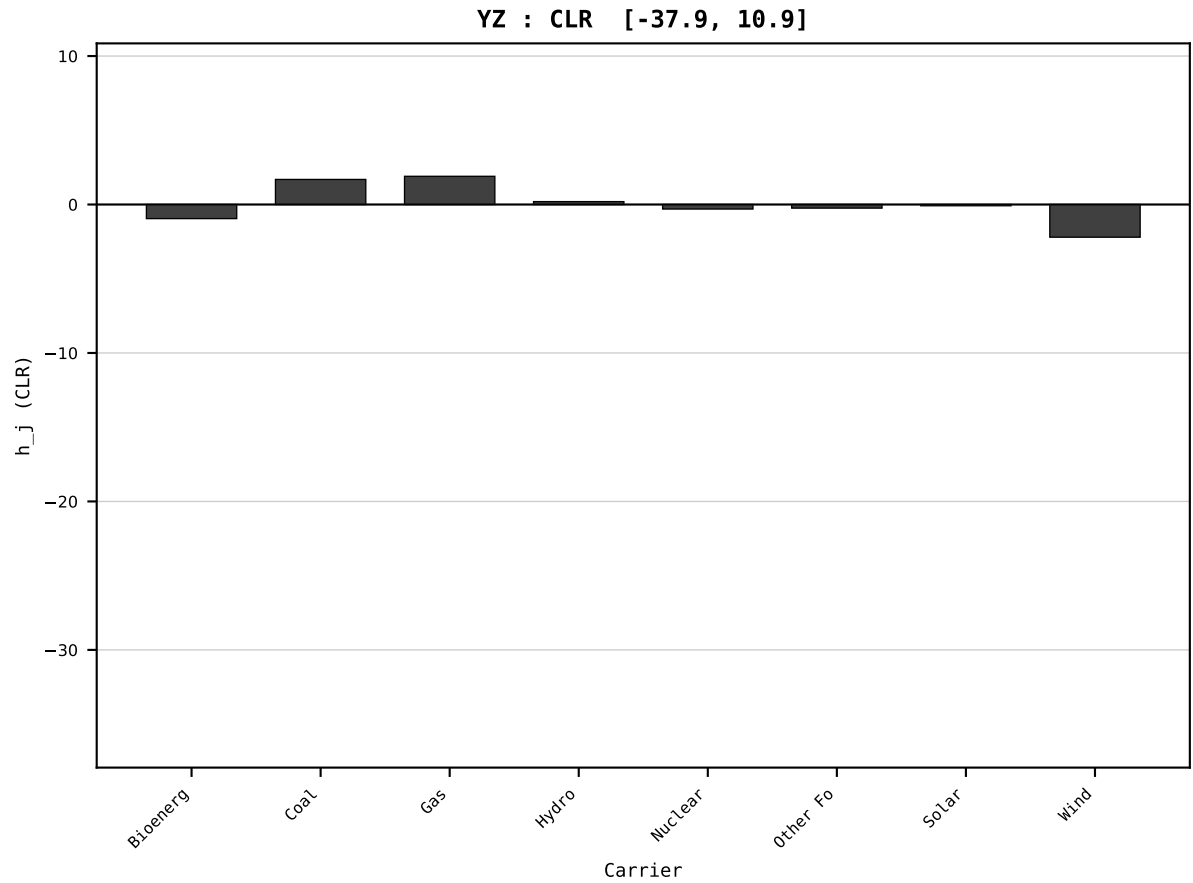
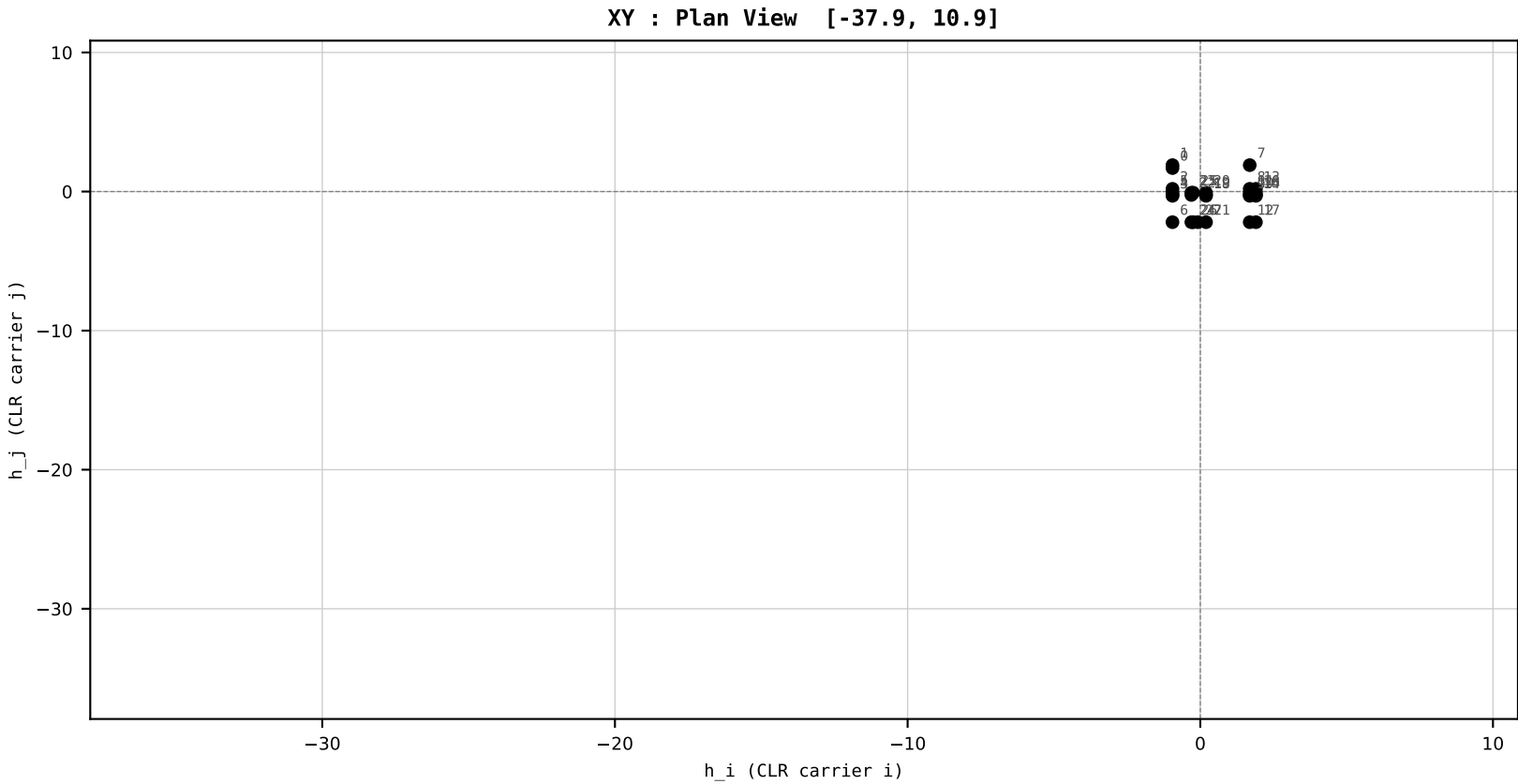
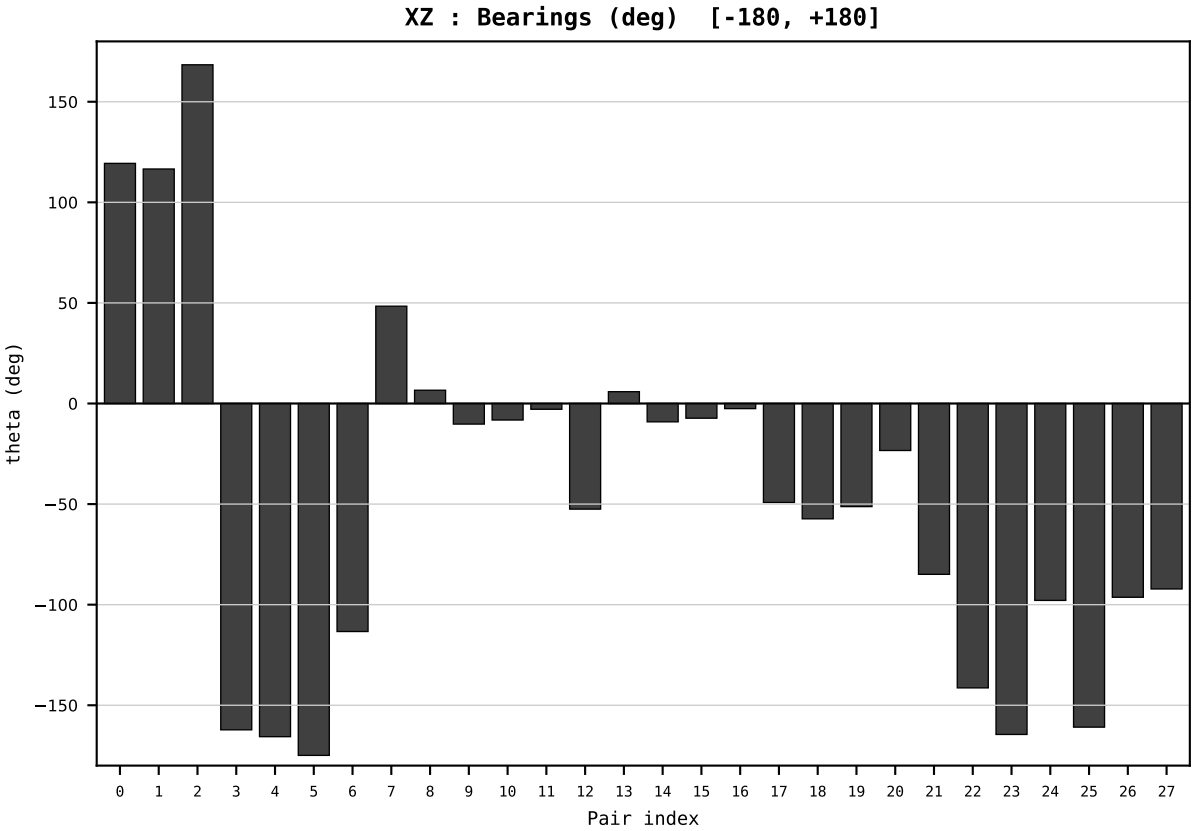
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.279970
Ring = Hs-2
E_metric= 12.4125
kappa_HS= 60.46
omega = 7.1966 deg
d_A = 0.507945
Helm = Nuclear
Helm d = +0.426213
DR = 4.102 HLR
DR ratio= 60.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

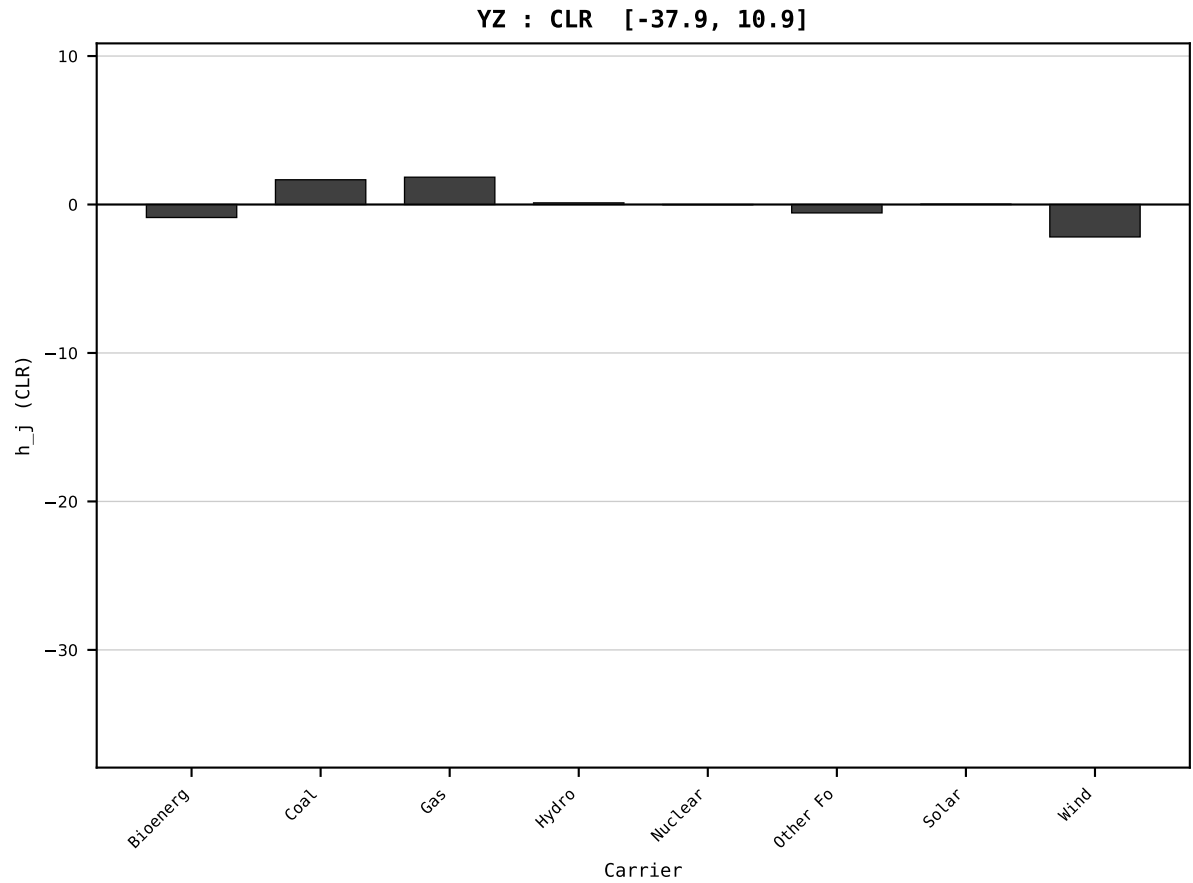
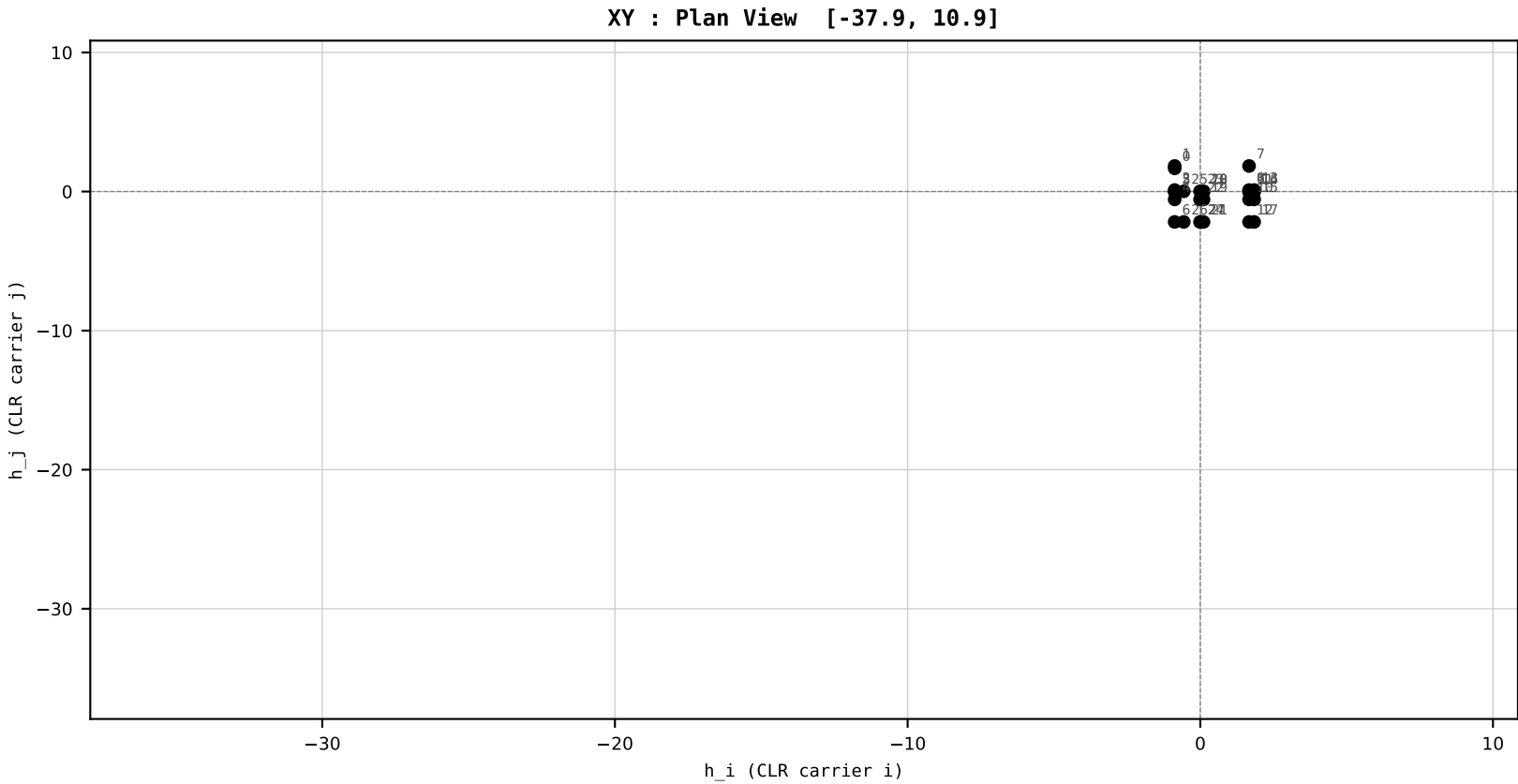
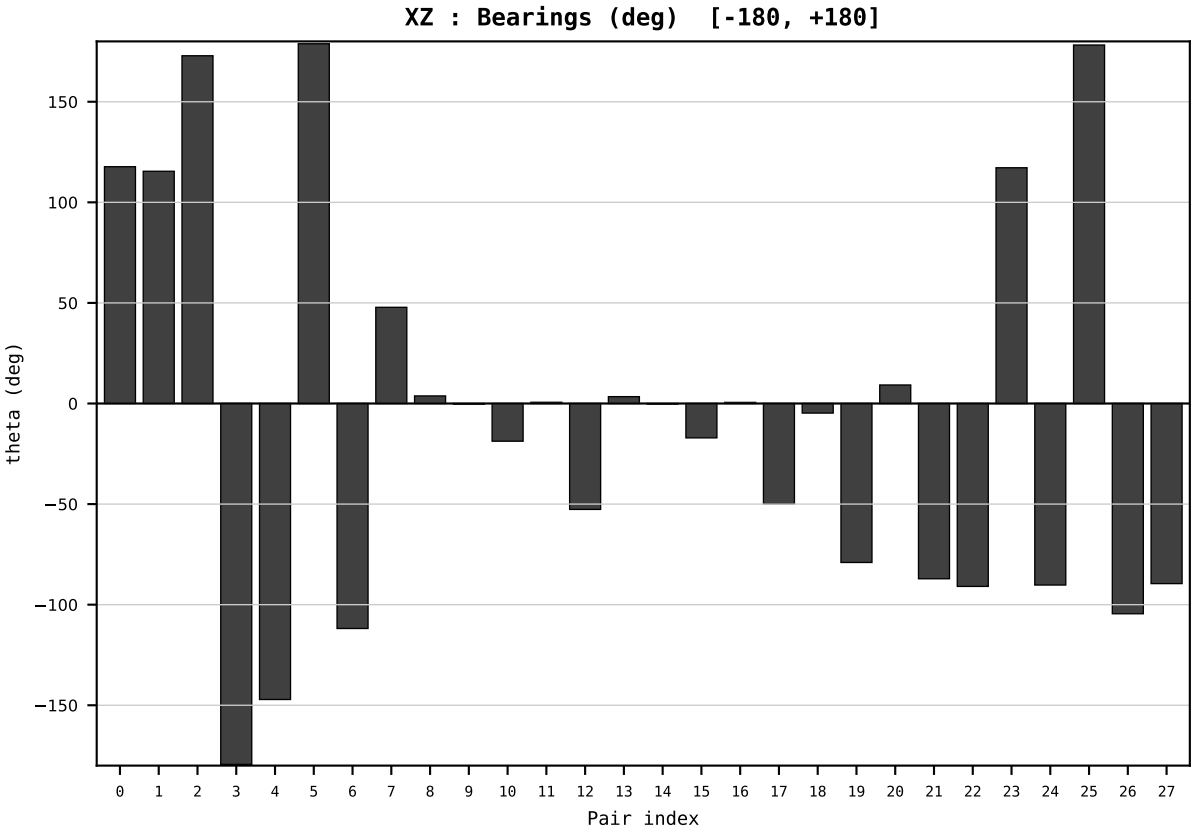
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.268630
Ring = Hs-2
E_metric= 12.0301
kappa_HS= 55.89
omega = 7.6236 deg
d_A = 0.467986
Helm = Other Fossil
Helm d = -0.321852
DR = 4.023 HLR
DR ratio= 55.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

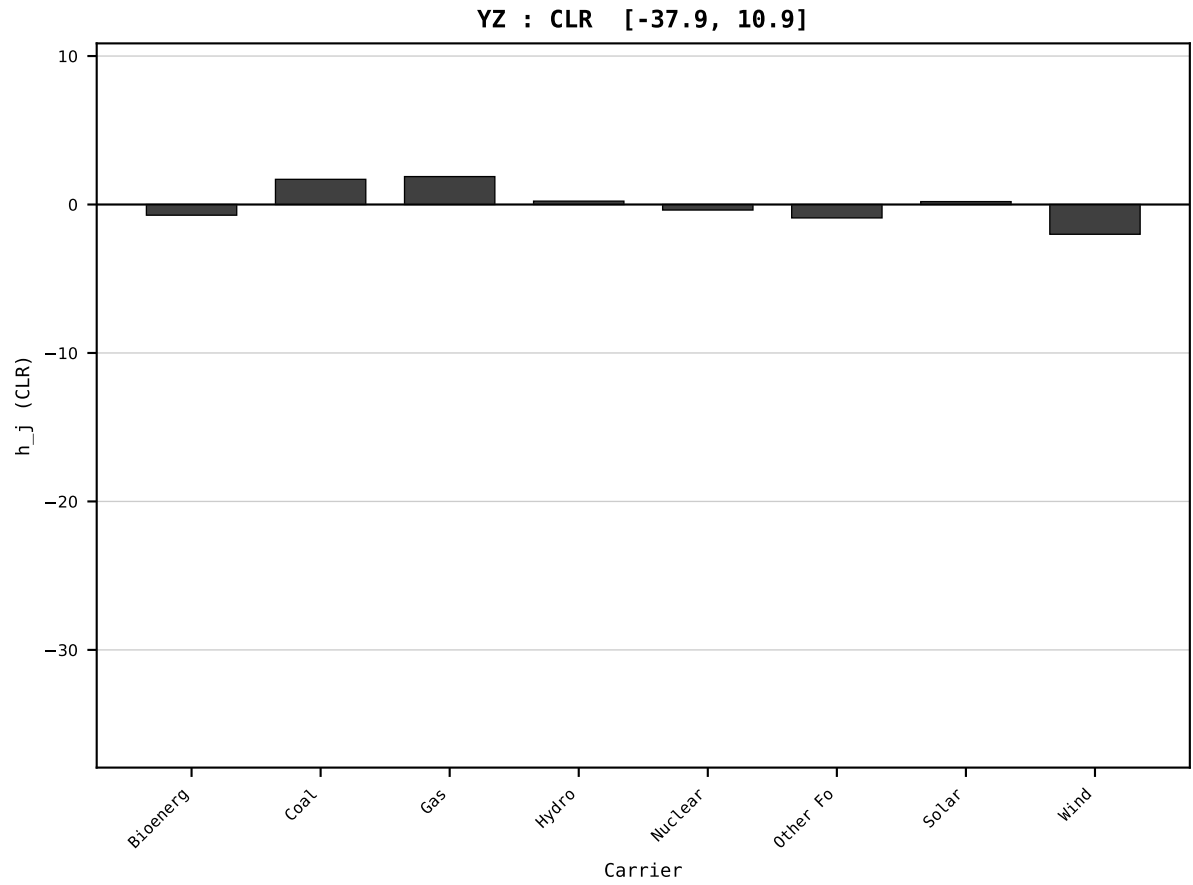
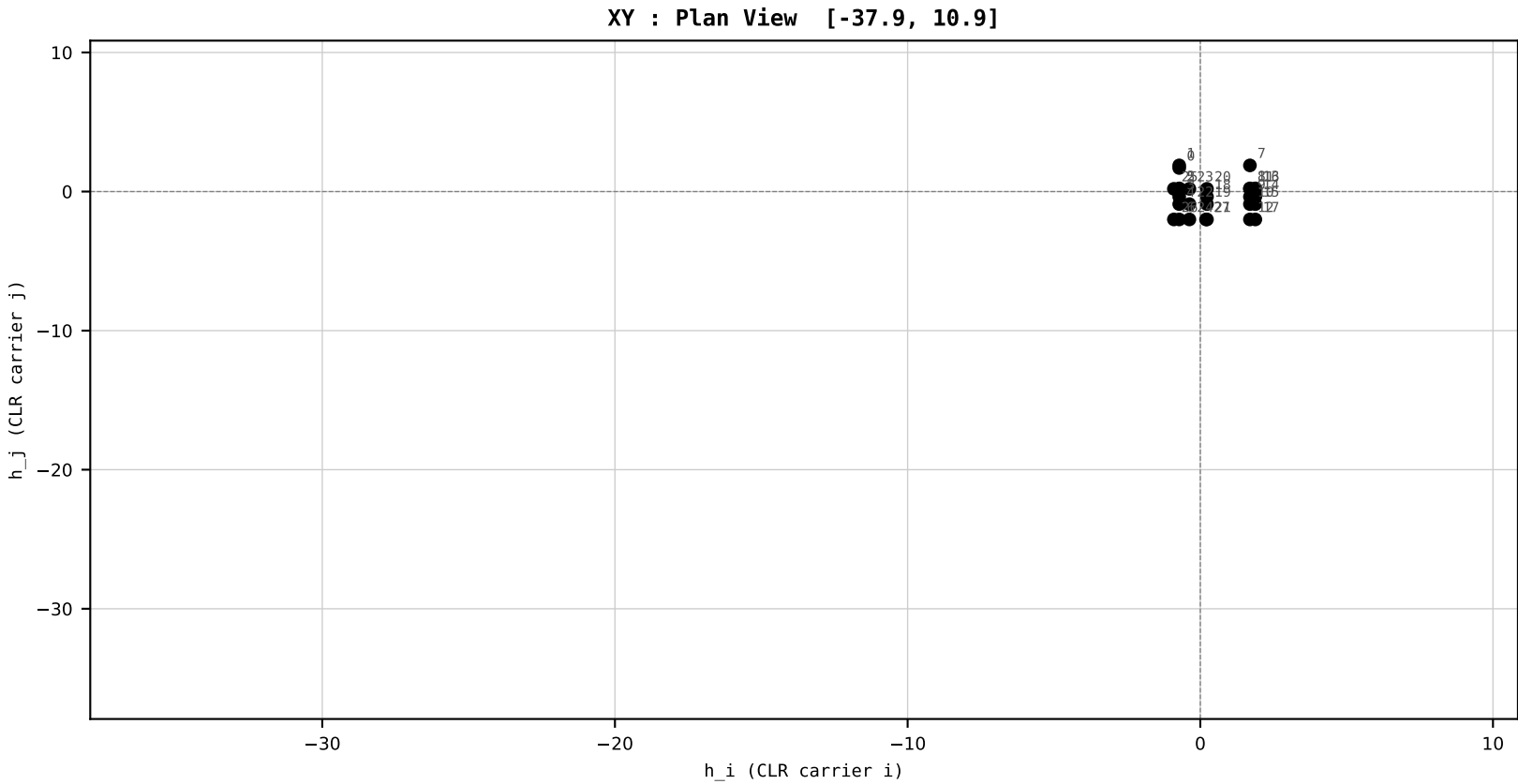
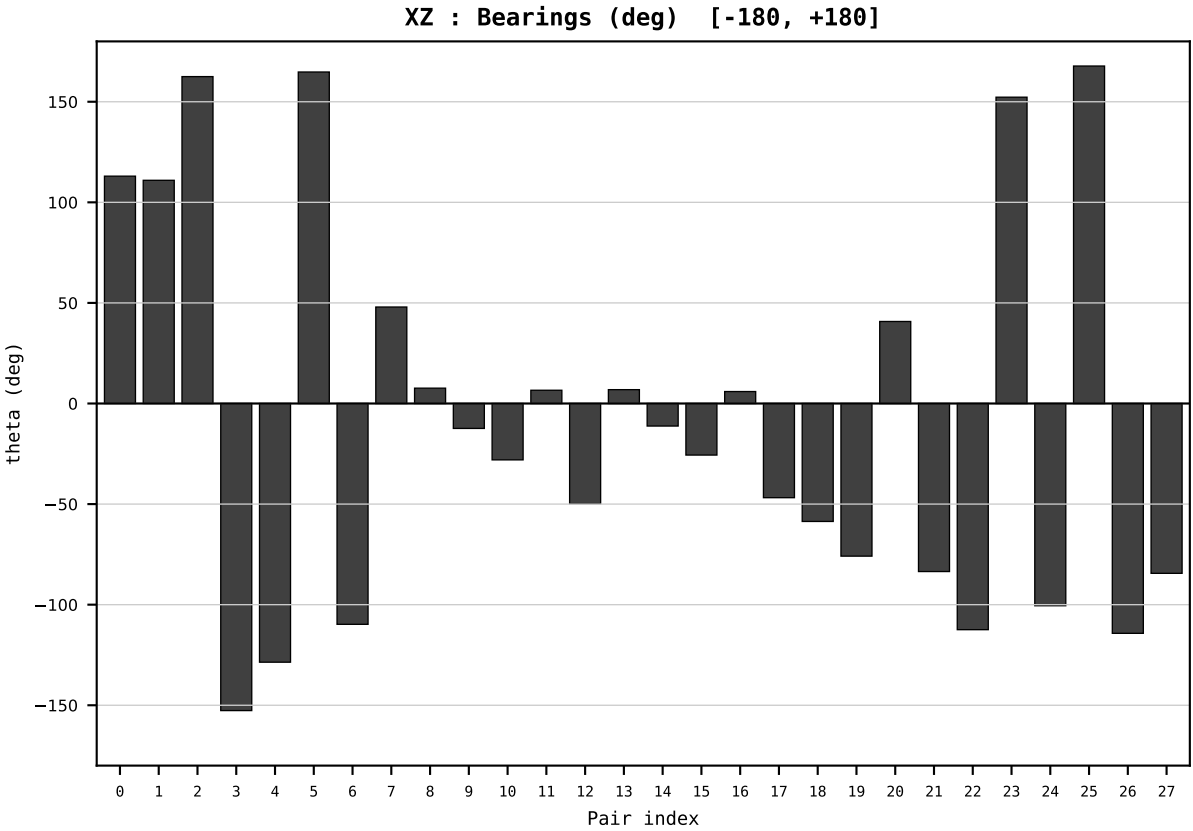
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.280146
Ring = Hs-2
E_metric= 11.9940
kappa_HS= 48.68
omega = 9.7926 deg
d_A = 0.591660
Helm = Nuclear
Helm d = -0.363464
DR = 3.885 HLR
DR ratio= 48.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

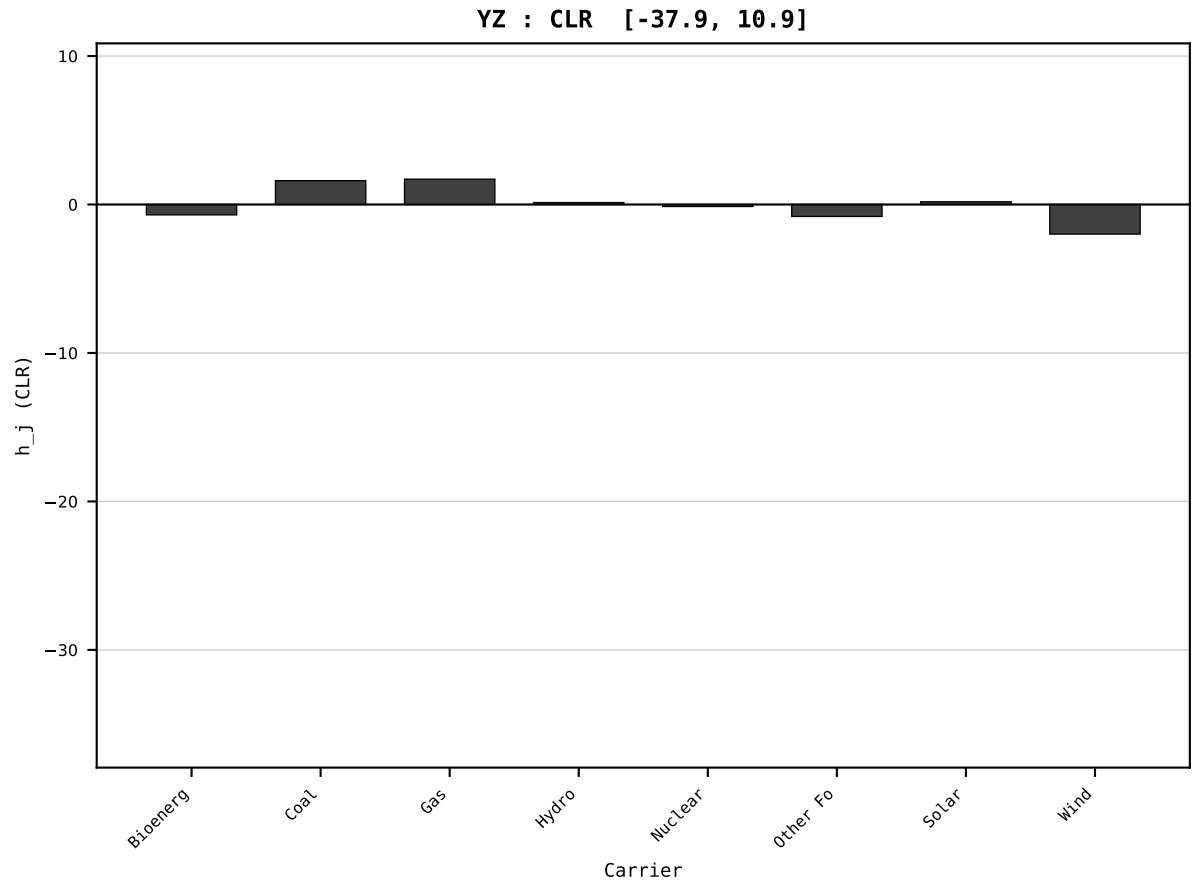
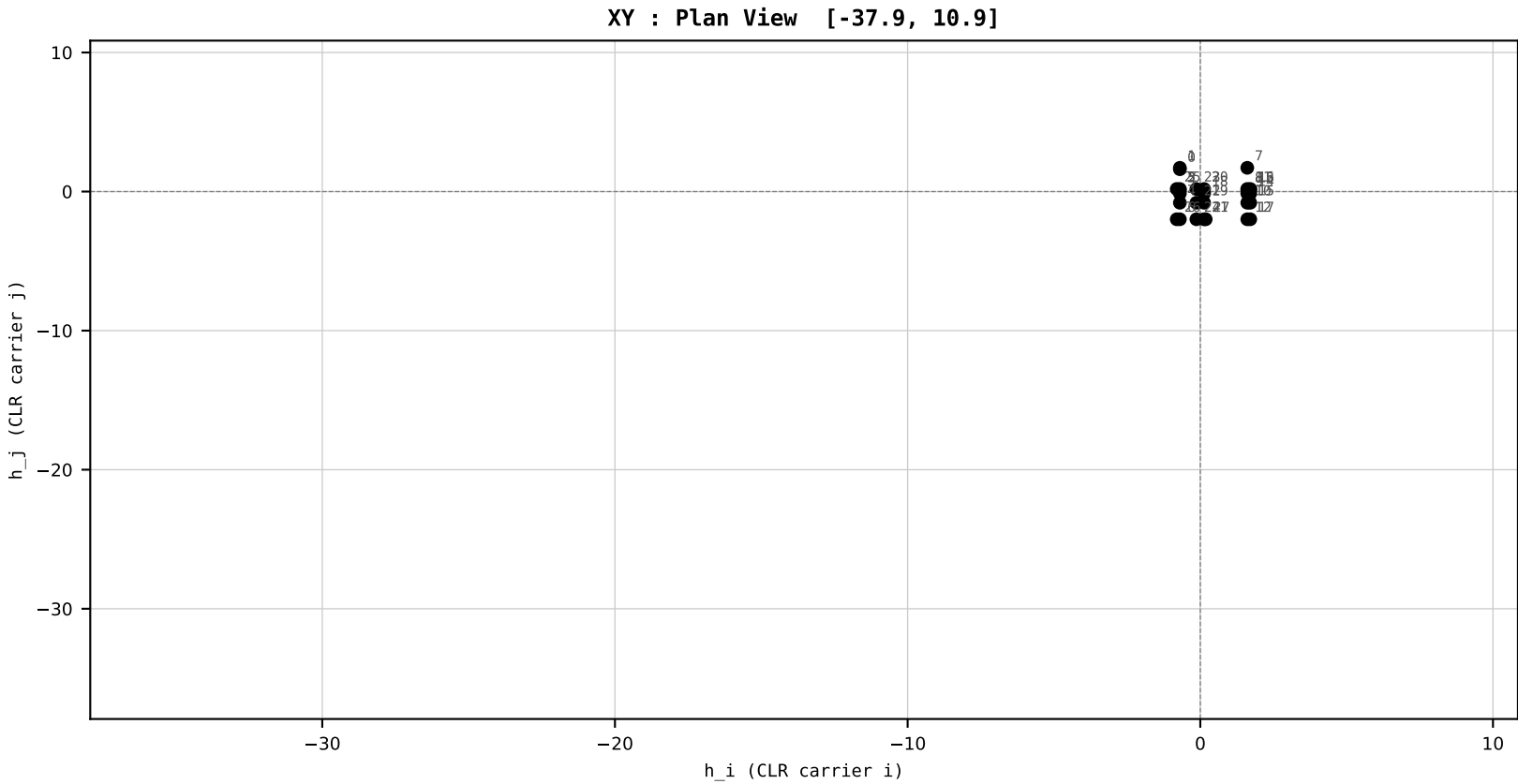
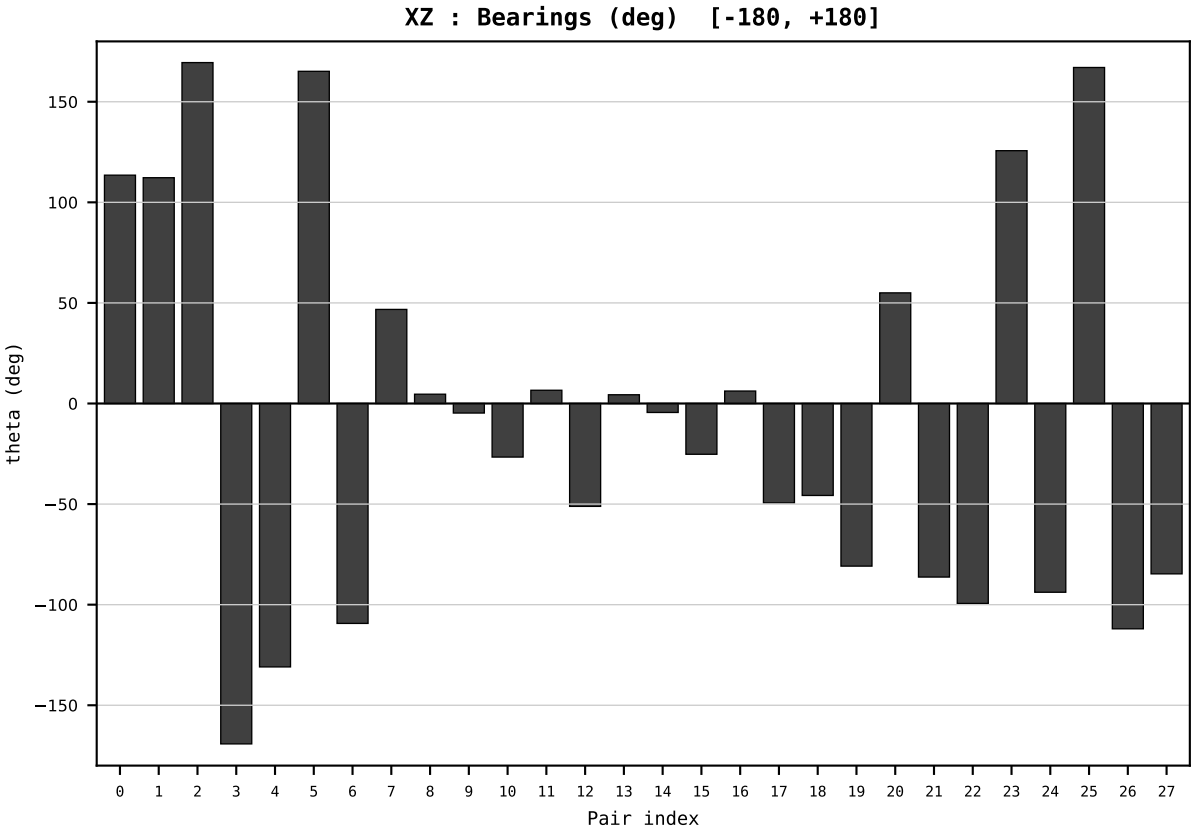
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.246834
Ring = Hs-2
E_metric= 10.6662
kappa_HS= 40.44
omega = 4.7105 deg
d_A = 0.339626
Helm = Nuclear
Helm d = +0.239503
DR = 3.700 HLR
DR ratio= 40.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

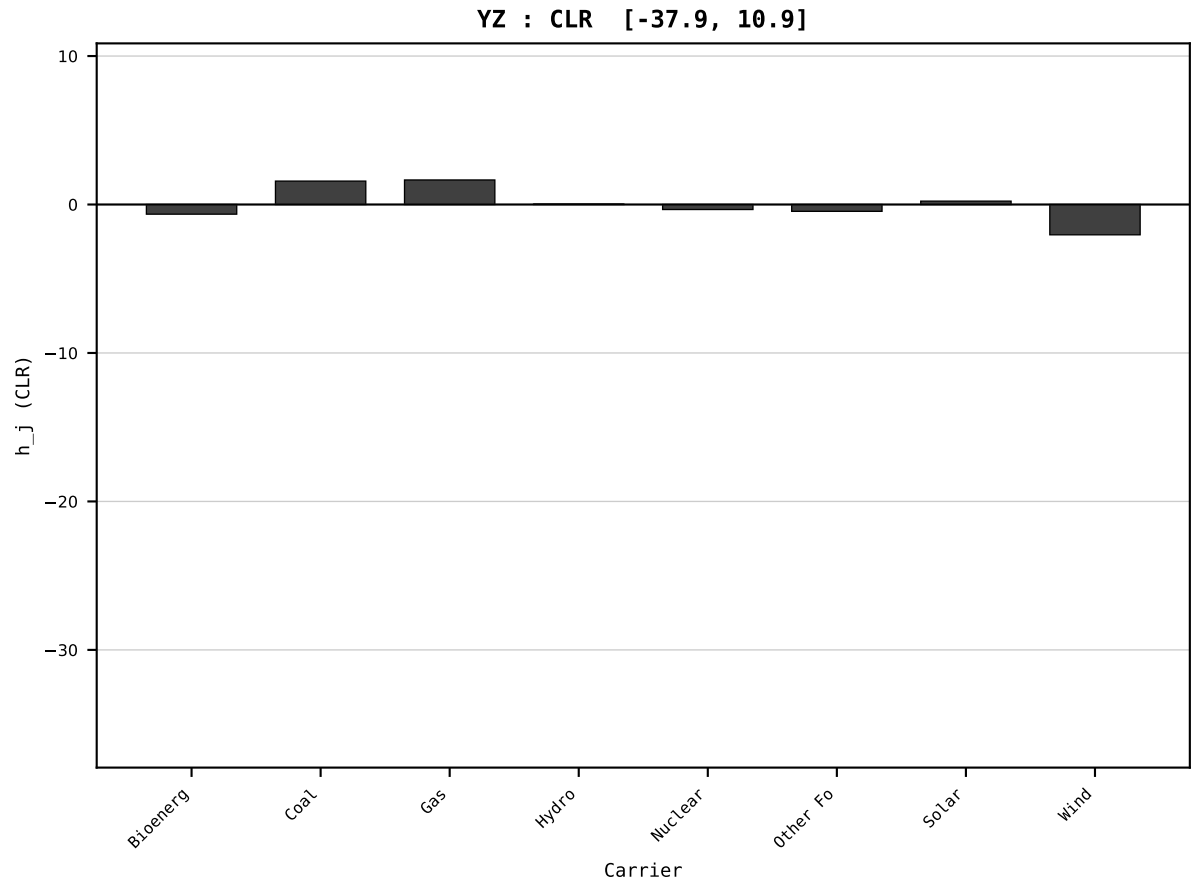
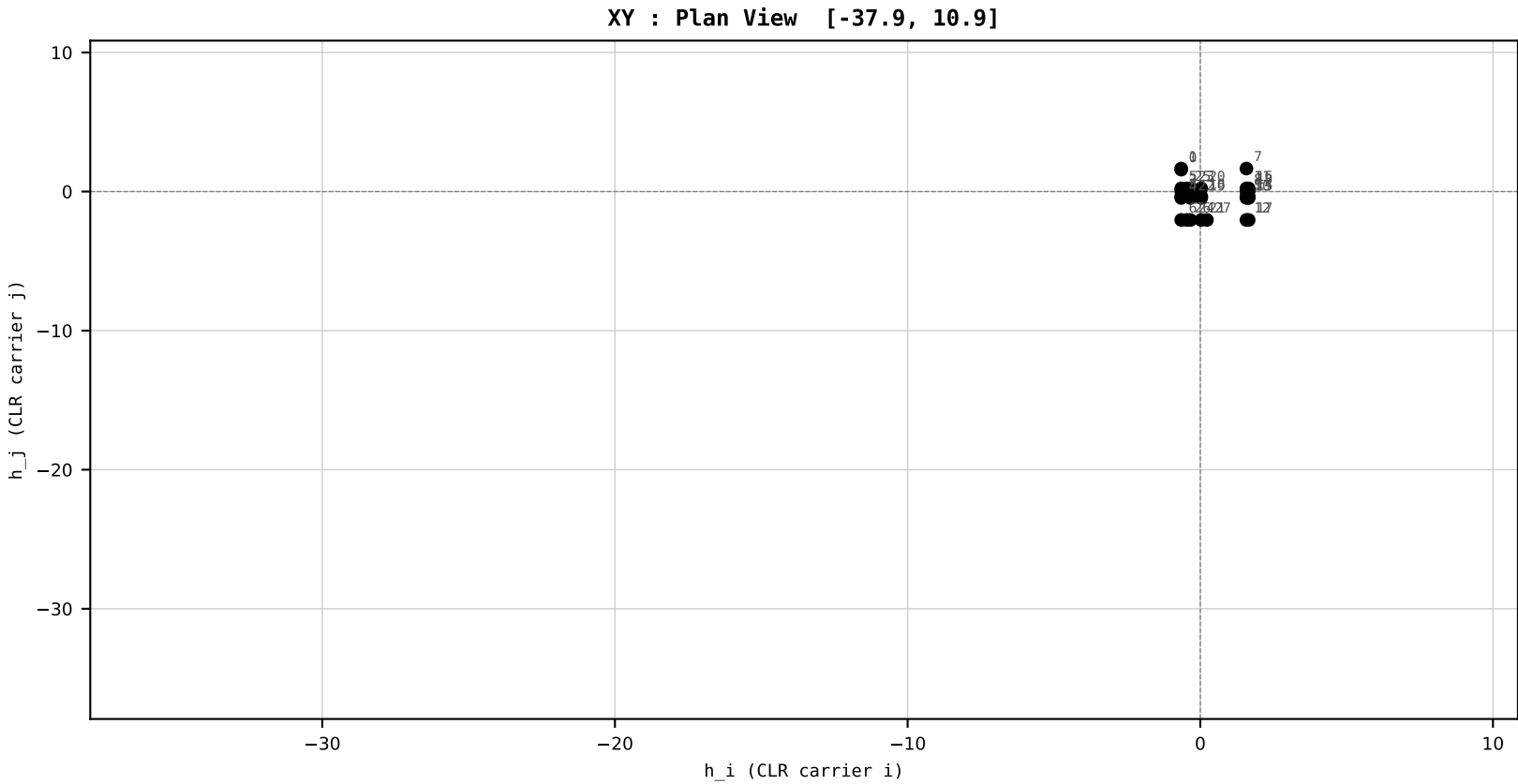
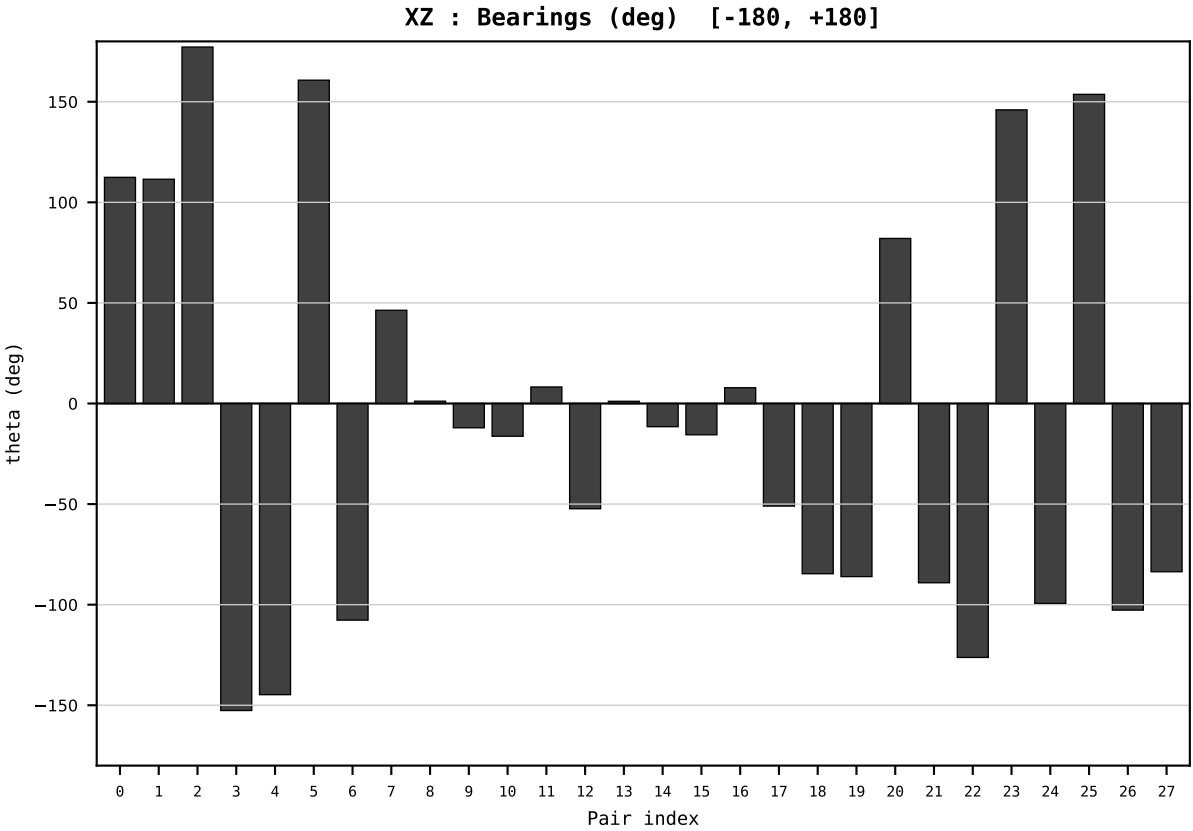
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.236919
Ring = Hs-2
E_metric= 10.1872
kappa_HS= 40.22
omega = 7.4261 deg
d_A = 0.424692
Helm = Other Fossil
Helm d = +0.344772
DR = 3.694 HLR
DR ratio= 40.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

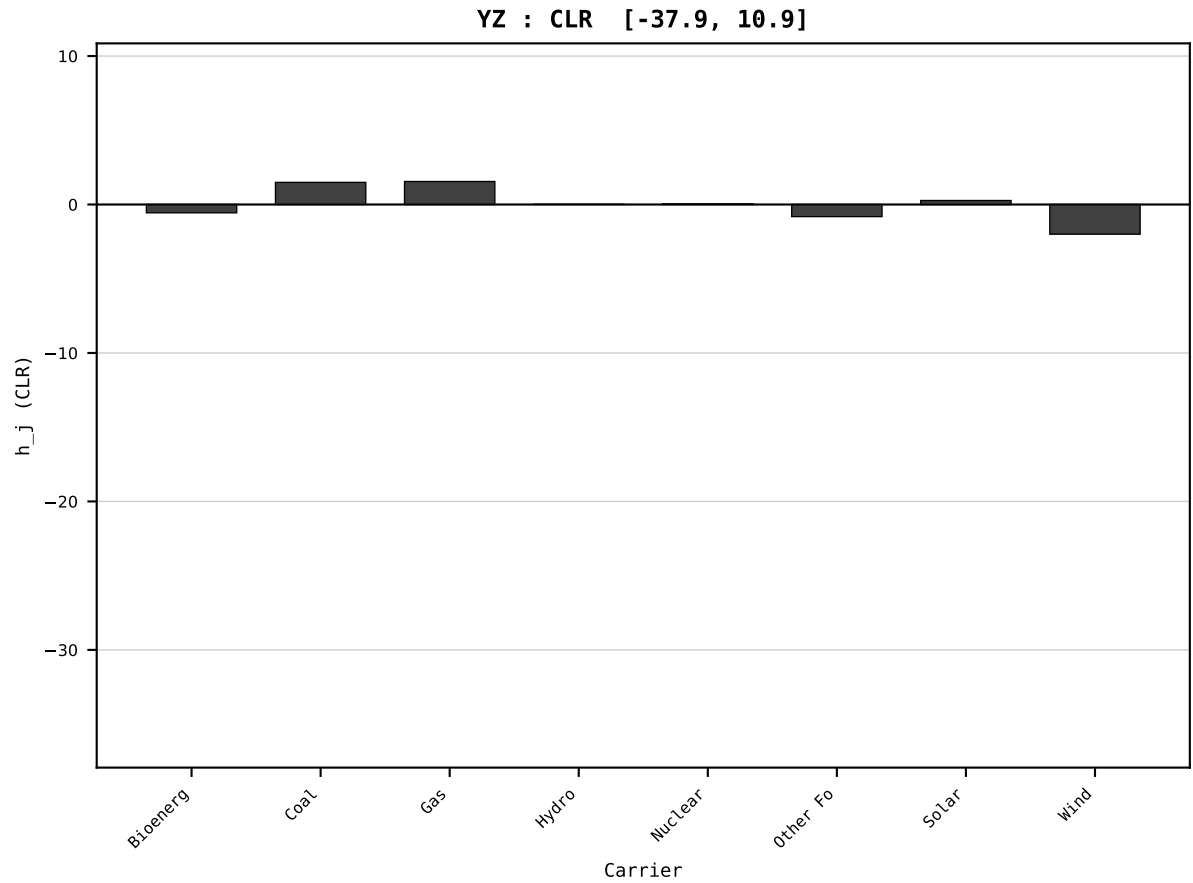
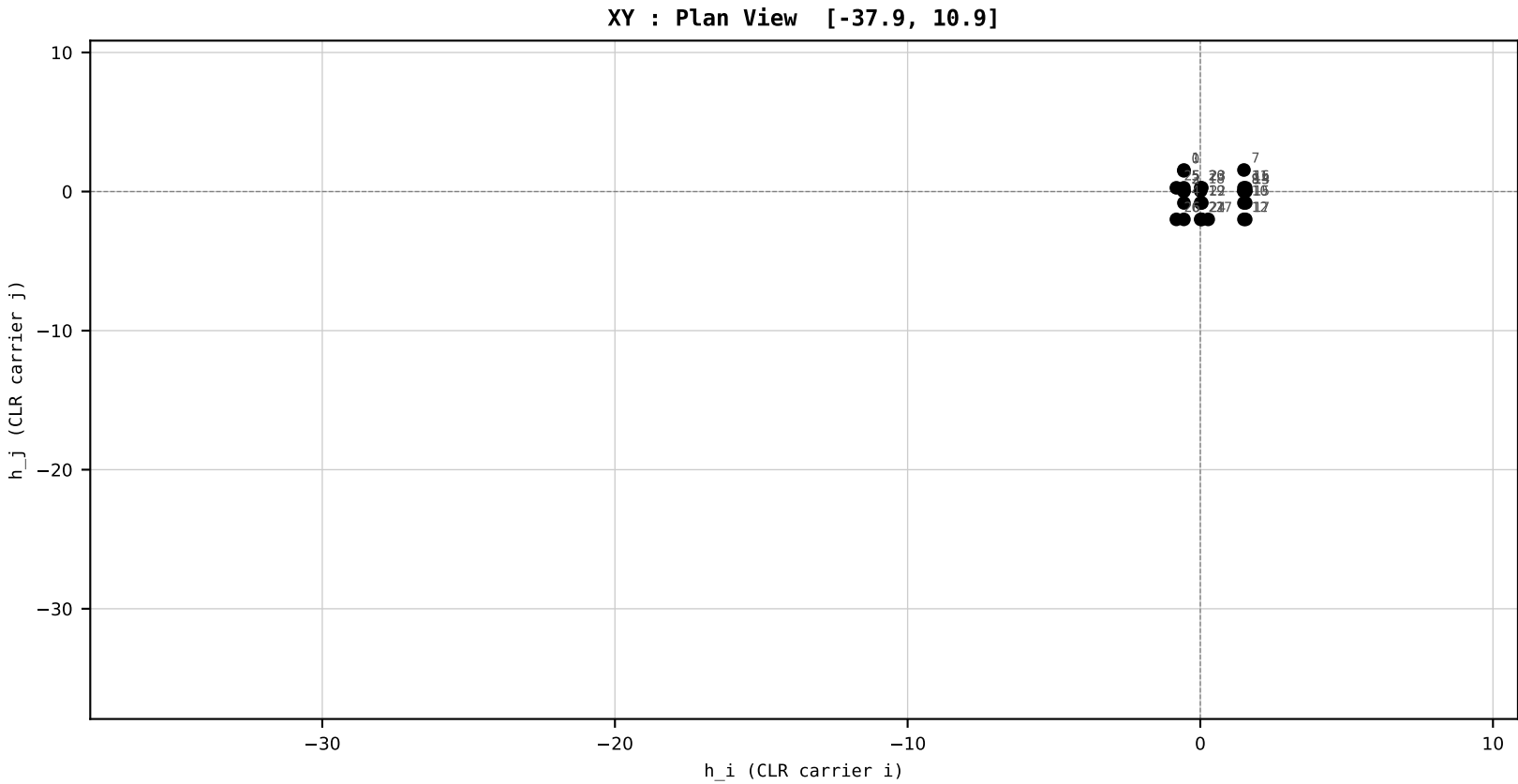
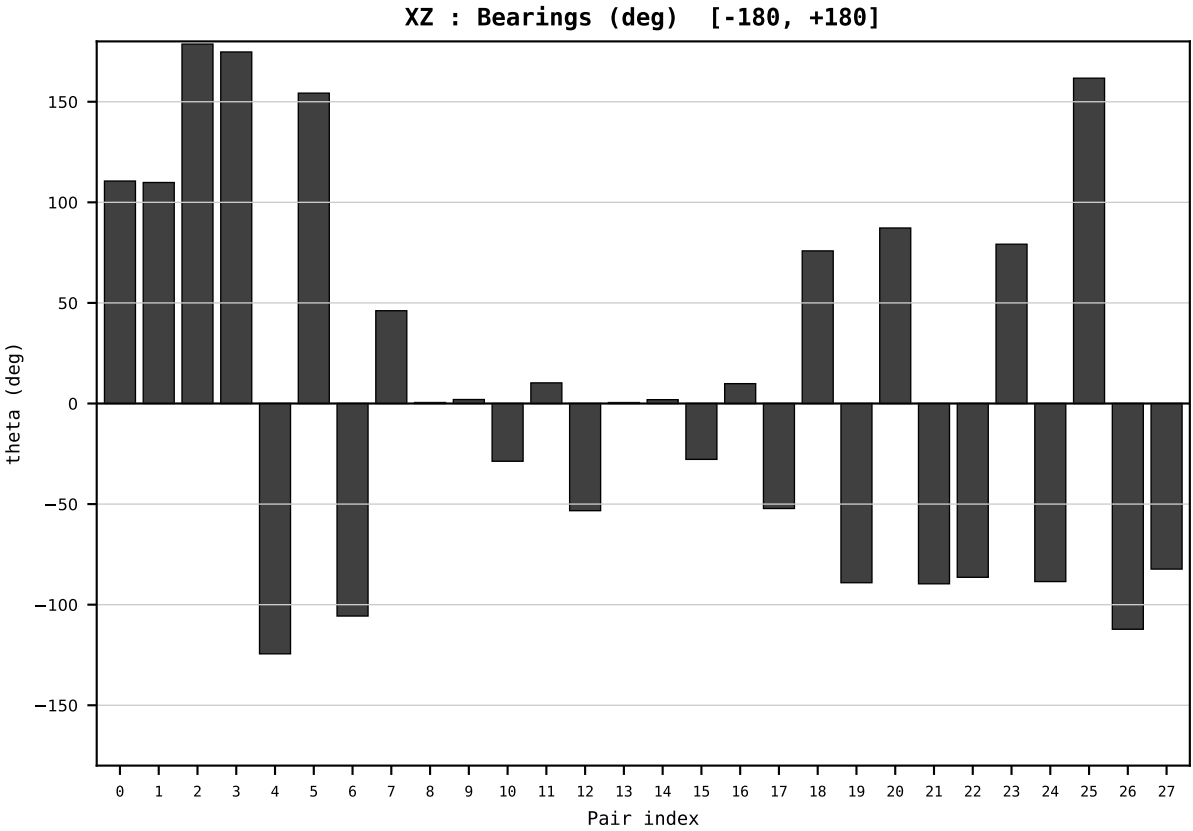
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.213214
Ring = Hs-2
E_metric= 9.6860
kappa_HS= 34.84
omega = 10.0007 deg
d_A = 0.555145
Helm = Nuclear
Helm d = +0.388708
DR = 3.551 HLR
DR ratio= 34.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

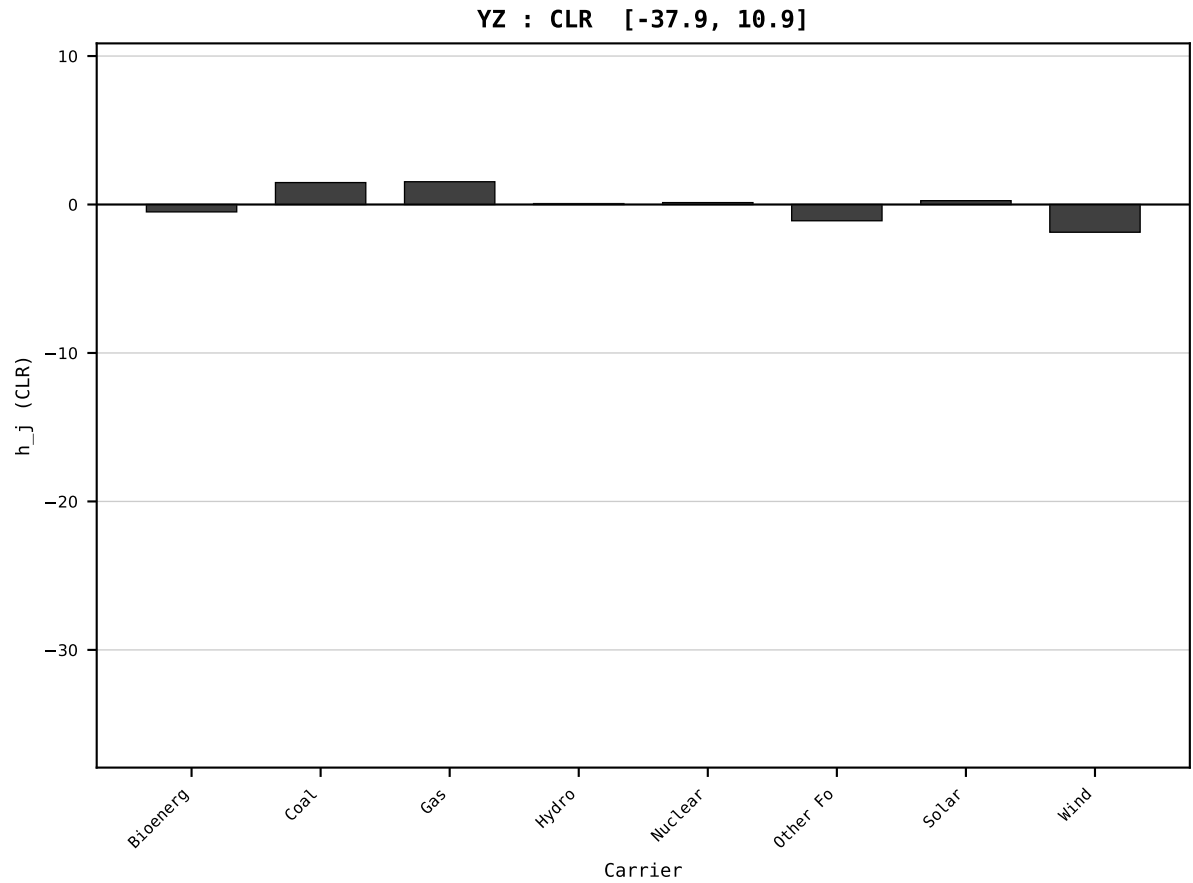
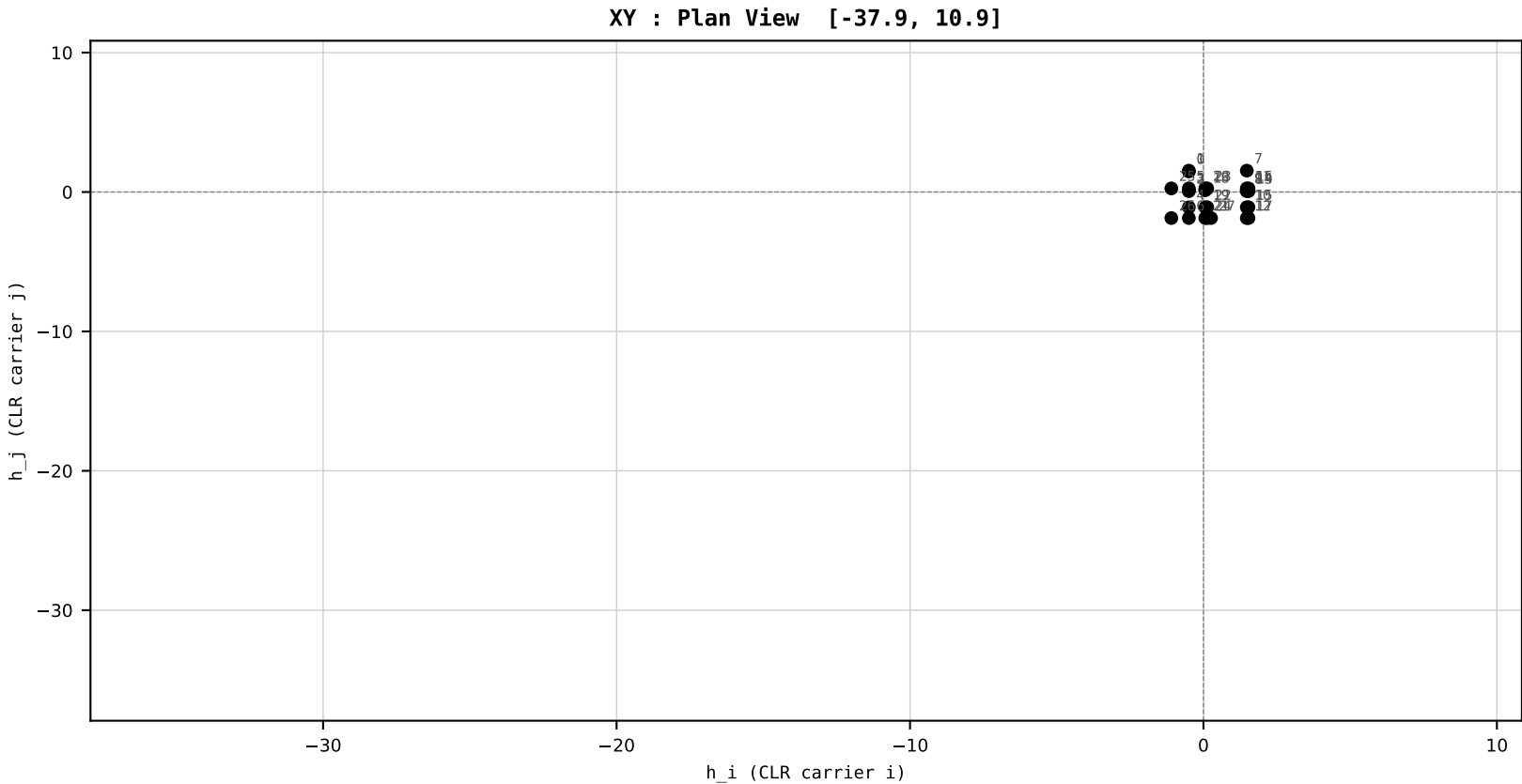
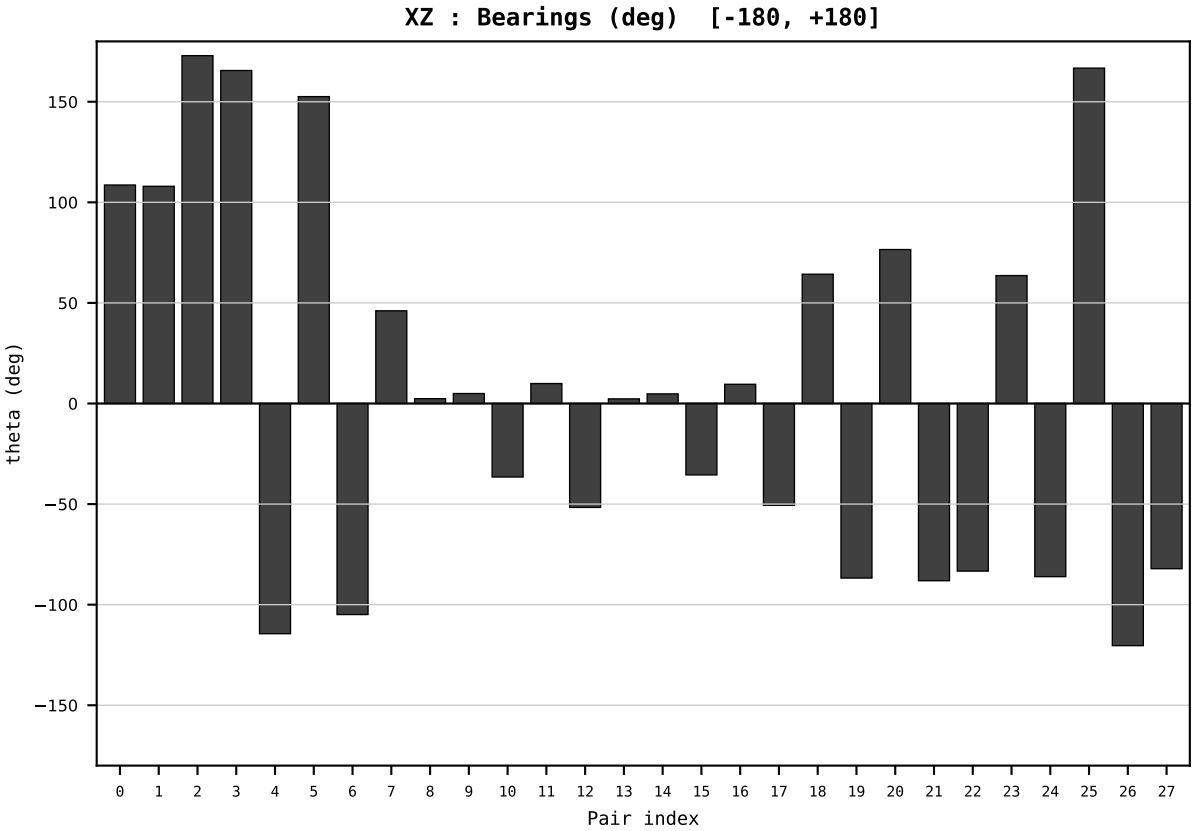
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.209872
Ring = Hs-2
E_metric= 9.5540
kappa_HS= 30.00
omega = 6.0418 deg
d_A = 0.327601
Helm = Other Fossil
Helm d = -0.277449
DR = 3.401 HLR
DR ratio= 30.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

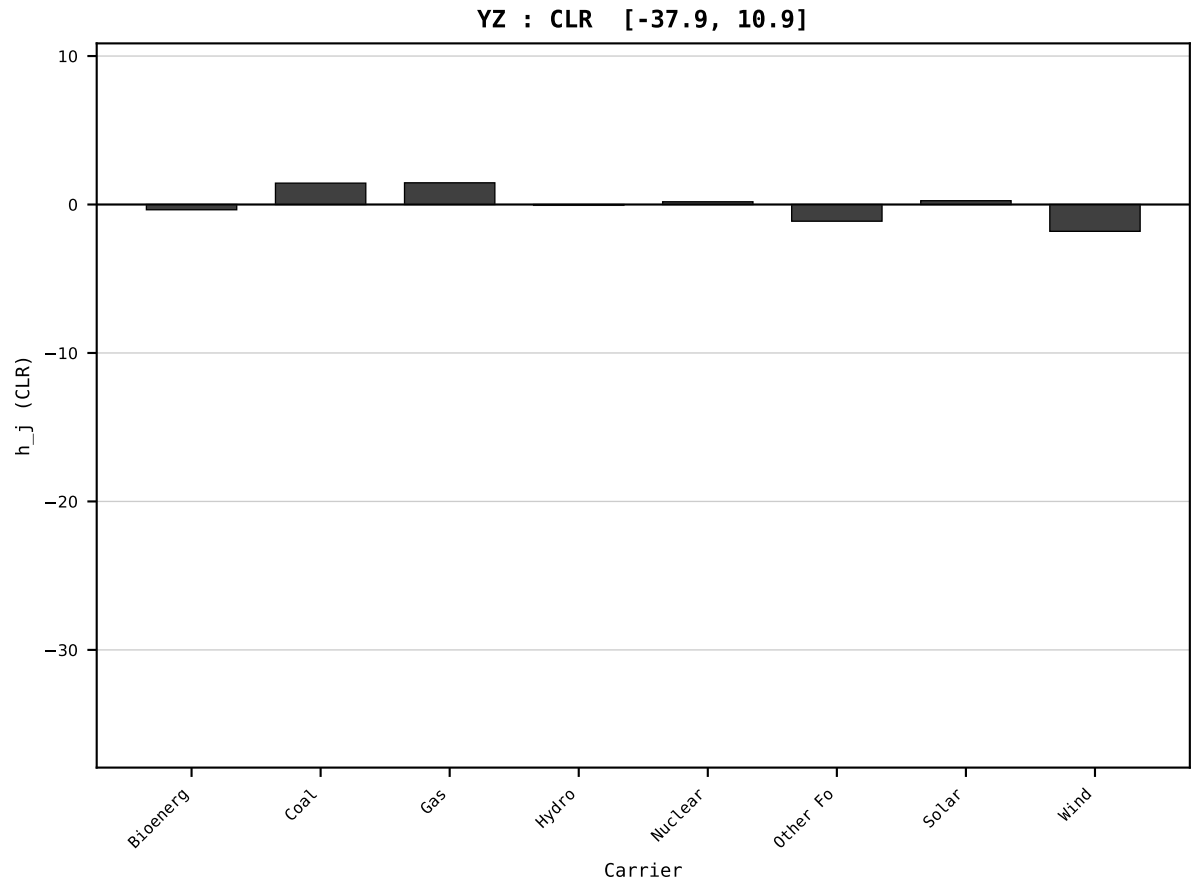
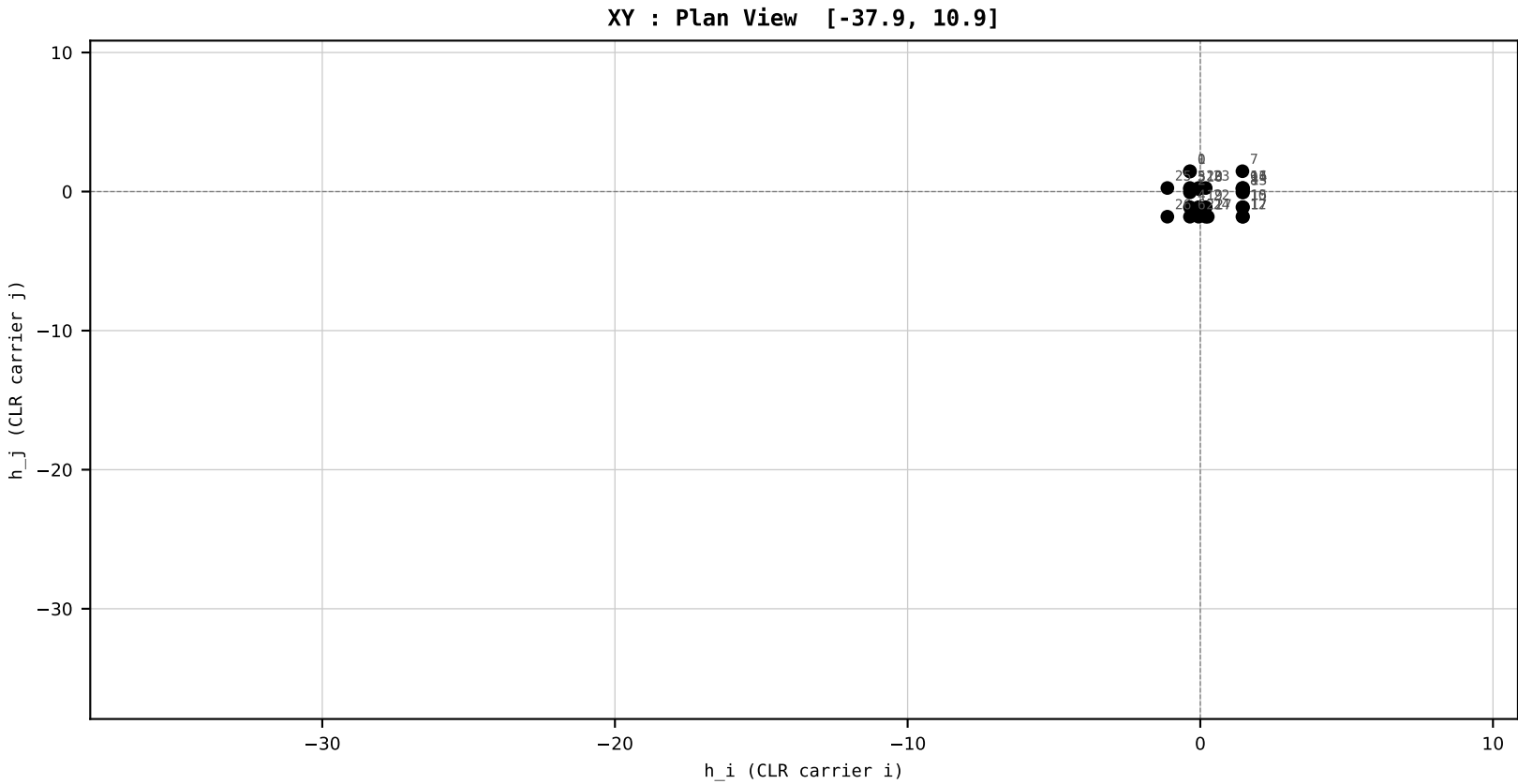
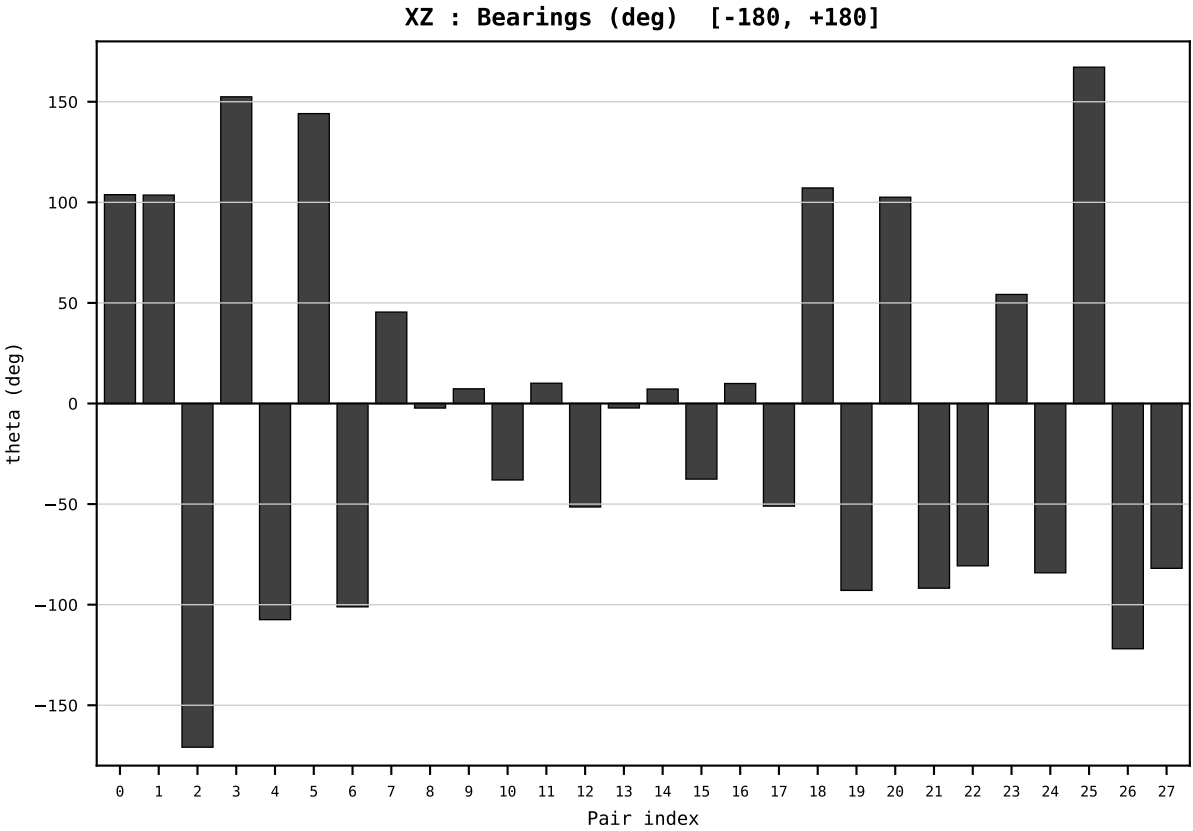
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

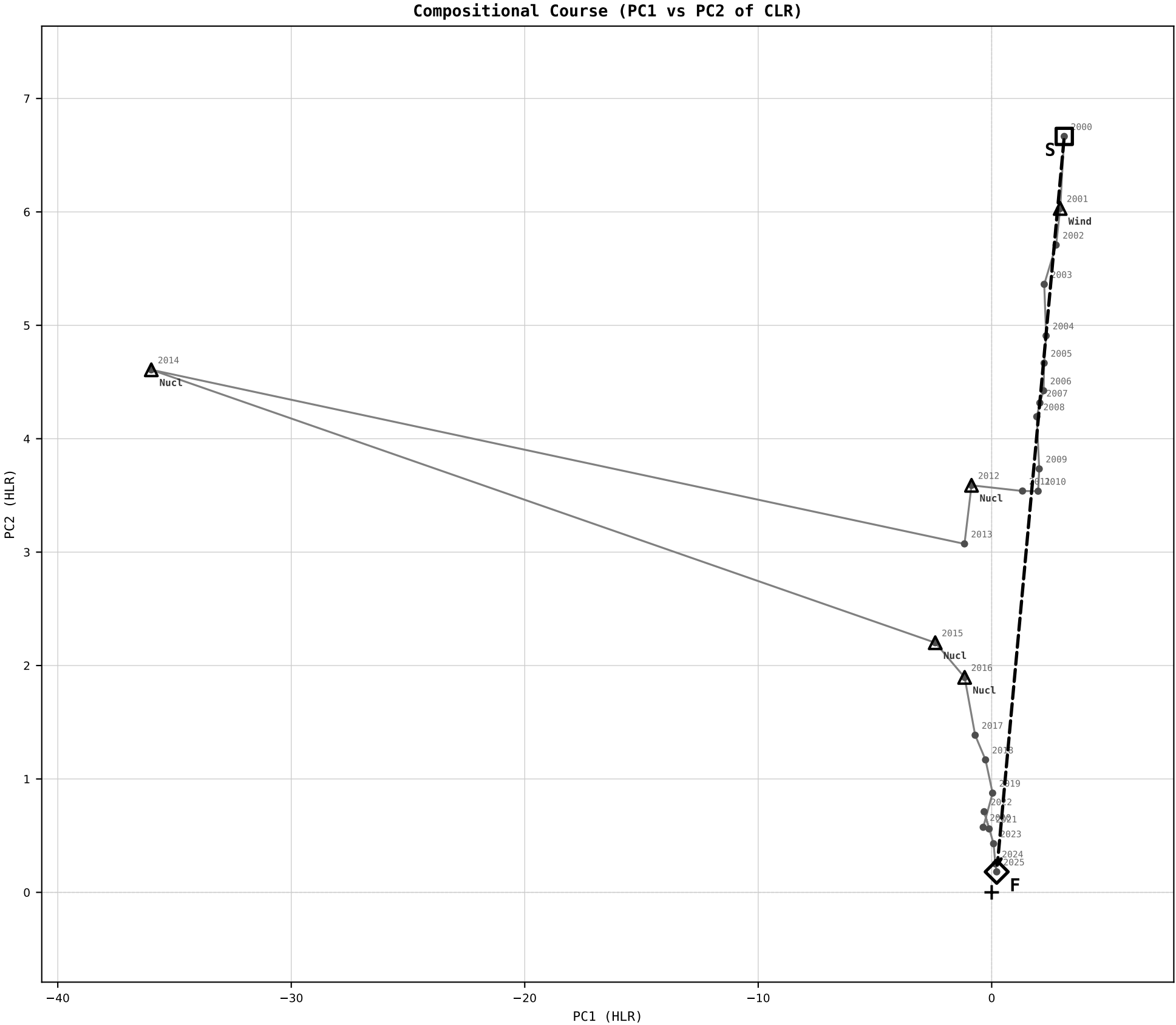
Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.197825
Ring = Hs-2
E_metric= 8.9676
kappa_HS= 26.27
omega = 3.7598 deg
d_A = 0.221651
Helm = Bioenergy
Helm d = +0.143563
DR = 3.268 HLR
DR ratio= 26.3x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 7.2233 HLR
Path length = 82.5468 HLR
Directness = 0.0875

DR start = 7.973 HLR
ratio = 2901.1x
DR final = 3.268 HLR
ratio = 26.3x

E_{metric} start = 70.5673
E_{metric} final = 8.9676

PC1 loadings:
Nuclear +0.934
Solar -0.184
Wind -0.152

PC2 loadings:
Solar -0.716
Other Fo +0.535
Wind -0.326

Top displacement events:
2014 Nuclear d=34.871
2015 Nuclear d=33.668
2012 Nuclear d=2.201
2016 Nuclear d=1.320
2001 Wind d=0.848

The instrument reads.
The expert decides.
The loop stays open.